

Affirmative Action in Centralized College Admissions Systems*

Nano Barahona[†]

Cauê Dobbin[‡]

Sebastián Otero[§]

October 4, 2026

Abstract

We study the consequences of affirmative action in college admissions. We develop an empirical framework that distinguishes the effects on marginally benefited and displaced students from the overall effects of the policy. The overall effects include equilibrium reallocations as benefited students leave seats that others fill and displaced students take seats elsewhere, displacing others in turn. We apply this framework to a large-scale policy in Brazil that required all federal institutions to reserve half their seats for racially and socioeconomically disadvantaged applicants. Exploiting admissions cutoffs, we find that marginally benefited targeted students attend and graduate from higher-quality programs. Their graduation gains are about twice the losses of marginally displaced non-targeted students, reflecting targeted students' weaker outside options. To evaluate the overall effects, we estimate a model of degree choices and potential outcomes. Among all reallocated students, benefited students still gain nearly twice what displaced students lose, but equilibrium reallocations make the policy displace about 30% more students than it benefits. Non-targeted students' losses therefore offset about two-thirds of targeted students' graduation gains, leaving a modest net gain across all students.

Keywords: Affirmative action; Higher education; Educational inequality; Centralized college admissions.

JEL codes: I23, I24, I28, D47, J15.

*We are indebted to Pascaline Dupas, Liran Einav, Matthew Gentzkow, Caroline Hoxby, and Paulo Somaini for invaluable guidance and support on this project. We also thank Claudia Allende, Luis Armona, Zachary Bleemer, Christopher Campos, José Ignacio Cuesta, Michael Dinerstein, Diego Jiménez, Patrick Kline, Tomás Larroucau, Neale Mahoney, Ursula Mello, Pablo Muñoz, Christopher Neilson, Cristóbal Otero, Pietro Tebaldi, Miguel Urquiola, Christopher Walters, and seminar participants at Stanford University for valuable comments and suggestions. We are grateful to Anna Popova for her insights, patience, and dedication to making this article better. We gratefully acknowledge financial support from the Stanford King Center on Global Development, Stanford Center for Computational Social Sciences, Stanford Institute for Economic Policy Research (SIEPR), Haley-Shaw Dissertation Fellowship, and NAEd/Spencer Dissertation Fellowship Program. [†]University of California, Berkeley and NBER. Email: nanobk@berkeley.edu. [‡]Georgetown University and NBER. Email: caue.dobbin@georgetown.edu. [§]Columbia University and NBER. Email: so2699@columbia.edu.

1 Introduction

About 25% of countries use some form of affirmative action (AA) to increase the representation of underrepresented groups in higher education (Jenkins and Moses, 2017). These policies give preferential treatment to applicants from marginalized backgrounds and are among the most controversial tools in education policy (Arcidiacono et al., 2015). The debate centers on a potential trade-off. AA can close existing gaps in access to selective colleges and promote equity and diversity, which are important pillars of a healthy democracy (Singer, 2011; Alon, 2015; Chetty et al., 2020a). At the same time, by prioritizing applicants from one group over more academically prepared applicants from another, AA may distort the allocation of students across colleges and move some away from the programs where they are best matched (Ellison and Pathak, 2021).

How these reallocations affect educational outcomes is theoretically ambiguous. If college selectivity and academic preparedness are complementary, lower-scoring students admitted under AA may struggle in the more demanding programs they enter—the central concern of the mismatch literature (e.g., Sander, 2004; Arcidiacono and Lovenheim, 2016).¹ Yet quasi-experimental evidence shows that academically marginal students can benefit substantially from attending selective colleges, with gains depending on the quality of their next-best alternative (Zimmerman, 2014; Bleemer, 2024; Mountjoy, 2026). The losses of displaced non-targeted students also depend on their alternatives. If targeted students have worse alternatives than the non-targeted students they displace, AA may produce large gains for targeted students who would otherwise attend lower-quality programs or not enroll at all, and smaller losses for displaced students with attractive alternatives, such as high-quality private colleges that many targeted students cannot afford.

The overall effects of AA, however, extend beyond the targeted students admitted through the additional reserved seats and the non-targeted students displaced by the corresponding reduction in open seats. Displaced students may enroll in their next-best alternatives and displace others in turn, so the gains and losses of the students directly affected need not equal the net effect of the policy. Evidence on these “knock-on” effects is limited, especially for large-scale policies.

We examine the overall effects of AA in Brazil’s centralized college admissions system, where applicants compete for seats at public institutions using scores from the national entrance exam. To do so, we study a large-scale regulation implemented between 2013 and 2016 that required all federal institutions to reserve 50% of seats in every degree program for students from public high schools (“targeted students”). Before the reform, targeted students were substantially underrepresented at federal universities relative to their 87% share of high school enrollment. Within the targeted group, the policy set aside most reserved seats for students from low-income households or marginalized racial groups.

Brazil is an ideal setting to study AA in college education for three reasons. First, researchers seldom observe how admissions decisions are made, let alone which applicants qualify for a reserved seat. Brazil’s centralized system overcomes this challenge: applicants compete in explicitly defined open or reserved tracks, each with its own cutoff, and we observe the applications and the assignment

¹A separate argument is that admissions test scores may be statistically biased against the targeted group, in which case AA could improve match quality by partly offsetting this bias (Jacob and Rothstein, 2016).

rules. Second, these applications and rules let us account for the indirect effects generated by displaced students by modeling how they cascade through the system. Third, the policy reallocates seats at highly selective and consequential institutions. Like flagship state universities in the United States, federal universities are among the most prominent in their states. They are also free of tuition and attract high-scoring applicants from across the socioeconomic distribution.

We combine administrative records from Brazil’s centralized admissions system (SISU), the national university entrance exam (ENEM), and the Higher Education Census. The SISU records cover about 2.4 million applicants per year in the 2013–2016 cohorts. For every applicant, we observe their application scores, ranked degree choices, admissions tracks, offers, and take-up decisions. ENEM provides exam scores and a registration questionnaire completed by every test-taker, which reports household income, parental education, race, and high school attended. Finally, the census allows us to follow applicants through 2019 and observe enrollment, persistence, and graduation across the universe of roughly 53,000 Brazilian degree programs, including those outside the centralized system.

We begin by documenting the college-quality gap between targeted and non-targeted students in 2012, the last pre-policy cohort. This gap summarizes differences in access to selective colleges and in the quality of the degrees students complete. Growing evidence suggests that college quality—not just college attainment—is an important driver of economic mobility (Zimmerman, 2019; Londoño-Vélez et al., 2026; Chetty et al., 2026). We proxy degree quality by ranking every degree program in Brazil from 1 to 100 based on the average entrance-exam score of incoming students (Black and Smith, 2006; MacLeod et al., 2017). We define the enrollment and graduation gaps as differences in average ranks between the two groups, assigning students who do not enroll or graduate to the bottom of the scale.² The pre-policy gaps were large: targeted and non-targeted students had average enrollment ranks of 17 and 48, respectively, a gap of 31 points. At graduation, the gap was 28 points.

Federal programs contributed disproportionately to these gaps. Although they represented only 16% of all programs, they accounted for about half of those in the top quality quartile. Admission to these programs depended heavily on entrance exam scores, on which targeted students scored 1.09 standard deviations below non-targeted students. By reserving seats for targeted students, AA substantially expanded their access to federal programs. Between 2012 and the policy’s full implementation in 2016, their share of enrollment in the most selective decile rose from 37% to 53%, while the share held by targeted students of color more than doubled, from 14% to 31%. The expansion came through lower admissions thresholds. In 2016, reserved-seat cutoffs for these degrees were on average 56.0 points, or 0.67 standard deviations (s.d.), below open-seat cutoffs.

Two main forces determine how much this expansion narrows the college-quality gap and how the gains and losses are distributed across groups: outside options and whether students persist to graduation. We show that, in this setting, outside options differ sharply between the two groups. For a given test score, targeted applicants are 17% more likely not to enroll at all and, when they enroll outside the federal system, attend programs 11 points lower on our quality measure.³ At the

²Rank-based outcomes are standard in the intergenerational-mobility literature, where mean percentile ranks summarize distributional outcomes and their differences across groups measure disparities (Chetty et al., 2014, 2020b).

³Among those enrolling in private programs, non-targeted students attend programs with tuition roughly 25% higher, consistent with targeted students’ limited access to high-quality private options.

same time, lower admissions cutoffs raise concerns about whether students persist in the degrees they enter. We show that, among students entering the same degree, those with lower ENEM scores are more likely to drop out. Consistent with this concern, dropout in the most selective decile rose from 16% to 23% between 2012 and 2016, with a larger increase among targeted students.

To quantify how these forces shape the policy’s overall effects, we model how AA operates within a centralized admissions system. In the model, students rank degree programs, each of which reserves a share of seats for targeted students. A deferred acceptance algorithm generates offers that can be summarized by admissions cutoffs that are weakly lower for targeted applicants (Dur et al., 2018; Rios et al., 2021). Because more than half of the offers are declined, we add a take-up stage and an aftermarket that reallocates declined seats through waitlists. The framework distinguishes two objects of interest. The first is the gain to a marginally benefited student and the loss to a marginally displaced student. Regression discontinuity (RD) designs at admissions cutoffs estimate these local effects and are widely used to study the returns to selective colleges (e.g., Hoekstra, 2009; Zimmerman, 2014; Mountjoy, 2026). The second, our main object of interest, is the overall effect of changing the reservation share, including gains and losses across all affected students. This distinction matters even for small policy changes, since an additional reserved seat can affect students far from the initial admissions cutoffs, including those enrolled in other degree programs.

We measure both objects of interest using enrollment and graduation ranks. Enrollment rank captures the initial reallocation induced by AA, while graduation rank also incorporates whether the gains persist through college graduation. College completion and persistence in more demanding fields, such as STEM, are standard outcomes in the AA literature because they capture the margins along which academic mismatch may arise (e.g., Arcidiacono et al., 2011, 2016; Kapor, 2025). Our graduation-rank outcome combines completion with the quality of the degree completed, providing a more granular measure than broad field categories typically used in the literature.

We estimate the effects of AA on marginal students by measuring the returns to enrolling in the submitted federal degree relative to students’ next-best alternatives. We compare applicants within a narrow window around their relevant cutoff and instrument enrollment in the submitted federal degree with crossing the admissions threshold. At enrollment, the marginal targeted student gains 30 rank points, while the marginal non-targeted displaced student loses 16 points. These differences are consistent with targeted students having weaker fallback options. Despite concerns about higher dropout, targeted students gain 9.8 rank points at graduation, roughly twice the 4.6-point loss for non-targeted students.

Moving from these local effects to the overall effect requires predicting student assignments under alternative reservation shares. To do so, we bring our model of centralized admissions to the data and estimate a joint model of degree preferences and offer take-up. Using preferences revealed by applicants’ rank-order lists, we estimate group-specific random utility models in which degree valuations depend on distance and on interactions of degree quality and field of study with students’ test scores. We estimate take-up from observed enrollment decisions by comparing the value of the offered degree to that of the outside option. We allow the value of the outside option to vary with student characteristics and the number of nearby degree options. Because the rules of the assignment mechanism are known, we can use these estimates to simulate admissions under alternative

reservation shares and recover the resulting cutoffs and assignments. The model closely reproduces the observed ordering of degree programs by their admissions cutoffs.

Fully characterizing these counterfactuals also requires estimating the graduation outcomes students would attain under these new assignments. Predicting them, however, creates an endogeneity challenge. Students who enroll in a program may have greater unobserved academic potential, so observed graduation outcomes may reflect selection into the program rather than the effect of attending it. Following [Abdulkadiroğlu et al. \(2020\)](#), we use a control-function approach that allows unobserved tastes to affect both sorting and outcomes, extending it to account for selection at the offer take-up stage. Identification of the model combines cutoff-induced variation in students' choice sets with variation in choices generated by distance to campus, conditional on a rich set of student characteristics ([Agarwal et al., 2025](#)).

We use our estimates to simulate assignments and outcomes as the reserved-seat share rises from zero to 50%. These counterfactuals hold degree capacities and preferences fixed, abstracting from behavioral responses such as changes in pre-college human capital ([Akhtari et al., 2024](#); [Tincani et al., 2026](#)) and strategic school choice for AA eligibility ([Mello, 2023](#); [Concha-Arriagada, 2026](#)). We also assume that the policy affects outcomes only through changes in students' assignments, ruling out channels such as peer effects and changes in grading standards. We discuss why these assumptions are reasonable in our setting, drawing on evidence from the law's staggered implementation and differential exposure to the policy across degree programs. We organize the counterfactuals around three exercises.

In the first exercise, we trace how increasing the reserved-seat share changes admissions cutoffs, choice sets, and assignments. The average targeted student's choice set expands from 175 to 315 degrees, while that of the average non-targeted student contracts from 463 to 320, out of 1,719 federal degrees in our sample. The resulting reallocation is far from one-for-one: about 34,000 targeted students benefit by entering the federal system or moving to a more-preferred degree, while the number of non-targeted students displaced from the system or to a less-preferred degree is 30% larger. This substantial excess displacement reflects meaningful knock-on effects.

In the second exercise, we measure gains and losses in enrollment and graduation ranks among reallocated students. At enrollment, benefited students gain 18.8 rank points on average, while displaced students lose 14.4, consistent with targeted students' weaker outside options. Seven years after application, benefited students gain 6.5 graduation-rank points on average, while displaced students lose 3.3. Thus, 34% of the enrollment gain among benefited students remains at graduation, but their average graduation gain is still 1.93 times the average loss among displaced students. These average graduation gains and losses are smaller than the corresponding regression discontinuity estimates of 9.8 and 4.6 rank points for marginal students.

In the third exercise, we aggregate gains and losses across all students, including those unaffected by the policy, to measure AA's net effect and how it changes the college-quality gap. Relative to no AA, a 50% reservation share closes 36% of the enrollment gap and 9% of the graduation gap. Averaged over all students in each group, graduation rank rises by 0.80 points for targeted students and falls by 0.56 points for non-targeted students. Both changes narrow the gap, but the gains and

losses partly offset each other in the aggregate. Total losses to non-targeted students offset 67% of total gains to targeted students, leaving a net increase of 0.13 points in average graduation rank across the full population. A one-for-one comparison of gains and losses, whether using average counterfactual effects or marginal RD estimates, overstates the net gain because it ignores the excess displacement generated by knock-on effects.

Our paper relates to several strands of the literature. A large body of work examines whether AA creates a mismatch between targeted students’ academic preparation and the demands of the programs they attend, potentially harming them (e.g., [Sander, 2004](#); [Alon and Tienda, 2005](#); [Arcidiacono et al., 2011](#)). Quasi-experimental evidence, however, indicates that students targeted by AA and similar policies benefit from expanded access to selective institutions, including public universities in California ([Bleemer, 2022, 2024](#)), engineering colleges in India ([Bertrand et al., 2010](#); [Bagde et al., 2016](#)), and a federal university in Brazil ([Francis-Tan and Tannuri-Pianto, 2018](#)). Recent work in Chile ([Carlana et al., 2026](#)) and China ([Chen et al., 2026](#)) suggests that the extent of the potential mismatch may depend on the generosity of the policy. While this literature informs our analysis of targeted students, we ask a different question, weighing their gains against the losses to non-targeted students to assess how AA changes inequality and aggregate educational outcomes.

A smaller set of papers speaks more directly to this comparison by studying the distributional consequences of AA and related admissions policies for students who gain and lose access to selective colleges. Using Texas’s Top Ten Percent rule, [Black et al. \(2023\)](#) find gains in enrollment and graduation among students pulled into selective public universities, but no detectable graduation or earnings losses among those pushed out. Studying California’s ban on race-based AA, [Bleemer \(2022\)](#) similarly finds large losses from reduced access among targeted students but little evidence of gains among White and Asian applicants near the admissions margin. We find the same asymmetry at the admissions margins and show that it reflects stronger outside options among students who lose access, a mechanism that may also explain the findings of these earlier studies. We also show, however, that marginal comparisons omit knock-on effects as displaced students alter other applicants’ admission opportunities.⁴ These effects are quantitatively important in our setting, substantially reducing the policy’s estimated aggregate gain. They may also be relevant in California, where the ban on race-based AA at public universities was followed by a cascade of reallocations across institutions ([Bleemer, 2022](#)).

Several papers are conceptually close to ours in using counterfactual analysis to study how changes in admissions policies affect student assignments and outcomes. In decentralized U.S. markets, [Arcidiacono \(2005\)](#) and [Howell \(2010\)](#) use partial-equilibrium models of applications, admissions, and enrollment to simulate the removal of AA, while [Chetty et al. \(2020a\)](#) reallocate students across fixed college seats to study mobility and earnings. Closest to ours, [Kapor \(2025\)](#) estimates how Texas’s percent plan affects GPA and persistence among students who gain or lose access, allowing applications and admissions standards to adjust in equilibrium. Relative to the decentralized settings examined in this literature, our centralized admissions system permits a simpler and more

⁴A related methodological literature develops nonparametric estimators of global treatment effects in centralized mechanisms with interference ([Karnani, 2023](#); [Munro, 2026](#)). Our model-based approach instead recomputes the assignment under each counterfactual and incorporates all resulting reassignments into the aggregate effect.

transparent market-wide analysis: because we observe applicants’ AA eligibility and the full assignment mechanism, including colleges’ admissions policies, we can evaluate any reservation share by rerunning the mechanism rather than modeling or estimating admissions behavior.⁵

Finally, we contribute to the literature on AA in Brazilian higher education. [Mello \(2022\)](#) studies the same nationwide reform, exploiting variation in the pace of its implementation to show that expanding reserved seats increased enrollment among public-school, non-White, and low-income students. Using a similar empirical strategy, [Vieira and Arends-Kuenning \(2019\)](#) reach comparable conclusions for universities that adopted AA policies before the federal mandate. Other studies examine outcomes beyond student composition but focus on individual universities. [Francis and Tannuri-Pianto \(2012\)](#) and [Francis-Tan and Tannuri-Pianto \(2018\)](#) study the effects of AA on academic performance and longer-run outcomes at the University of Brasília, while [Melo \(2025\)](#) examines its effects on major choice at the Federal University of Espírito Santo. We complement this work by quantifying both the marginal and system-wide effects of the nationwide reform, including knock-on reallocations across degrees.

The remainder of the paper is organized as follows. Sections 2 and 3 describe the setting and data, document the college-quality gap, and show how AA changed access to federal universities. Section 4 distinguishes marginal from overall policy effects, and Section 5 estimates the gains and losses at the admissions cutoffs. Section 6 estimates the model of preferences, offer take-up, and potential outcomes. Section 7 quantifies gains and losses under alternative reservation shares, including knock-on effects and changes in the college-quality gap. Section 8 concludes.

2 Institutional background, regulation, and data

2.1 Higher education in Brazil

Brazil’s higher education system is highly liberalized. Roughly 77% of incoming students enroll in private, fee-charging colleges, and the remaining 23% attend public institutions, which charge no tuition and are run by the federal, state, or municipal government.⁶ Federal universities are typically the most prominent universities in their state and play a role similar to that of flagship state universities in the United States. Because they are tuition-free and prestigious, they are highly oversubscribed, admitting about 6% of applicants. In 2016, 107 federal institutions served nearly 1.2 million students across more than 7,300 degrees, and enrolled 15% of in-person incoming students.⁷

As is common in many countries, students apply directly to a degree program: a specific major at a specific institution (e.g., Economics at the University of São Paulo). Bachelor’s programs typically last three to five years, teaching programs three to four years, and vocational programs two to three years. Some bachelor’s programs, including medicine, take six years. In addition to selecting a degree, students must choose an instructional period, or “shift”: morning, afternoon, night, or full-

⁵Related work studies other design features of centralized admissions systems. In Chile, [Kapor et al. \(2024\)](#) examine the integration of previously off-platform options, while [Larroucau and Rios \(2026\)](#) study reapplication rules.

⁶The sector shares reported here refer to in-person degrees. Online degrees, which are almost entirely private, account for 28% of incoming students. [Barahona et al. \(2025\)](#) study the expansion of this sector in Brazil.

⁷These comprise 63 universities (about 87% of federal enrollment) and 44 vocational institutions.

time. Finally, dropout rates are high. About a third of federal entrants drop out within four years of admission (Appendix Figure B.1).

2.2 College admissions: ENEM and SISU

The national university entrance exam (ENEM) is a highly competitive standardized exam consisting of four multiple-choice tests—natural science, social science, language, and math—and one written essay. Several million students in Brazil take this exam every year. Students can use ENEM test scores to gain access to public institutions through a centralized admissions system (SISU).⁸ In 2013, about 52% of the 3.6 million ENEM takers applied to a degree program in SISU.⁹

Students participating in SISU are matched to degree programs following an iterative deferred acceptance mechanism.¹⁰ The system assigns students to degree programs based on their application score, which is a weighted average of their test scores based on degree-specific weights for each ENEM component. In contrast to the standard deferred acceptance mechanism, students are sequentially asked to submit up to two ranked program choices throughout several “trial” days. At the end of each day, the system produces a cutoff score that represents the lowest score necessary to be accepted into a specific program, and students are allowed to change their degree program preferences given the new information. The results of the last day are final and determine the initial offers for every degree program. Individuals who are not offered admission to their top preference can opt in to be waitlisted in that degree program.

2.3 The affirmative action regulation

In August 2012, the Brazilian federal government passed the *Law of Social Quotas* (Lei de Cotas, no. 12.711/2012), requiring all federal institutions to reserve half of their admission spots in each degree program for students from public high schools. This group accounted for 87% of high school enrollment in 2012 (INEP, 2013). Within this group, most seats are reserved for low-income students and for Black, Brown, and Indigenous students. The regulation sought to reverse racial and income inequality in university access under the rationale that disadvantaged groups should not be underrepresented relative to their proportion in the population. The law received strong endorsement in Congress, with just one dissenting vote, and is supported by both targeted and non-targeted populations in Brazil (Folha de S.Paulo, 2022).

Under the regulation, only students who completed high school entirely in public schools are eligible to compete for the reserved vacancies, while the other half of the vacancies remain open for broad competition. We refer to students eligible for reserved seats as *targeted* and to the rest as *non-targeted* throughout the paper. Targeted students can be admitted through either a reserved or

⁸SISU was introduced in 2010, and federal institutions adopted it quickly. In 2013, the federal seats offered through SISU amounted to about 55% of incoming federal students admitted through a selection process, rising to 83% by 2016 (Appendix Table A.1).

⁹The academic year runs from March to December. Students take the ENEM in late October or early November, apply to SISU in January, and begin their studies in March. Some degrees also open additional seats for the second round of admissions through SISU in June. Students taking the ENEM in a given year can only use their score to participate in the SISU process in the following year, in either the first or second round of applications.

¹⁰See Bó and Hakimov (2022) for formal properties of the mechanism.

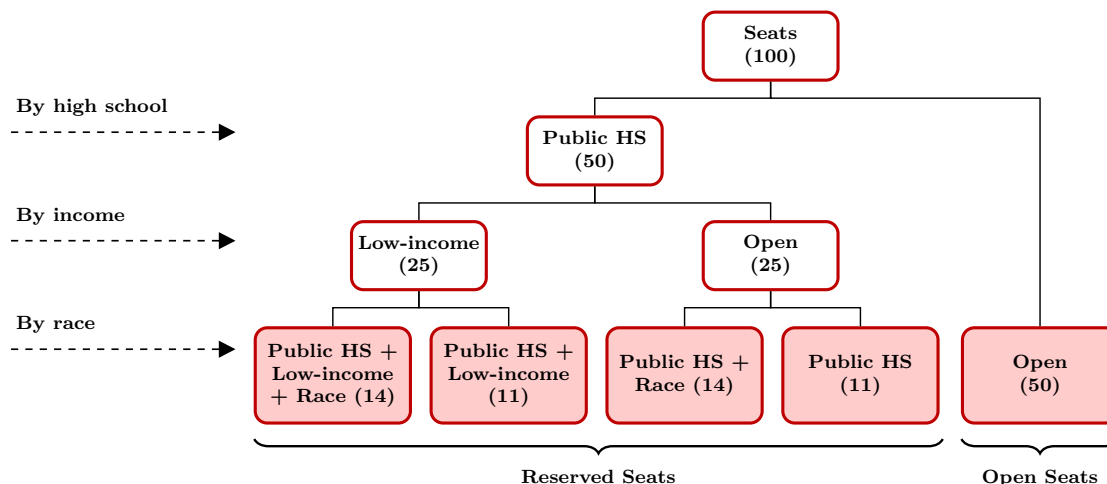


Figure 1: Affirmative action regulation

Notes: This figure shows how the regulation allocates 100 seats in a degree program once the policy was fully phased in. *Low-income* denotes students with monthly household income per capita at or below 1.5 times the minimum wage, and *Race* denotes Black, Brown, and Indigenous students. The number of seats reserved by race depends on the racial makeup of the state. The figure uses the racial shares for Minas Gerais in 2012.

an open seat, while non-targeted students compete only for open seats.

To address persistent disparities in college access, many federal universities had adopted AA policies of their own before the law.¹¹ In 2012, 24% of federal seats offered through SISU were reserved, with reservation shares varying widely across degree programs. The law established a common rule for all federal institutions, phased in over four years. Starting in 2013, institutions were required to reserve at least 12.5% of their seats for eligible students, with the minimum rising by 12.5 percentage points per year until it reached 50% in 2016. Compliance was virtually perfect (Appendix Figure A.1).

Figure 1 summarizes the allocation of seats within each degree program once the policy was fully phased in. For every 100 seats, 50 are AA seats for public high school graduates. Of those reserved seats, 25 are set aside for students with monthly household income per capita at or below 1.5 times the minimum wage, a threshold of about 1,000 BRL in 2013 (roughly US\$470). The remaining 25 are open to public high school graduates regardless of income. Within each category, seats are further reserved for Black, Brown, and Indigenous students based on the racial makeup of the state where the institution is located. Together, these racial groups accounted for about 51% of Brazil’s population in the 2010 Census.¹² These rules create four AA admissions pools and one open pool to which students can apply. Applicants receive an offer if their application score exceeds the cutoff for their chosen degree program and admissions pool.

¹¹These policies date to the 2000s, at state as well as federal universities, and were based on socioeconomic and racial criteria. See Francis and Tannuri-Pianto (2012) and Estevan et al. (2019) for evidence from the University of Brasília and the University of Campinas, and Francis-Tan and Tannuri-Pianto (2024) for an overview of AA in Brazil.

¹²In 2010, the shares were 47.5% White, 43.4% Brown, 7.5% Black, 1.1% Asian, and 0.42% Indigenous (IBGE, 2011).

2.4 Data

We use three different data sources, which we combine using individual-level unique identifiers.

Entrance exam scores. We have access to test-score data for all students taking the national university entrance exam, ENEM. We observe these data for 2009–2015, corresponding to the 2010–2016 admissions cycles. The data include scores on each component of the exam and answers to a detailed demographic and socioeconomic questionnaire. The ENEM is scored on a common scale anchored to the 2009 cohort, which makes performance comparable across years (INEP, 2011). We summarize a student’s performance by averaging the five components of the exam: the four multiple-choice areas and the essay. Throughout the paper, we scale test-score differences by the s.d. of the 2012 distribution, the last exam taken before the policy’s first admissions cycle. One s.d. is about 82 points.

Centralized admissions process. We complement test scores with records from the centralized admissions system, SISU. These data cover applications to federal and state institutions between 2010 and 2016. We focus on applications to federal institutions, since state institutions were not required to comply with the AA regulation. For every applicant, we observe the rank-order list, the admissions pool requested for each option, the program they were offered admission to, if any, and whether they took up the offer. For every combination of degree program and admissions pool, we observe the admissions cutoff.

Higher Education Census. The third dataset we use in our analysis is the Brazilian Higher Education Census. It covers every enrollment in any degree program, including private ones, so we can follow each student’s educational path between 2009 and 2019. The data are at the degree-student level and include graduation, dropout, or current enrollment status at the end of the academic year, as well as the exact semester of enrollment in the program. The data are of high quality, as most institutions report directly from their administrative records.

3 The college-quality gap and the role of affirmative action

The stated aim of the AA regulation was to reduce racial and socioeconomic inequality in higher education by changing admissions to federal universities (Brasil, 2012; Tribunal de Contas da União, 2022). Before estimating the effects of the policy, we document this inequality: the gap in college quality between targeted and non-targeted students, and the role of federal degrees in sustaining it. We then show how the policy changed who enters federal universities and present evidence on the two forces that shape its consequences: academic preparedness and outside options.

3.1 The college-quality gap at enrollment and graduation

College quality has several dimensions, including instructional quality, prestige, peer composition, and labor-market networks. To measure the disparity in access to high-quality colleges between targeted and non-targeted students, we proxy for these dimensions using the average entrance-exam score of a program’s incoming students, a standard approach in the literature (Black and Smith, 2006; MacLeod et al., 2017). To put this measure on an interpretable scale, we rank all in-person

and online programs in the country from 1 to 100 based on this score. In Appendix C, we validate this measure against other quality proxies, including graduates’ wages, the Ministry of Education’s quality index, and tuition fees for private degrees.

We define the *college-quality gap* as the difference in average degree ranks between the two groups.¹³ We measure the gap at enrollment using the rank of the degree students enter, and at graduation using the rank of the degree they complete. Students who do not enroll or graduate are assigned the bottom of the scale for the corresponding outcome. Each measure therefore combines the extensive margin of enrollment or completion with the intensive margin of degree quality.

Among students finishing high school who took the entrance exam for the 2012 admissions cycle, the last before the policy, the average *college enrollment rank* was 17 for targeted students and 48 for non-targeted students, a gap of roughly 31 rank points. On the extensive margin, targeted students were much less likely to enroll in college: 67% did not enroll in any degree program in the year of application, compared with 38% of non-targeted students. On the intensive margin, conditional on enrollment, non-targeted students enrolled in programs ranked about 25 rank points higher.

We measure graduation within seven years of application.¹⁴ The average *college graduation rank* was 13 for targeted students and 41 for non-targeted students, a gap of 28 points. On the extensive margin, 72% of targeted students did not graduate from a degree program, compared with 46% of non-targeted students. Dropout is part of the reason: among those who enrolled in the year of application, 38% of targeted students did not graduate within seven years, compared with 30% of non-targeted students. On the intensive margin, conditional on graduating, non-targeted students graduated from programs ranked about 28 rank points higher.

3.2 The role of federal degrees in sustaining the gap

Federal colleges contribute to the college-quality gap through the combination of two features of the setting. First, federal degrees are among the highest-quality degrees in the country. Figure 2, Panel A plots the distribution of program quality by sector. Federal programs make up 16% of all degree programs but 48% of those in the top quartile of the quality distribution. Within the federal sector, 74% of programs are in the top quartile, compared with 40% of state programs and 12% of private programs.

Second, without any AA, federal admissions are based almost exclusively on exam scores, and ENEM scores are strongly correlated with socioeconomic status. Figure 2, Panel B plots the score distribution by group on the exam taken ahead of the 2012 admissions cycle. The average targeted student scored 495 points, compared with 585 points for the average non-targeted student—a difference of 89 points, or 1.09 s.d. of exam performance. As a result, targeted students accounted for

¹³Our use of rank-based outcomes follows the intergenerational-mobility literature, where average percentile ranks are a standard way to summarize distributional outcomes (Chetty et al., 2014). For example, Chetty et al. (2020b) measure racial disparities in economic opportunity using differences in mean income ranks between Black and White children.

¹⁴Seven years after application, some students are still enrolled in the degree they entered without having completed a degree of equal or higher rank (about 1.7% of students, or 5.2% of those who enrolled). We treat them as having graduated, assigning them the rank of the degree in which they are still enrolled.

only 30% of test-takers above the median admissions cutoff for federal open seats, even though they represented 84% of all test-takers. The dashed line in Panel B marks this cutoff.

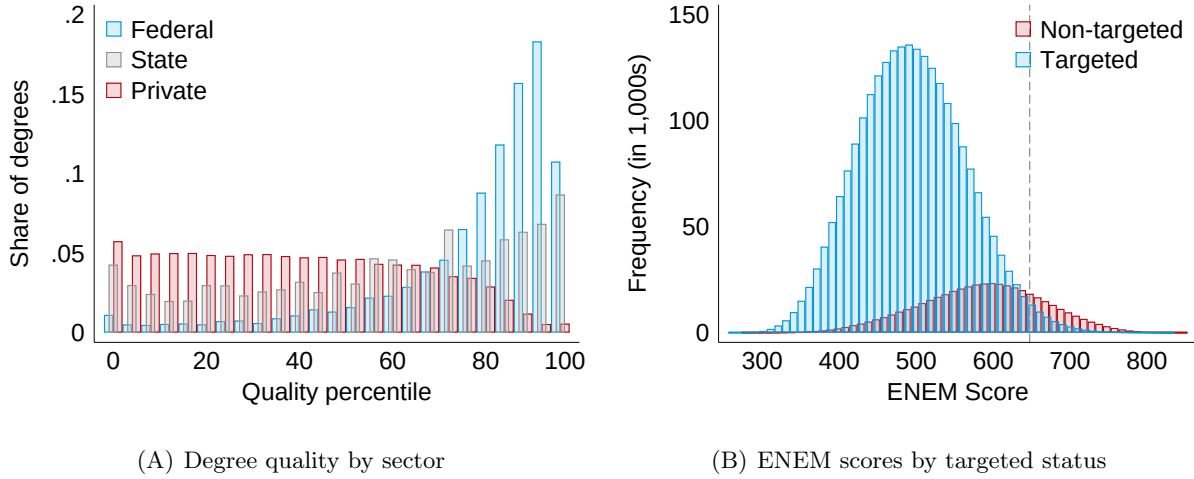


Figure 2: Degree quality and the test-score difference

Notes: Panel A shows the national distribution of degree quality—the average incoming-student ENEM score, ranked on a 1–100 scale—across federal, state, and private programs. An observation is a degree program, and the sample includes both in-person and online degrees. Panel B shows the distribution of average ENEM scores by targeted status among students who took the exam ahead of the 2012 admissions cycle. Targeted students are those who attended only public high school. The dashed line marks the median admissions cutoff for federal open seats (648 points).

3.3 How affirmative action changed federal admissions

After the policy was introduced, targeted students faced lower admissions cutoffs than applicants competing for open seats. The difference between open- and reserved-seat cutoffs averaged 29.8 points (0.36 s.d.) in 2013 and widened to 40.2 points (0.48 s.d.) by 2016, as the policy was phased in. For the most restrictive reserved track, which sets aside seats for low-income public high school graduates of color, the difference grew from 37.7 to 50.6 points over the same period (from 0.45 to 0.61 s.d.). Appendix Figure A.2 shows that the gap between open- and reserved-seat admissions cutoffs widens with program selectivity. In the top 10% most selective programs, the gap in 2016 averaged 56.0 points (0.67 s.d.) across all reserved tracks and 73.8 points (0.89 s.d.) for the most restrictive track.¹⁵

These lower cutoffs helped reshape the composition of federal enrollment. Figure 3, Panel A shows targeted students' share of incoming enrollment by degree selectivity, before and after the policy. In 2012, public high school graduates accounted for 59% of admissions to federal universities. By 2016, when the policy was fully in place, this share had risen to 68% but remained below their 84% share of test-takers.¹⁶ In the top 10% most selective federal programs, targeted students' share rose from 37% in 2012 to 53% in 2016, while the share of Black, Brown, and Indigenous targeted

¹⁵ Appendix Table A.2 reports the cutoff differences by seat type and year.

¹⁶ These results are in line with Mello (2022), who exploits variation across public institutions and over time in the adoption of AA. Moving from zero to 50% reserved seats increased the share of incoming students from public high schools by 9.9 percentage points on average.

students more than doubled, from 14% to 31%.

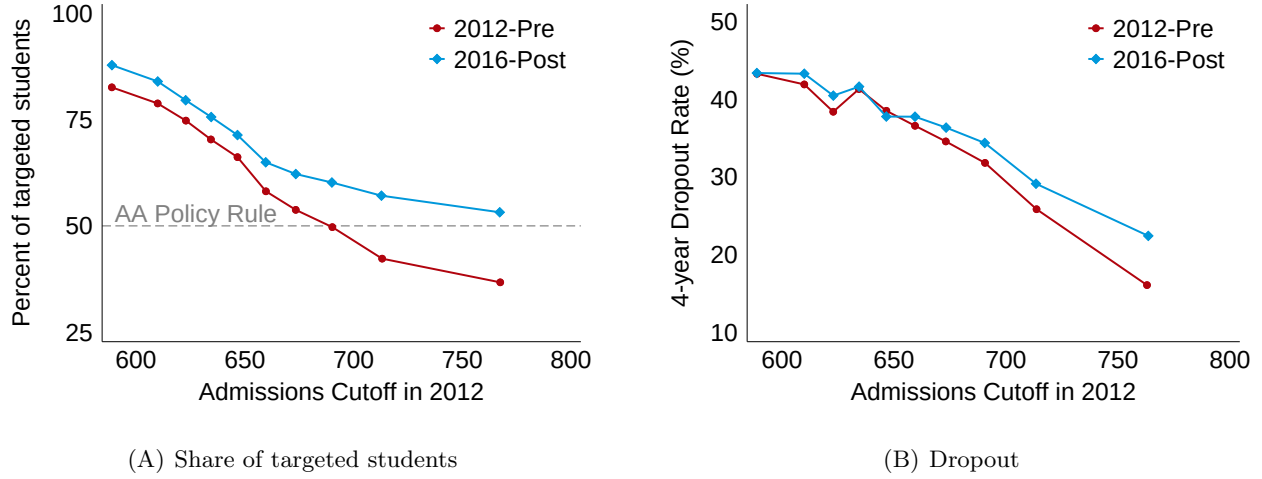


Figure 3: Targeted-student share and dropout by degree selectivity, before and after AA

Notes: Both panels plot outcomes by degree selectivity, measured by the seat-weighted average admissions cutoff across all tracks in the 2012 SISU. Panel A shows targeted students' share of incoming enrollment in federal programs. The dashed line marks the 50% reserved-seat requirement. Panel B shows the four-year dropout rate. Each observation is a degree program in a given shift. The sample includes programs that participated in SISU in both 2012 and 2016.

3.4 Academic preparedness and outside options

AA can narrow the college-quality gap by giving targeted students access to more selective degrees through lower admissions cutoffs. How much it narrows the gap depends on two forces. First, the gains depend on whether students persist in these degrees. AA expands access for students with lower test scores, who may be less prepared and more likely to drop out (Arcidiacono and Lovenheim, 2016). Second, these gains depend on students' outside options (their next-best alternatives without the seat). The weaker the outside option, the higher the return to a seat. The same forces determine the policy's cost to non-targeted applicants displaced from federal degrees. We document each in turn.

Academic preparedness. Whether students are academically prepared for the degrees they enter is an important concern in our setting. We define dropout as leaving college altogether or moving to a degree ranked below the one initially entered. Among incoming students at federal universities in 2012, 32% dropped out within four years.¹⁷ In Appendix B.2, we show that, within a degree, this risk is higher for students who enter with lower admissions scores, especially in math-intensive programs, consistent with scores capturing academic preparation. Figure 3, Panel B shows that dropout rose to 35% among entrants in 2016, with the increase concentrated among the most selective degrees. In the most selective decile, dropout rose from 16% to 23% between 2012 and 2016, with a larger increase among targeted students (Appendix Figure A.3).

¹⁷We track each entrant across the entire higher-education system, including private institutions. Leaving the initial degree is not the relevant measure here because moving across degrees is common, and moving to a higher-ranked degree is not evidence of poor academic preparation (Larroucau and Rios, 2026). Among 2016 entrants, 44% left their initial degree within four years. Of these, 54% left college entirely, 25% moved to a lower-ranked degree, and 21% moved to a degree ranked as high or higher.

Outside options. Targeted applicants have weaker outside options than non-targeted applicants. Among non-targeted applicants in 2012 who had finished high school the previous year and did not enroll in a federal degree, 10% attended another public institution, 40% enrolled at a private one, and 50% did not attend college. Among these applicants, 53% also reapplied in a later admissions cycle. Compared with non-targeted applicants with the same test scores, targeted applicants were 4% less likely to attend another public institution, 21% less likely to enroll at a private one, 17% more likely not to enroll at all, and 7% less likely to reapply. Income may help explain these differences. Private institutions charge tuition, and reapplying requires forgoing earnings for another year. Both costs weigh more heavily on targeted applicants, who come from lower-income households. The two groups also differed in the quality of the programs they attended outside the federal system. Among enrollees with the same test scores, non-targeted applicants attended programs ranked about 11 rank points higher on our quality ranking and 0.12 s.d. higher on the Ministry of Education’s quality index. Among private enrollees, they attended programs with tuition fees 25% higher.

4 A model of affirmative action in a centralized system

In this section, we formalize how AA operates within a centralized admissions system. As in [Dur et al. \(2018\)](#) and [Rios et al. \(2021\)](#), we model the regulation as reserving seats for targeted students in each degree. Because many students decline their offers in our setting, we extend this framework to include a take-up stage and a waitlist that reallocates declined seats. We then map the resulting allocation into student outcomes and use the model to define the gains and losses from AA.

4.1 Environment

We consider a set of students indexed by $i \in \mathcal{I} = \{1, \dots, n\}$ who apply to a finite set of selective degree programs through a centralized platform. Let $\mathcal{J} = \{1, \dots, J\}$ denote the degrees offered across all federal institutions. The outside option, denoted by $j = 0$, corresponds to not enrolling in a federal degree and is always available. The admissions process has two stages: an initial-offer stage and an aftermarket that reallocates declined seats through a waitlist.¹⁸

Applicants. Students have preferences $\succ_i$ over the degree programs in $\mathcal{J}$. Each student submits a rank-order list, or report, R_i , to the platform. Each student also has a take-up rule $T_i = (T_i^1, T_i^2)$, where $T_i^h = (T_{i1}^h, \dots, T_{iJ}^h)$ and $T_{ij}^h \in \{0, 1\}$ indicates whether the student accepts an offer from degree j in stage $h \in \{1, 2\}$. The stage-specific rules allow a student’s willingness to accept a given degree to depend on when the offer arrives. We assume that take-up is monotone in preferences within each stage: a student who accepts an offer from degree j also accepts an offer from any degree they prefer to j .¹⁹ In addition, students have two characteristics that set their priority in the mechanism:

¹⁸This structure mirrors the sequential nature of the Brazilian college admissions process and is common in the recent literature on educational matching markets (e.g., [Walters, 2018](#); [Kapor et al., 2024](#); [Kapor, 2025](#)).

¹⁹Formally, $j \succ_i j'$ implies $T_{ij}^h \geq T_{ij'}^h$ for each stage h . Thus, the offers a student accepts in each stage form an upper contour set of $\succ_i$. Equivalently, take-up can be represented by placing the outside option at a stage-specific position in the student’s ranking. Appendix Figure A.4 shows that take-up rates are higher for programs ranked higher in students’ submitted lists and for more selective programs, consistent with this assumption.

their application scores $s_i = (s_{i1}, \dots, s_{iJ})$ and their AA status $t_i \in \mathcal{T}$.²⁰ For ease of exposition, let $\mathcal{T} = \{\text{AA}, \text{NA}\}$, where $t_i = \text{AA}$ denotes targeted students and $t_i = \text{NA}$ non-targeted students. Let $R = (R_i)_{i \in \mathcal{I}}$, $T = (T_i)_{i \in \mathcal{I}}$, $\mathbf{s} = (s_i)_{i \in \mathcal{I}}$, and $\mathbf{t} = (t_i)_{i \in \mathcal{I}}$ denote the profiles of reports, take-up rules, application scores, and AA statuses. Student i 's type is $\theta_i = (\succ_i, T_i, s_i, t_i)$, and $\Theta = (\theta_i)_{i \in \mathcal{I}}$ denotes the profile of student types.

Degrees. Each degree $j \in \mathcal{J}$ has capacity $q_j > 0$. Let $q = (q_j)_{j \in \mathcal{J}}$ denote the profile of degree capacities. An AA policy is described by a reservation schedule $\omega = (\omega_j)_{j \in \mathcal{J}}$, where $\omega_j \in \{0, \dots, q_j\}$ is the number of seats in degree j reserved for targeted students. The remaining $q_j - \omega_j$ seats are open to all applicants. A common reservation share $x \in [0, 1]$ induces the schedule $\omega(x) = (\text{round}(xq_j))_{j \in \mathcal{J}}$. Degree j ranks each applicant i by the application score s_{ij} defined above. We assume these scores are sufficiently fine that no tie-breaking rule is needed. As in our empirical application, where all degrees are oversubscribed, we also assume that every degree has more applicants than seats in both its open and reserved pools and fills its capacity by the end of the admissions process.

Mechanism. Given the reports R , take-up rules T , priority profiles $(\mathbf{s}, \mathbf{t})$, capacities q , and reservation schedule ω , the mechanism φ generates a sequence of offers and a final allocation, which we denote jointly by $\mu = \varphi(R, T, \mathbf{s}, \mathbf{t}; q, \omega)$. The final allocation assigns each student to a degree or the outside option, and no degree j fills more than q_j seats. The mechanism proceeds in two stages. First, a student-proposing deferred acceptance algorithm with reserved seats makes initial offers, which students accept or decline according to T^1 .²¹ Second, an aftermarket sequentially offers declined seats to students on a waitlist, who accept or decline according to T^2 , until no further seats can be filled.

Offers and admissions cutoffs. We summarize the offer process by the last offer a student receives in each stage, $D_i^h(\mu) \in \mathcal{J} \cup \{0\}$ for $h \in \{1, 2\}$, with $D_i^h(\mu) = 0$ if the student receives none. The corresponding offer indicator is $D_{ij}^h(\mu) \equiv \mathbb{1}\{D_i^h(\mu) = j\}$. In a large matching market, offers generated by deferred acceptance can be represented by admissions cutoffs (Azevedo and Leshno, 2016). Specifically, we write $c_j^{h,t}(\mu)$ for the minimum application score a student with AA status t needs to receive an offer from degree j in stage h under μ . Because non-targeted students compete only for open seats while targeted students are eligible for both, $c_j^{h,\text{NA}}(\mu) \geq c_j^{h,\text{AA}}(\mu)$ for every j and h .

Enrollment and outcomes. After the offer process is complete, let $A_i(\mu) \in \mathcal{J} \cup \{0\}$ denote student i 's realized enrollment, where $A_i(\mu) = j$ if the student enrolls in degree j and $A_i(\mu) = 0$ if the student receives no offer or declines every offer received. Student i 's realized outcome under μ is $Y_i(\mu) = \sum_{j=0}^J A_{ij}(\mu) \cdot Y_{ij}$, where $A_{ij}(\mu) \equiv \mathbb{1}\{A_i(\mu) = j\}$ indicates enrollment in j and Y_{ij} denotes student i 's potential outcome under option j . Potential outcomes depend only on a student's own

²⁰We allow for degree-specific scores, which arise when degrees assign different weights to the components of the entrance exam (as in our setting) or use degree-specific exams.

²¹Reservations can be processed in different orders, depending on whether open or reserved seats are filled first (Sönmez and Yenmez, 2022). Our framework accommodates both. In our empirical application, open seats go first to the highest-scoring applicants, and reserved seats then go to the highest-scoring targeted students not already admitted. The implementation of deferred acceptance with reserved seats follows Rios et al. (2021) and is detailed in Appendix D.

assignment.²²

4.2 Gains and losses from affirmative action

We use the framework to define two objects of interest. The first is the marginal student's gain from enrollment relative to their fallback option. A large literature recovers this object using regression discontinuity designs that compare applicants just above and below an admissions cutoff, either to estimate the returns to a more selective college (e.g., [Hoekstra, 2009](#); [Zimmerman, 2014](#); [Mountjoy, 2026](#)) or to measure the effects of AA on marginally admitted and displaced students ([Bertrand et al., 2010](#); [Bleemer, 2022](#); [Machado et al., 2025](#)). The second is the overall effect of moving between AA schedules, together with how its gains and losses are distributed across groups. It captures the gains and losses of all students, including those far from any admissions cutoff and those reallocated in equilibrium by the change in ω .

Effects on marginal students. To define the first object, we fix a degree j , an AA status t , and a stage h . Let $\tilde{\mu}_{jt}$ denote the counterfactual matching obtained by marginally increasing the number of seats available at j to status- t applicants, while holding capacities at other degrees fixed. We call a student h -*marginal* if the marginal increase in capacity allows them to clear the stage- h cutoff at j and receive an offer from it in that stage. These students satisfy $D_{ij}^h(\mu) = 0$ and $D_{ij}^h(\tilde{\mu}_{jt}) = 1$, which implies $c_j^{h,t}(\tilde{\mu}_{jt}) \leq s_{ij} < c_j^{h,t}(\mu)$. In a large market, this marginal increase in capacity shifts the cutoff only slightly, so marginal students lie just below the original cutoff. Because take-up is imperfect, a stage- h offer need not lead to eventual enrollment in j . We call a marginal student an h -*complier* if gaining the stage- h offer changes their enrollment in j , so that $A_{ij}(\mu) = 0$ and $A_{ij}(\tilde{\mu}_{jt}) = 1$.²³

The return to enrollment in degree j for h -compliers of status t is

$$\tau_{jt}^{M,h} = \mathbb{E} \left[Y_i(\tilde{\mu}_{jt}) - Y_i(\mu) \mid t_i = t, D_{ij}^h(\mu) < D_{ij}^h(\tilde{\mu}_{jt}), A_{ij}(\mu) < A_{ij}(\tilde{\mu}_{jt}) \right]. \quad (1)$$

Thus, $\tau_{jt}^{M,h}$ measures the return to enrolling in j relative to the student's fallback assignment, $A_i(\mu)$. Pooling these returns across degrees and stages gives

$$\tau_t^M = \sum_h \sum_j w_{jt}^h \tau_{jt}^{M,h}, \quad (2)$$

where w_{jt}^h are positive weights that sum to one.

With total capacity at j fixed, a marginal increase in reserved seats reduces the number of open seats by the same amount. When reserved seats bind, converting one open seat into a reserved seat admits a targeted complier at the reserved-seat cutoff and displaces a non-targeted complier at the open-seat cutoff. We therefore define the difference in marginal returns as

$$\Delta^M = \tau_{AA}^M - \tau_{NA}^M,$$

which measures how much more a marginal seat is worth to a targeted complier than to a non-targeted one, on average across degrees.

²²This rules out peer effects and other spillovers in outcomes, which lie outside the scope of our framework.

²³We assume that a stage- h offer cannot reduce enrollment in j , ruling out defiers.

Although $\tilde{\mu}_{jt}$ is generated by marginally increasing the seats available to status- t applicants at degree j , assignments are determined in equilibrium. Students who gain access to j may free seats in other degrees, which other students take. Conversely, when open seats are reduced, a student who loses a seat moves to their fallback assignment and may displace others in turn. We call these additional assignment changes *knock-on* effects. The gains and losses of students reallocated through these knock-on effects do not enter $\tau_{jt}^{M,h}$, which records only the marginal compliers' outcome changes.

Overall effect. The marginal comparison concerns returns for compliers at the admissions cutoffs. We are interested, however, in the full policy change, from schedule ω to an alternative schedule ω' , including every reallocation it induces in equilibrium. Holding reports, take-up rules, priorities, and capacities fixed, the alternative schedule generates the counterfactual matching $\mu' = \varphi(R, T, \mathbf{s}, \mathbf{t}; q, \omega')$.

We call student i a *switcher* if the policy changes their enrollment, so that $A_i(\mu') \neq A_i(\mu)$. Student i 's policy effect is

$$\tau_i(\mu', \mu) = Y_i(\mu') - Y_i(\mu). \quad (3)$$

Because potential outcomes depend only on a student's own enrollment, $\tau_i(\mu', \mu) = 0$ for non-switchers.

The overall effect differs from the effect on marginal students in two ways. First, the average effect among switchers may differ from the return for marginal compliers. For each group t , we define

$$\tau_t^S = \mathbb{E} [Y_i(\mu') - Y_i(\mu) \mid t_i = t, A_i(\mu') \neq A_i(\mu)]. \quad (4)$$

Switchers include students far from the original cutoffs whose offers change when the policy moves the cutoffs substantially, as well as students reallocated through knock-on effects. Appendix E illustrates how the average return for switchers, τ_t^S , can differ from that for marginal compliers, τ_t^M , in a single-degree example.

Second, the marginal comparison is one-for-one, pairing one benefited targeted complier with one displaced non-targeted complier. For the full policy change, the numbers of benefited and displaced students may differ. Let

$$N_t^S = \sum_{i:t_i=t} \Pr(A_i(\mu') \neq A_i(\mu)) \quad (5)$$

denote the expected number of switchers in group t . Knock-on effects can make N_{AA}^S and N_{NA}^S differ even when capacities are fixed and each additional reserved seat replaces an open seat. For example, a targeted student whose fallback is the outside option frees no federal seat when admitted. A displaced non-targeted student may instead take a seat in another federal degree and displace another non-targeted student. A single reserved seat can therefore change one targeted assignment and several non-targeted ones.

Multiplying the two terms gives the total gain or loss to each group t ,

$$\Delta_t(\mu', \mu) = N_t^S \tau_t^S, \quad (6)$$

and summing across groups gives the policy's overall effect,

$$\Delta(\mu', \mu) = \Delta_{AA} + \Delta_{NA}. \quad (7)$$

Equations (6) and (7) show that the overall effect depends on the number of benefited and displaced students and their average gains and losses. A policy can lower the average outcome even when benefited students gain more on average than displaced students lose, because it displaces more students than it benefits.

5 The impact of affirmative action on marginal students

In this section, we estimate the effects of AA on marginal students, as defined in Section 4.2. Because reserved and open seats clear at different cutoffs, we can estimate separately the gains to marginally benefited targeted students and the losses to marginally displaced non-targeted students. We first describe the sample and outcomes, then lay out the empirical design, which compares applicants just above and below each admissions cutoff. We present reduced-form estimates of the effects of crossing the cutoff and 2SLS estimates of the returns to enrolling in the submitted federal degree relative to students' next-best alternatives. We use these estimates to weigh the gains against the losses and recover Δ^M , the difference in marginal returns defined above.

5.1 Sample and outcome variables

Sample. We concentrate our analysis on students applying to federal universities in SISU during the first semester of 2013, whom we can follow for up to seven years after application. We exclude art programs that require additional admissions exams beyond ENEM, as well as military programs that are not obligated to follow the Law of Social Quotas. We limit the sample to open seats and the four quotas mandated by the regulation, removing students who applied through other university-specific AA initiatives (7% of federal seats that year). Our final dataset covers 83 institutions and 2,987 degrees. It consists of 1,755,296 applicants, of whom 147,881 are admitted to a federal degree through one of these pools. Because many universities moved ahead of the phase-in (Section 2.3), 31.8% of the seats in our sample were reserved for AA, well above the 12.5% minimum for 2013.

Table 1 reports summary statistics for our sample. Columns 1 and 2 report mean characteristics for non-targeted (NA) and targeted (AA) applicants, respectively.²⁴ AA applicants come from poorer municipalities and from lower-income households, with average monthly household income of 1,251 BRL (about US\$640 in 2012 U.S. dollars), compared with 4,086 BRL (US\$2,090) for NA applicants, equivalent to about 2.0 and 6.6 times the 2012 monthly minimum wage. AA applicants are more likely to identify as Black, Brown, or Indigenous (65% versus 39%) and are 2.5 years older on average (23.5 versus 21.0). Their average ENEM score is 1.01 s.d. below that of NA applicants. These differences highlight the socioeconomic disparities that the AA policy is designed to address.

Columns 3 and 4 report the same characteristics for NA and AA students who received an initial offer. Compared with applicants of the same type, admits come from higher-income households, have higher application scores, are younger, and are less likely to identify as Black, Brown, or Indigenous. Columns 5–9 separate admits by admissions pool. Students admitted through open seats (column

²⁴Some students could also qualify for public-school quotas through high-school equivalency certificates or court rulings that extended eligibility to applicants from tuition-free private charitable schools. This explains why not all AA applicants had attended public high school.

5) score higher than those admitted through any of the four reserved-seat pools (columns 6–9). On average, reserved-seat admits are older (21.8 versus 20.7) and more likely to have been enrolled in college in 2012 (24% versus 19%). Their characteristics also line up with the eligibility criteria of each reserved admissions pool. Nearly all (98%) attended a public high school. Over 90% of admits through race-restricted pools (columns 8 and 9) identify as Black, Brown, or Indigenous, while admits through income-restricted pools (columns 7 and 9) have substantially lower household incomes than those in the other pools.²⁵

Outcomes. We track applicants’ educational outcomes at three horizons after application. In the application year, we record four outcomes: enrollment in the submitted degree program (our first-stage outcome), enrollment in any federal institution, enrollment in any college, and access to high-quality programs as measured by college enrollment rank. The rank, defined in Section 3.1, combines whether a student enrolls with the quality of the program they attend. Four years after application, we record dropout among those who enrolled, defined as leaving college or moving to a lower-ranked degree. Seven years after application, by which point most students have completed or left higher education, we record graduation from any program and from a federal institution, and use college graduation rank to capture completion of high-quality programs.²⁶ Graduation rank is our preferred outcome because it combines access to higher-quality degrees with persistence through graduation. The Higher Education Census covers all degree programs in Brazil, so we observe applicants’ outcomes wherever they enroll, including outside the federal system.

5.2 Research design

Empirical setting. As described in Section 2, students submit a rank-order list of up to two degree programs. Let $R_i = (R_{i1}, R_{i2})$ denote student i ’s report, with R_{ip} denoting their p th choice and $R_{i2} = 0$ if only one degree is listed. For each option R_{ip} , let k_{ip} denote the selected admissions pool. Each degree has one open and four reserved-seat pools, and we refer to each degree-pool pair (j, k) as an *admissions list*. The admissions process has two stages, the initial-offer stage, $h = 1$, and the waitlist, $h = 2$. For each list, let $c_j^{h,k}$ denote the minimum application score required to receive an offer in stage h . In the initial-offer stage, the centralized platform offers each student their highest-ranked feasible option. Students who do not receive an initial offer from their first choice can be considered for a later offer through its waitlist. After the acceptance deadline, the Ministry of Education forwards the waitlists to participating institutions, which use them to fill remaining vacancies.

2SLS estimation. We use this setting to estimate the marginal returns defined in Section 4.2. These measure the effect of enrolling in a submitted degree relative to the student’s next-best alternative (Kirkeboen et al., 2016; Zimmerman, 2019; Mountjoy, 2026). This alternative may be another federal degree, a program outside the federal system, or no college enrollment. We estimate the

²⁵These variables are self-reported in ENEM and may contain measurement error.

²⁶About 2% of applicants, or 6.5% of those who entered a degree, are still enrolled in their initial degree seven years after application without having completed a degree of equal or higher rank. We assign these students the rank of their initial degree.

Table 1: Summary statistics

	Applicants		Admits						
			By AA Status		By Admissions Pool				
					Reserved: All Require Public High School				
	NA	AA	NA	AA	Open	Public high school	+ Income	+ Race	+ Income + Race
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Traits targeted by the policy</i>									
Public high school	0.00	0.99	0.00	0.98	0.28	0.99	0.99	0.99	0.95
Black, Brown, Indig.	0.39	0.65	0.35	0.60	0.38	0.24	0.25	0.94	0.90
Household income	4,086	1,251	5,169	1,998	4,392	2,778	1,330	2,053	1,286
<i>SISU applications</i>									
# options ranked	1.96	1.95	1.98	1.98	1.98	1.97	1.97	1.98	1.98
Application score	595.9	511.9	677.5	638.4	669.7	652.2	636.1	628.2	623.0
<i>ENEM scores</i>									
Natural science	555.1	476.4	622.2	575.0	610.8	592.3	575.2	566.7	561.9
Social science	599.8	532.7	662.8	633.3	656.6	644.7	631.8	627.4	621.2
Language	557.7	501.6	608.0	582.5	602.8	592.7	581.4	576.4	571.4
Math	629.4	515.5	721.3	659.2	707.1	684.5	659.6	646.7	637.6
Essay	626.5	527.4	752.6	720.1	748.9	727.2	712.4	705.4	703.1
<i>Other student characteristics</i>									
Female	0.55	0.58	0.50	0.44	0.49	0.45	0.48	0.42	0.44
Age	21.0	23.5	20.0	22.1	20.7	21.5	21.2	22.6	21.7
HS graduate in $t - 1$	0.37	0.29	0.43	0.32	0.39	0.35	0.35	0.30	0.32
GED-equivalent	0.03	0.06	0.01	0.05	0.03	0.05	0.05	0.05	0.05
Married	0.06	0.14	0.04	0.10	0.06	0.08	0.07	0.12	0.08
Municipal income	2,280	1,893	2,307	2,001	2,219	2,256	2,123	1,930	1,858
<i>College enrollment in $t - 1$</i>									
Private	0.10	0.10	0.06	0.11	0.07	0.09	0.13	0.11	0.13
Federal	0.09	0.05	0.10	0.12	0.11	0.10	0.09	0.12	0.11
Any college	0.19	0.16	0.17	0.24	0.19	0.21	0.23	0.25	0.25
Number of students	426,127	1,329,169	69,574	78,307	96,638	9,360	10,348	14,807	16,728

Notes: Elaboration based on SISU microdata from applications in the first semester of 2013. We restrict the data to federal universities, as described in the main text. This table shows the demographic characteristics of applicants (columns 1 and 2), admits (columns 3 and 4), and admits by admissions pool (columns 5–9). We classify a student as AA if they graduated from a public high school, as reported in ENEM, or applied through a reserved pool, the same definition we use in the model. All other students are classified as NA. Public-high-school attendance, race, and monthly household income are self-reported in the ENEM questionnaire. “HS graduate in $t - 1$ ” is based on expected completion in 2012 reported at ENEM registration. “GED-equivalent” indicates students who used ENEM to obtain a high-school equivalency certificate. Rows marked $t - 1$ refer to 2012, the year before application.

marginal return for targeted students, τ_{AA}^M , from discontinuities at reserved-list cutoffs, and that for non-targeted students, τ_{NA}^M , from discontinuities at open-list cutoffs. Targeted students may also apply through an open pool, but we exclude these applications from the 2SLS sample.

Following [Abdulkadiroğlu et al. \(2014\)](#), we also restrict the sample to applications for which crossing the cutoff determines whether the student receives an offer. For initial offers, we include all first-choice applications. We include second-choice applications only when the student does not

qualify for their first choice, since students who qualify receive their first-choice offer instead. At the waitlist stage, we include all applications on the degree’s waitlist. Within these samples, crossing the relevant cutoff is equivalent to receiving an offer. We stack the resulting observations across submitted options and stages and use crossing the relevant cutoff as an instrument for enrollment in the submitted degree.

Specifically, for applicants of AA status t , we estimate:

$$Y_i = \tau_t^M A_{ij} + \mu_{jk_{ip}}^h + g_{jt}^h(r_{ij}^h) + X_i' \delta^t + \varepsilon_{ij}^h, \quad (8)$$

$$A_{ij} = \pi_{k_{ip}}^h D_{ij}^h + \phi_{jk_{ip}}^h + f_{jt}^h(r_{ij}^h) + X_i' \alpha^t + u_{ij}^h, \quad (9)$$

where $j = R_{ip}$ is student i ’s p th-choice degree, and k_{ip} is the selected admissions pool. The superscript $h \in \{1, 2\}$ indexes the initial-offer and waitlist stages. The running variable $r_{ij}^h = s_{ij} - c_j^{h,k_{ip}}$ is student i ’s score centered at the relevant cutoff, and $D_{ij}^h = \mathbb{1}\{r_{ij}^h \geq 0\}$ indicates crossing that cutoff. The variable A_{ij} indicates actual enrollment in degree j and does not vary across stages, while Y_i is an outcome of interest and does not vary across submitted options or stages. The terms $\mu_{jk_{ip}}^h$ and $\phi_{jk_{ip}}^h$ are admissions-list–stage fixed effects. The functions $g_{jt}^h(\cdot)$ and $f_{jt}^h(\cdot)$ allow the relationship between the outcome or enrollment and the running variable to differ across degrees, AA statuses, stages, and sides of the cutoff. The specification also includes student covariates X_i to improve precision.²⁷ The coefficient τ_t^M measures the effect of enrolling in the submitted degree relative to the fallback assignment for marginal applicants of AA status t . The first-stage coefficients $\pi_{k_{ip}}^h$ vary by admissions pool and stage. We cluster standard errors by student because students may contribute observations for multiple degrees and stages. We select bandwidths separately by outcome, seat type, and stage following [Calonico et al. \(2017\)](#).

While we estimate the gains and losses for marginal students with the 2SLS specification described above, we also present a graphical representation of the first-stage and reduced-form discontinuities in [Figure 4](#). Each panel plots an outcome against the application score centered at the cutoff, with linear fits on each side. The figure notes describe the construction.

5.3 Results

We present the first stage, then summarize the reduced-form and 2SLS estimates in four findings.

First stage. The first stage measures the effect of crossing a submitted option’s cutoff on enrollment in that degree. [Figure 4](#), Panel A plots enrollment against the application score centered at the cutoff, and [Table 2](#), Panel A reports the estimate. Crossing the cutoff raises enrollment by about 18 percentage points for open seats and 19 for reserved seats. Enrollment among non-offered applicants is well above zero because many who fall just short still enroll through the waitlist. Take-up of initial offers is low: only about a third of students offered a seat enroll in it (32% for open seats

²⁷The covariates include race, gender, age, household income, average household income in the student’s municipality, marital status, an indicator for having attended a public high school, the student’s high-school completion status at ENEM registration, an indicator for using ENEM to obtain a high-school equivalency certificate, scores on the five ENEM components, and indicators for enrollment in any college and in a federal institution in the year prior to application.

and 31% for reserved). On average, take-up is higher for initial offers than for waitlist offers. It also rises with preference and program selectivity and falls with distance to the offered degree (Appendix Figure A.4). Finally, enrollment declines with the score above the cutoff and rises with it below the cutoff. Above the cutoff, outside options improve with the score, so stronger admits decline their offers more often. Below it, applicants closer to the cutoff are called from the waitlist more often.

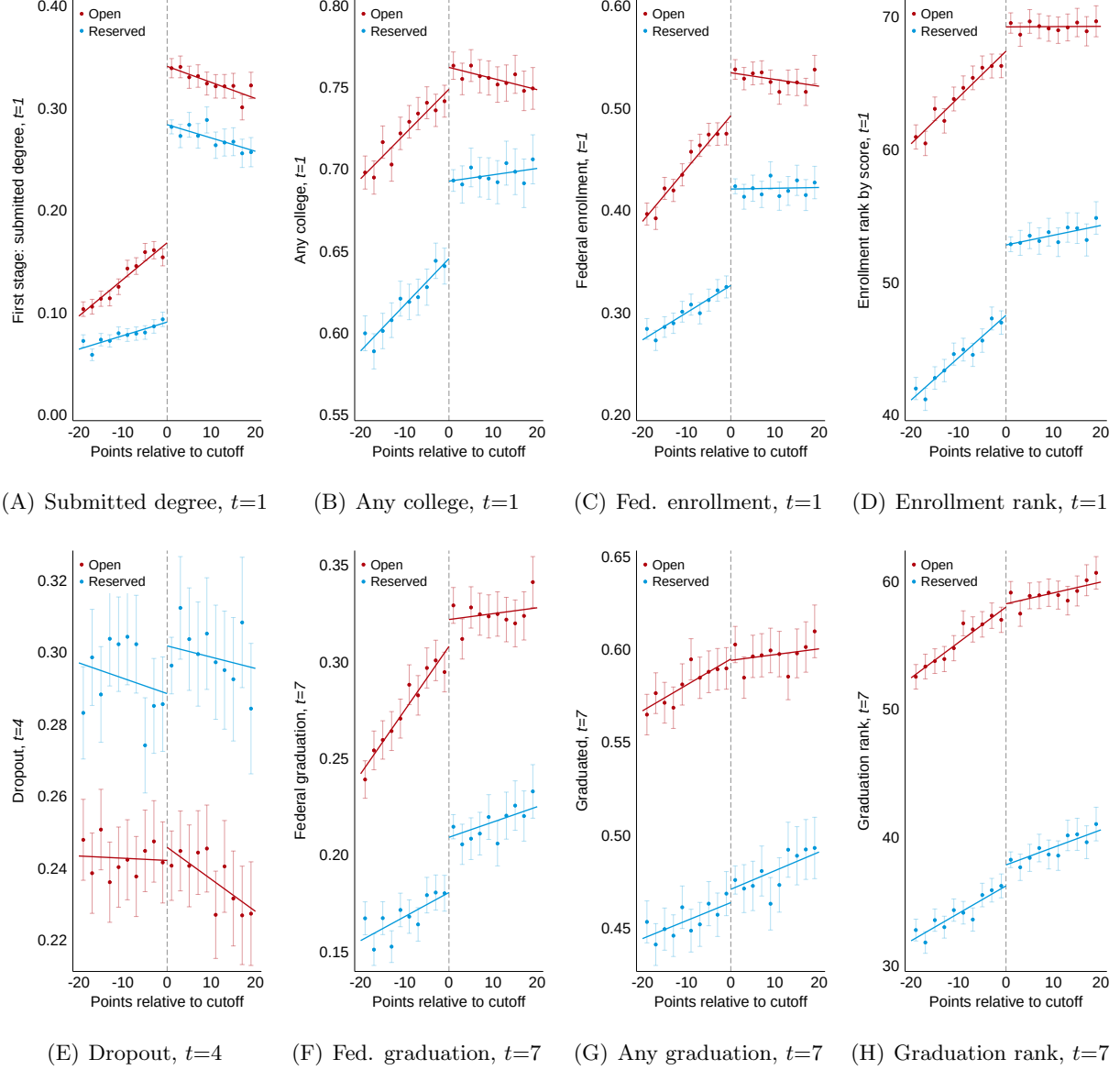


Figure 4: First-stage and reduced-form effects of crossing an admissions cutoff

Notes: We pool applications across admissions lists and the initial-offer and waitlist stages, combining the four reserved-seat categories. Scores are centered at the relevant cutoff. Dots show bin means for open (red) and reserved (blue) seats after removing admissions-list-stage fixed effects. Error bars show 95% confidence intervals. We include applications within 20 points of the cutoff. Panels A–D report application-year outcomes, Panel E reports dropout within four years, and Panels F–H report graduation outcomes after seven years. In the panel titles, t counts years from application, with $t = 1$ the application year (2013).

Table 2: Reduced-form and 2SLS estimates

	Non-offered mean		Reduced-form			2SLS			Observations	
	NA	AA	NA	AA	Diff	NA	AA	Diff	NA	AA
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Panel A. First stage</i>										
Degree enrollment	0.120	0.043	0.180 (0.004)	0.189 (0.004)	0.009 (0.006)	-	-	-	163,182	153,350
<i>Panel B. Academic outcomes in $t = 1$</i>										
Federal enrollment	0.439	0.252	0.055 (0.005)	0.103 (0.004)	0.048 (0.007)	0.301 (0.025)	0.493 (0.016)	0.192 (0.030)	147,040	194,648
Any enrollment	0.725	0.572	0.019 (0.005)	0.055 (0.004)	0.036 (0.006)	0.115 (0.024)	0.264 (0.020)	0.149 (0.031)	147,452	165,430
Enrollment rank	64.36	39.84	2.64 (0.423)	6.20 (0.346)	3.55 (0.547)	15.67 (2.17)	29.50 (1.53)	13.83 (2.65)	149,497	175,926
<i>Panel C. Academic outcomes in $t = 4$ and $t = 7$</i>										
Dropout by year 4	0.236	0.293	0.003 (0.005)	0.016 (0.005)	0.013 (0.007)	0.011 (0.023)	0.061 (0.018)	0.051 (0.029)	125,194	135,409
Federal graduation	0.275	0.146	0.019 (0.005)	0.030 (0.004)	0.011 (0.006)	0.115 (0.026)	0.142 (0.017)	0.027 (0.031)	160,547	194,291
Any graduation	0.588	0.440	0.001 (0.005)	0.008 (0.005)	0.007 (0.007)	0.004 (0.029)	0.038 (0.023)	0.034 (0.037)	149,300	161,434
Graduation rank	56.05	30.98	0.684 (0.477)	2.07 (0.352)	1.38 (0.592)	4.59 (2.56)	9.84 (1.66)	5.25 (3.05)	150,950	198,463

Notes: This table reports reduced-form and 2SLS estimates for SISU applicants in the first semester of 2013. NA denotes non-targeted students at open-seat cutoffs, and AA denotes targeted students at reserved-seat cutoffs. Differences are AA minus NA. In the panel headings, t counts years from application, with $t = 1$ the application year. We pool admissions lists across the initial-offer and waitlist stages, combining the four reserved-seat categories. The 2SLS estimates instrument enrollment in the submitted federal degree with crossing the relevant admissions cutoff. The specification includes admissions-list-stage fixed effects, student covariates, and local-linear functions of the running variable that vary by degree, seat type, stage, and side of the cutoff. We use triangular weights and select bandwidths separately by outcome, seat type, and stage following [Calonico et al. \(2017\)](#). The final two columns report the number of observations for each outcome. Standard errors (in parentheses) are clustered by student.

Finding 1: Crossing the cutoff improves access to higher-quality degrees. Figure 4, Panels B and C show that crossing the cutoff raises enrollment in any college and in a federal university. Table 2, Panel B reports the corresponding 2SLS estimates. Enrollment in any college rises by 12 percentage points for marginal non-targeted students and 26 for marginal targeted students, while enrollment in a federal university rises by 30 and 49 percentage points, respectively. Federal universities tend to offer higher-quality programs than students’ alternatives. These enrollment gains therefore improve access to higher-quality degrees, as reflected in higher college enrollment ranks (Figure 4, Panel D). The 2SLS estimates show that enrolling in the submitted degree raises enrollment rank by 16 points for non-targeted students and 30 for targeted students, relative to non-offered means of 64 and 40. Both groups benefit from access to federal degrees, but the gains are larger for targeted students. Appendix Figure A.5 reports similar results using alternative quality rankings based on graduates’ wages and the Ministry of Education’s quality index.

Finding 2: Non-targeted students have better outside options. To understand why targeted students gain more from admission, in Table 3 we report estimates of marginal enrollees’ next-best

Table 3: Quality of the next-best alternative

	NA	AA
	(1)	(2)
Not enrolled anywhere (rank = 0)	0.15 (0.02)	0.31 (0.02)
Fallback rank, conditional on enrolling	88.48 (0.79)	74.12 (1.14)
Fallback enrollment rank	75.02 (2.14)	50.97 (1.51)

Notes: This table reports the next-best alternatives of marginal enrollees, defined as the compliers of the 2SLS design in Table 2. We estimate these fallback outcomes following Abadie (2002), using the specification and enrollment-rank bandwidths in Table 2. Standard errors (in parentheses) are clustered by student.

alternatives, following Abadie (2002) and Mountjoy (2026). Without the seat, 15% of non-targeted students would not enroll anywhere, compared with 31% of targeted students. Among those who would enroll, non-targeted students would attend programs ranked at the 88th percentile of quality on average, compared with the 74th for targeted students. College enrollment rank combines these differences in enrollment and program quality by assigning zero to those who would not enroll. The average fallback enrollment rank is the 75th percentile for non-targeted students and the 51st for targeted students. These results are consistent with evidence from admissions discontinuities at two Brazilian federal universities, where Francis-Tan and Tannuri-Pianto (2018) find gains for racially marginalized applicants and Duryea et al. (2023) find gains for low-income applicants but little benefit for higher-income students, who may have access to private programs of similar quality to the federal alternative.

Finding 3: Higher dropout does not translate into lower college completion. As discussed in Section 3.4, one concern is that students admitted with lower test scores may be more likely to drop out. To investigate this concern, we compare dropout rates of marginally admitted and rejected students. Conditional on enrolling, students above the cutoff enter more challenging degrees than those below it. Dropout four years after admission rises for targeted students, with no clear increase for non-targeted students (Figure 4, Panel E). The 2SLS estimates are 1.1 and 6.1 percentage points for open and reserved seats, over non-offered means of 24% and 29%. The estimated increase is larger for targeted students. Because crossing the cutoff affects enrollment, students who enroll on either side need not be comparable. We therefore interpret the dropout estimates as descriptive. Higher dropout need not imply lower college completion, since students who move to a lower-ranked degree may still graduate. We find no evidence that admission lowers the probability of graduating from any program within seven years (Figure 4, Panel G).

Finding 4: The quality gains persist to graduation and are larger for targeted students.

Figure 4, Panel H shows that the gains in access to higher-quality degrees persist through graduation for both groups, despite higher dropout among targeted students. College graduation rank captures both whether students complete a degree and the quality of the degree they complete. Table 2, Panel C reports the corresponding 2SLS estimates. Seven years after application, enrolling in the submitted degree raises college graduation rank by 4.6 points for non-targeted students and 9.8 for

targeted students, over non-offered means of 56 and 31. Panel F of the figure also shows that crossing the cutoff makes both groups more likely to graduate from a federal university, mirroring the gains in federal enrollment at entry.

Taken together, these results show that targeted students gain more from admission at both enrollment and graduation, reflecting their weaker outside options. This suggests that AA may help narrow the college-quality gap discussed in Section 3.1. We can also use these estimates to assess the direct effect of converting an open seat into a reserved seat. The gain in graduation rank for the benefited targeted student is about twice the loss for the displaced non-targeted student. The implied difference in marginal returns, Δ^M , defined in Section 4.2, is about 5.2 rank points.

Design validity. Our regression discontinuity design requires continuity in potential outcomes and potential enrollment choices at each admissions cutoff. Appendix Figure A.6 shows no bunching in application scores at either open- or reserved-seat cutoffs. The local-polynomial manipulation tests of Cattaneo et al. (2020) likewise find no evidence of a discontinuity in either density.

We also test whether predetermined characteristics are balanced across the cutoffs. For each covariate W_{ij} , we estimate

$$W_{ij} = \gamma_t D_{ij}^h + \alpha_{jk_{ip}}^h + q_{jt}^h(r_{ij}^h) + \nu_{ij}^h, \quad (10)$$

where $\alpha_{jk_{ip}}^h$ denotes admissions-list-stage fixed effects and $q_{jt}^h(\cdot)$ is defined analogously to the running-variable functions in the 2SLS specification. The coefficient γ_t measures the covariate discontinuity for AA status t , pooling the initial-offer and waitlist stages and averaging over the admissions pools belonging to that status. We test whether $\gamma_t = 0$ for each AA status. Consistent with a valid research design, Appendix Table A.3 and Appendix Figure A.7 show small differences across the eighteen predetermined characteristics reported in Table 1. To summarize balance across these covariates, Appendix Figure A.8 plots predicted federal enrollment and enrollment rank around the cutoffs using the same student characteristics. Neither prediction shows a visible discontinuity.²⁸

6 The overall effects of affirmative action: Model estimation

Section 5 uses admissions discontinuities to estimate Δ^M , the difference in returns for marginal targeted and non-targeted students. Our main object of interest, however, is $\Delta(\mu', \mu)$, the overall effect of changing the AA schedule from ω to ω' . It aggregates outcome changes across all students whose assignments change, including those reallocated through knock-on effects, and thus allows us to measure how much AA closes the enrollment and graduation gaps in Section 3.1 and how its gains and losses are distributed across groups.

To recover this overall effect, we need to estimate the policy effect for each student. Using the

²⁸A potential concern is that SISU's iterative structure may allow students to anticipate the provisional cutoffs. However, take-up of initial offers is low, so waitlists move substantially. Falling below a provisional cutoff therefore neither rules out admission nor removes the incentive to apply. Additionally, this concern does not apply to the waitlist cutoffs, which are unknown to applicants when they submit their choices. Consistent with this argument, predetermined covariates are very well balanced across these cutoffs.

notation from Equation (3) in Section 4, we write this individual effect as

$$\tau_i(\mu', \mu) = \sum_{j=0}^J A_{ij}(\mu') \cdot Y_{ij} - \sum_{j=0}^J A_{ij}(\mu) \cdot Y_{ij},$$

where μ and μ' are the matchings under each schedule, $A_{ij}(\cdot)$ indicates enrollment in j , and Y_{ij} is student i 's potential outcome under assignment j . Computing these individual effects requires two objects: the allocation under each schedule and the potential outcome under each possible assignment. Since the mechanism φ is known, computing the allocation under any schedule requires only recovering its inputs. In practice, however, preferences and take-up rules are only partially observed: students submit at most two options rather than a complete ranking over all degrees, and take-up decisions are observed only for offers students actually receive. Potential outcomes are likewise observed only under each student's realized assignment. The next three subsections recover preferences, take-up rules, and potential outcomes in turn.

6.1 Preferences over degree programs

We start by recovering preferences with a random utility model. Let student i 's indirect utility from receiving an offer from degree $j \in \mathcal{J}$ be

$$u_{ij} = V_{ij} + \eta_{ij}, \quad (11)$$

where V_{ij} denotes the systematic component of utility and η_{ij} captures unobserved tastes. This utility governs the preference ordering of degree programs offered in the centralized platform. It does not compare the value of an offer with that of the outside option, a comparison we model separately as a take-up decision in Section 6.2.

We parameterize the systematic component as

$$V_{ij} = \delta_j^t + \kappa_d^t \mathbf{d}_{ij} + \kappa_s^t \mathbf{s}_{ij}, \quad (12)$$

where δ_j^t is a degree fixed effect. The vector $\mathbf{d}_{ij}$ collects distance and geographic covariates. We include the log distance between the student's residence and the campus to capture commuting time, and geographic indicators to allow for relocation costs when the campus lies outside the student's commuting zone or state. We also allow distance costs to vary with household income and ENEM scores.²⁹ The vector $\mathbf{s}_{ij}$ captures how preferences vary with students' academic profiles. It includes the degree-specific application score and interactions of students' ENEM scores with degree quality and field of study. The application-score term allows for heterogeneity in preferences by academic performance. The interaction between average ENEM score and degree quality captures differences in how much students value higher-quality programs (vertical preferences), while the field interactions capture their tastes for particular fields (horizontal preferences). For instance, students with stronger math scores may sort more strongly toward quantitatively oriented programs.³⁰ We assume that

²⁹Specifically, $\mathbf{d}_{ij} = (\ell_{ij}, M_{ij}, \ell_{ij}C_{ij}, \ell_{ij}S_{ij}, \ell_{ij}\tilde{\ell}_i, \ell_{ij}\tilde{s}_i)$, where ℓ_{ij} is the log of one plus the distance in kilometers between the student's and the campus's municipalities, which is zero within the same municipality; M_{ij} , C_{ij} , and S_{ij} indicate the same municipality, commuting zone, and state; and $\tilde{\ell}_i$ and $\tilde{s}_i$ are centered log household income and average ENEM score. Commuting zones are IBGE microregions.

³⁰Specifically, $\mathbf{s}_{ij} = (s_{ij}, r_j \tilde{s}_i, \mathbf{f}_j s_i^L, \mathbf{f}_j s_i^M)$, where s_{ij} is the degree-specific application score, r_j is degree j 's quality

the taste shocks η_{ij} are i.i.d. Type I Extreme Value, conditional on observed student and degree characteristics. To allow preferences to vary flexibly by AA status, we estimate the model separately for each status t .

Under the mechanism described in Section 4, we assume that initial offers form the student-optimal stable matching. SISU’s design supports this assumption: under standard conditions, the iterative mechanism produces the student-optimal stable matching (Bó and Hakimov, 2022).³¹ Each student is therefore offered their most preferred degree among those for which their score clears the relevant admissions cutoff. As formalized by Fack et al. (2019), stability implies that each student effectively faces an observable, personalized choice set defined by all ex post feasible degrees.

Let $\Omega_i = \{j \in \mathcal{J} : s_{ij} \geq c_j^{1,t_i}\}$ denote student i ’s initial-offer feasible set, where c_j^{1,t_i} is the cutoff at degree j for students of AA status t_i . Let $R_i = (R_{i1}, R_{i2})$ denote the student’s submitted rank-order list. Because students may anticipate a waitlist offer, their submitted options need not belong to Ω_i . We therefore define the comparison set as $C_i \equiv \Omega_i \cup \{R_{i1}, R_{i2}\}$. Stability yields the following revealed-preference restrictions:

$$R_{i1} = \arg \max_{j \in C_i} u_{ij}, \quad R_{i2} = \arg \max_{j \in C_i \setminus \{R_{i1}\}} u_{ij}. \quad (13)$$

The first restriction implies that R_{i1} is preferred to R_{i2} and to every other degree in the initial-offer feasible set Ω_i . The second implies that, after removing R_{i1} , R_{i2} is preferred to every remaining degree in Ω_i . Together, these restrictions reveal how the two submitted options rank relative to all degrees the student could have obtained at the initial-offer stage. Under the logit assumption, the conditional probability of observing the two-option list R_i is the product of the two corresponding choice probabilities:

$$p_i^R(\boldsymbol{\kappa}^t) \equiv \Pr(R_i \mid C_i, \mathbf{d}_i, \mathbf{s}_i; \boldsymbol{\kappa}^t) = \left(\frac{\exp(V_{iR_{i1}})}{\sum_{k \in C_i} \exp(V_{ik})} \right) \left(\frac{\exp(V_{iR_{i2}})}{\sum_{k \in C_i \setminus \{R_{i1}\}} \exp(V_{ik})} \right).$$

6.2 Offer take-up

Conditional on receiving an offer from degree j in stage h (i.e., $D_i^h = j$), the student decides whether to enroll. In our setting, accepting an initial offer does not prevent a student from accepting a later waitlist offer. Declining an offer therefore reveals that the student prefers the outside option to the offered degree, consistent with the take-up rule of Section 4. Because students face no trade-off between accepting the current offer and retaining access to later offers, we also model take-up

rank, $\bar{s}_i$ is the average ENEM score, s_i^L and s_i^M are language and math scores, and $\mathbf{f}_j$ contains indicators for STEM and medicine.

³¹The two main potential threats to this stability result appear unlikely to matter in our setting. First, Bó and Hakimov (2022) note that groups of high-scoring students could, in principle, manipulate admissions cutoffs by coordinating changes to their reported options during the final days of the mechanism. Such manipulation would require coordination among many applicants without any means of enforcing compliance, and the daily cutoffs reported in Appendix F show no sign of such behavior. Second, because stability is an ex post optimality condition, imperfect information about final admissions cutoffs could prevent students from reaching a stable allocation. Appendix F shows that SISU’s iterative structure leaves students with little uncertainty about these cutoffs. Lab experiments also find that iterative versions of deferred acceptance generate stable allocations more frequently than the direct mechanism (Klijn et al., 2019; Bó and Hakimov, 2020).

separately in each stage.

To parameterize the stage-specific take-up rules T^1 and T^2 defined in Section 4, let $\tilde{\eta}_{ij} \equiv \eta_{ij} - \bar{\eta}$, where $\bar{\eta}$ is the mean of a standard Type I Extreme Value random variable. Centering the preference shocks does not change the ranking of degrees, so we write the take-up rule in terms of the centered shocks:

$$T_{ij}^h = \mathbb{1} \left\{ V_{ij} + \tilde{\eta}_{ij} > g_i^h \right\}, \quad g_i^h = \nu_i^h + \sigma_e^t \epsilon_i^h. \quad (14)$$

To decide whether to take up an offer from degree j in stage h , students compare the utility from the degree with the value of their outside option, g_i^h . The deterministic component ν_i^h captures observed differences in take-up across students and stages. The shocks $\epsilon_i^h \sim N(0, 1)$ represent unobserved outside opportunities, changes in plans, and constraints that arise after an offer. We assume these shocks are independent across stages and of the degree-preference shocks η_{ij} . The parameter σ_e^t allows their dispersion to vary by AA status.

We specify the deterministic component as

$$\nu_i^h = \pi_w^t \mathbf{w}_i + \pi_z^t \mathbf{z}_i + \pi_h^t \mathbb{1}\{h = 2\}, \quad (15)$$

where $\mathbf{w}_i$ includes a constant, student characteristics, and state-of-residence indicators.³² The vector $\mathbf{z}_i$ captures how local college supply and competition affect the value of the outside option. We allow the effect of degree availability to vary with student achievement.³³ Waitlist offers arrive later, often after students have made other plans. The indicator $\mathbb{1}\{h = 2\}$ allows take-up to differ at this stage.

Conditional on $\tilde{\eta}_{ij}$, the take-up probability for an offer from degree j is

$$\Pr \left(T_{ij}^h = 1 \mid V_{ij}, \nu_i^h, \tilde{\eta}_{ij} \right) = \Phi \left(\frac{V_{ij} + \tilde{\eta}_{ij} - \nu_i^h}{\sigma_e^t} \right). \quad (16)$$

In the data, however, we observe take-up only for the degree offered to student i in each stage. Because offers depend on submitted lists, students who receive an offer from degree j tend to have stronger unobserved tastes for it. We account for this selection by conditioning the distribution of $\tilde{\eta}_{ij}$ on the student's initial-offer feasible set Ω_i and submitted list R_i .

Let $\Gamma^t \equiv (\boldsymbol{\kappa}^t, \boldsymbol{\pi}^t, \sigma_e^t)$. Integrating over the conditional distribution of the preference shock gives the take-up probability for the observed offer in stage h :

$$p_{ih}^T(\Gamma^t) \equiv \Pr \left(T_{iD_i^h}^h = 1 \mid \Omega_i, R_i; \Gamma^t \right) = \int \Phi \left(\frac{V_{iD_i^h} + \tilde{\eta} - \nu_i^h}{\sigma_e^t} \right) f_{\boldsymbol{\kappa}^t}(\tilde{\eta} \mid \Omega_i, R_i) d\tilde{\eta}, \quad (17)$$

where $f_{\boldsymbol{\kappa}^t}(\cdot \mid \Omega_i, R_i)$ is the conditional density of $\tilde{\eta}_{iD_i^h}$ implied by the revealed-preference restrictions in Equation (13). Appendix G derives this density.

³²The constant also absorbs $\bar{\eta}$, the mean subtracted from the preference shocks. The student characteristics include gender, log age, log household and municipal average income, an indicator for belonging to an AA-targeted racial group (Black, Brown, or Indigenous), indicators for prior enrollment in a federal or private higher education institution and high-school graduation in the year before application, and standardized average ENEM score.

³³Specifically, $\mathbf{z}_i$ includes the log numbers, within the student's commuting zone, of federal degrees, non-federal degrees in the top half of the quality distribution, and applicants with an average ENEM score above 550. We interact the two degree counts with indicators for the student's average ENEM score falling in the 75th–90th percentile, the 90th–95th percentile, or the top 5% of the applicant score distribution.

We estimate the preference and take-up parameters jointly by maximum likelihood, separately by AA status $t \in \{\text{AA}, \text{NA}\}$, stacking the offers each student receives in both stages. The likelihood is

$$\mathcal{L}(\Gamma^t) = \left[\prod_{i:t_i=t} p_i^R(\boldsymbol{\kappa}^t) \right] \left[\prod_{\substack{(i,h): t_i=t \\ D_i^h > 0}} [p_{ih}^T(\Gamma^t)]^{T_{D_i^h}^h} [1 - p_{ih}^T(\Gamma^t)]^{1-T_{D_i^h}^h} \right]. \quad (18)$$

6.3 Potential outcomes

The remaining object is the potential outcome Y_{ij} under each assignment. We focus on the college graduation rank defined in Section 3.1, which we observe only under each student's realized assignment. The key selection concern is that unobserved factors affecting preferences and offer take-up may also predict these outcomes. We therefore estimate potential outcomes with a control-function approach that corrects for selection into offers and enrollment, extending [Abdulkadiroğlu et al. \(2020\)](#) to allow for non-random take-up of offers.

Let Y_{ij} denote student i 's potential outcome if enrolled in degree $j \in \{1, \dots, J\}$, and let Y_{i0} denote their potential outcome under the outside option. We specify

$$Y_{ij} = \alpha_j^t + \beta_s^t \mathbf{s}_{ij} + \beta_x^t \mathbf{x}_i + \varepsilon_{ij}, \quad (19)$$

$$Y_{i0} = \alpha_0^t + \beta_{0x}^t \mathbf{x}_i + \varepsilon_{i0}, \quad (20)$$

where α_j^t is a degree fixed effect and α_0^t is the outside-option intercept. The inside-option equation includes $\mathbf{s}_{ij}$, the degree-specific score covariates from the preference model in Section 6.1. These covariates do not enter the outside-option equation, which has no degree-specific quality, field of study, or application weights. The vector $\mathbf{x}_i$ includes $\mathbf{w}_i$ and $\mathbf{z}_i$, as defined in Section 6.2, together with additional demographic controls, the five ENEM component scores, and quadratic and cubic terms in average ENEM performance. It enters the inside- and outside-option equations with separate coefficient vectors, β_x^t and β_{0x}^t .

As in the multinomial logit selection model of [Dubin and McFadden \(1984\)](#), we assume that the conditional means of the outcome residuals are linear in the preference and take-up shocks.³⁴ To simplify notation, we suppress the observed covariates from the conditioning sets below.

$$\mathbb{E}[\varepsilon_{ij} \mid \eta_i, \epsilon_i^1, \epsilon_i^2] = \sum_{k=1}^J \psi_k^t \tilde{\eta}_{ik} + \rho^t \tilde{\eta}_{ij} + \xi^t \sum_{h=1}^2 \epsilon_i^h, \quad (21)$$

$$\mathbb{E}[\varepsilon_{i0} \mid \eta_i, \epsilon_i^1, \epsilon_i^2] = \sum_{k=1}^J \psi_{0k}^t \tilde{\eta}_{ik} + \xi_0^t \sum_{h=1}^2 \epsilon_i^h. \quad (22)$$

This error structure allows for three sources of selection. The parameters ψ_k^t capture selection on levels: students with strong unobserved preferences for particular degrees may also have higher unobserved ability, which shifts their outcomes under every assignment. Students who prefer medicine,

³⁴Nonparametric identification of selection models becomes demanding as the number of alternatives grows, and linearity in the unobserved shocks is the standard restriction ([Lee, 1983](#); [Dahl, 2002](#); [Heckman et al., 2008](#)).

for example, may also perform better everywhere else. The coefficient ρ^t captures selection on gains, since students who prefer degree j may also gain more from attending it. The coefficient ξ^t captures selection through take-up. Among students with the same offer, those who accept may be more likely to graduate or have weaker outside options than those who decline. The parameters ψ_{0k}^t and ξ_0^t play the same roles for outside-option outcomes. We impose the same take-up selection parameters on the initial-offer and waitlist shocks.

Let Y_i denote the realized outcome and $A_i \in \mathcal{J} \cup \{0\}$ the student's final enrollment assignment. To account for selection into A_i , we condition on the feasible set Ω_i , the submitted list R_i , and the observed offer and take-up history $\mathcal{H}_i \equiv \{(D_i^h, T_{iD_i^h}^h) : h = 1, 2, D_i^h > 0\}$. Applying the law of iterated expectations to Y_i and using the error structure in Equations (21)–(22) gives the estimating equations

$$\mathbb{E}[Y_i \mid \Omega_i, R_i, \mathcal{H}_i, A_i = j] = \alpha_j^t + \beta_s^t \mathbf{s}_{ij} + \beta_x^t \mathbf{x}_i + \sum_{k=1}^J \psi_k^t \lambda_{ijk} + \rho^t \lambda_{ijj} + \xi^t \sum_{h=1}^2 m_{ij}^h, \quad (23)$$

$$\mathbb{E}[Y_i \mid \Omega_i, R_i, \mathcal{H}_i, A_i = 0] = \alpha_0^t + \beta_{0x}^t \mathbf{x}_i + \sum_{k=1}^J \psi_{0k}^t \lambda_{i0k} + \xi_0^t \sum_{h=1}^2 m_{i0}^h, \quad (24)$$

where, for any $a \in \mathcal{J} \cup \{0\}$, $\lambda_{iak} \equiv \mathbb{E}[\tilde{\eta}_{ik} \mid \Omega_i, R_i, \mathcal{H}_i, A_i = a]$ and $m_{ia}^h \equiv \mathbb{E}[\epsilon_i^h \mid \Omega_i, R_i, \mathcal{H}_i, A_i = a]$. The λ terms correct for selection on unobserved preferences over degree programs, while the m^h terms correct for selection through take-up in both stages. Appendix G derives these control functions using the estimated preference and take-up models. We estimate Equations (23)–(24) by OLS.

Identification. The identification of potential outcomes under endogenous application choices and take-up decisions requires two sources of variation: variation in choice sets and a shifter of students' choices that is excluded from potential outcomes (Agarwal et al., 2025; Kapor et al., 2024). In our model, admissions cutoffs generate variation in the choice sets students face, while the distance between a student's home and each campus provides the excluded choice shifter.

Admissions cutoffs, the first source, provide two experiments on different groups of students: those who choose the same degree on either side of the cutoff, and those whose degree changes when they cross it. Students in the first group who are close to a cutoff but on opposite sides of it face discretely different choice sets. This generates variation in their control functions, λ_{ijk} , since a student who chooses degree k when degree j is also available is likely to have a stronger unobserved preference for k than a student who chooses k without having j available. For the second group, once we condition on unobserved preferences via the control functions, the cutoff provides a second experiment: they enroll in different degrees because they receive different offers.

We combine cutoff discontinuities with a second source of variation that shifts both choices and take-up decisions. As in Walters (2018) and Mountjoy (2022), we use distance as an excluded choice shifter. The distance covariates $\mathbf{d}_{ij}$ enter the offer utility and the take-up equation but are excluded from the potential outcome Equations (19)–(20). The identifying assumption is that distance has no direct effect on outcomes conditional on enrollment and observables. Similar to Mountjoy (2022), our controls include the average income of the student's municipality and state-of-residence indicators to make distance comparisons among students living in comparable areas. This exclusion generates

variation in the control functions λ_{ijk} and m_{ij}^h that is independent of potential outcomes. Together, the two sources identify the selection parameters in Equations (21)–(22). Given these parameters, the second cutoff experiment identifies the degree effects α_j^t .

Formally, identification relies on three conditions. First, cutoffs cannot be manipulated and unobservables are smooth at the thresholds, so that cutoff-crossing provides quasi-experimental variation in Ω_i . These assumptions are supported by the density and balance tests of Section 5. Second, distance is excluded from potential outcomes. Third, we rely on the functional-form assumptions of the model to extrapolate the degree effects away from the cutoffs.

6.4 Estimates

We estimate the model using applicants in the first-semester 2013 SISU edition who applied to a federal institution and had an average ENEM score above 550. We classify students as targeted if they graduated from a public high school, as reported in ENEM, or applied through a reserved-seat pool, and as non-targeted otherwise. We restrict the degree universe to programs with at least five targeted and five non-targeted enrollees and positive capacity in both open and reserved seats, which allows us to estimate program-level fixed effects in the preference and outcome models separately by AA status. The final sample contains 543,814 students applying to 1,719 degrees across 73 federal institutions.

Preferences over degree programs. Panel A of Table 4 reports selected preference estimates, with the full coefficient sets in Appendix Tables H.1–H.3. We find large negative coefficients on log distance for both targeted and non-targeted students, indicating strong preferences for nearby programs. Distance matters less, however, for higher-income students, consistent with their greater ability to afford travel and relocation costs. We next turn to the score interactions that capture horizontal preferences over fields of study and vertical preferences over degree quality. Our estimates show that students display horizontal preferences over fields. Those with stronger math scores place more value on STEM and medicine. The positive coefficient on quality $\times$ score also indicates vertical preferences, with higher-scoring students placing more value on higher-quality programs.

Enrollment take-up model. Panel B reports the take-up estimates. Students offered an initial-offer seat enroll at rates of 39.4% for targeted and 36.1% for non-targeted students. Two patterns stand out. First, among otherwise similar students, receiving the offer through the waitlist lowers this probability by 6.8 percentage points for targeted students and 0.4 for non-targeted students. The larger penalty is consistent with targeted students facing greater difficulty enrolling after a late offer. Second, take-up varies with the strength of students’ outside options. Students are more likely to decline an offer when they have higher scores, were previously enrolled at a private institution, or have more high-quality non-federal degrees in their commuting zone. Older students are also more likely to accept, consistent with having fewer alternatives.

Outcomes model. Panel C of Table 4 reports the outcome-model parameters, with empirical-Bayes posterior means for the degree-level parameters α_j^t , ψ_j^t , and ψ_{0j}^t . The outcome is the college graduation rank defined in Section 3.1. We estimate the program fixed effects using empirical-Bayes shrinkage toward a covariate-assisted prior, following Abdulkadiroğlu et al. (2020). In Appendix I,

Table 4: Model parameters

	Targeted				Non-targeted			
	Mean	SD/(SE)	P10	P90	Mean	SD/(SE)	P10	P90
<i>Panel A.</i> Degree preferences								
Degree FE (δ_j^t)	1.172	2.522	-2.085	4.267	1.633	3.377	-2.660	5.540
Distance (κ_d^t)								
Log distance	-1.219	(0.005)			-1.015	(0.005)		
Log distance $\times$ same state	0.224	(0.002)			0.182	(0.002)		
Log distance $\times$ income	0.054	(0.002)			0.055	(0.002)		
Log distance $\times$ score	-0.088	(0.005)			-0.106	(0.004)		
Score (κ_s^t)								
Application score	-3.326	(0.062)			-4.221	(0.060)		
Quality $\times$ score	29.187	(0.250)			42.742	(0.291)		
STEM $\times$ math	0.574	(0.013)			0.530	(0.014)		
STEM $\times$ language	-0.237	(0.012)			-0.382	(0.012)		
<i>Panel B.</i> Enrollment take-up model								
Outside-option value (π^t)								
Log age	-5.739	(0.548)			-15.621	(1.008)		
Prior private enrollment	4.258	(0.368)			1.130	(0.446)		
Average score	25.136	(0.616)			39.424	(0.596)		
Log non-federal degrees in CZ	3.494	(0.290)			5.364	(0.493)		
Waitlist shift (π_h^t)	3.957	(0.248)			0.270	(0.263)		
Utility scale (σ_e^t)	19.898	(0.869)			23.432	(1.084)		
Students offered		75,125				83,926		
Initial offer		38,179				44,605		
Waitlist		36,946				39,321		
Enrollment rate		34.6%				35.7%		
Waitlist offers		29.7%				35.2%		
<i>Panel C.</i> Outcomes model for graduation rank after 7 years								
Degree FE (α_j^t)	721.7	9.8	711.2	732.1	733.3	8.8	724.4	741.8
Selection on levels (ψ_j^t)								
Inside option	-0.548	0.651	-1.315	0.268	-0.230	0.300	-0.389	-0.043
Outside option	-0.485	0.501	-1.105	0.154	-0.612	0.657	-1.484	0.187
Selection through take-up (ξ^t)								
Inside option	0.883	(1.421)			3.900	(7.600)		
Outside option	1.296	(0.380)			0.475	(0.391)		
Selection on gains (ρ^t)	0.527	(0.267)			-0.124	(0.314)		
Share of students		51.0%				49.0%		
Number of students				543,814				
Number of degrees				1,719				

Notes: This table presents parameter estimates from the offer-choice, take-up, and outcomes models. Standard errors are in parentheses. Panel A reports offer-choice coefficients from Equation (12), including the degree fixed effects δ_j^t and a subset of the distance-block parameters κ_d^t and score-block parameters κ_s^t . Distance enters as log kilometers, scores are in s.d., and the quality rank is rescaled to $[0, 1]$. Panel B reports take-up parameters from Equations (14)–(15) for the outside-option value π^t , including demographics and measures of local college supply and competition in the student’s commuting zone (CZ); the waitlist shift π_h^t ; and the utility scale σ_e^t . A larger π raises the outside-option value and lowers take-up. Panel C reports the outcomes model, including degree fixed effects α_j^t ; selection on levels ψ_j^t , ψ_{0j}^t and through take-up ξ^t , ξ_0^t (inside and outside options); and selection on gains ρ^t . The degree-level parameters α_j^t , ψ_j^t , and ψ_{0j}^t are empirical-Bayes posterior means. Full coefficient sets are displayed in Appendix Tables H.1–H.3.

we describe the procedure.

We find substantial variation in program fixed effects. The standard deviation across programs is 9.8 for targeted students and 8.8 for non-targeted students. The gap between the 90th- and 10th-percentile program effects is 21 and 17 points, respectively. These magnitudes are large on a 0–100 outcome scale, indicating that the degree a student attends is strongly related to graduation quality. The remaining estimates describe how unobserved heterogeneity that drives degree choices and offer take-up relates to graduation outcomes. The selection-on-gains parameter ρ^t is positive and marginally significant for targeted students but indistinguishable from zero for non-targeted students. For selection on levels, the estimates $\hat{\psi}_j^t$ and $\hat{\psi}_{0j}^t$ have means near zero and standard deviations across programs below one, which are small relative to the variation in program fixed effects. The take-up selection parameter ξ^t is imprecisely estimated for both groups. For targeted students, however, the estimate of ξ_0^t suggests that those with a higher unobserved propensity to decline offers have better outside-option outcomes. Overall, most systematic variation in graduation quality is captured by student characteristics and program fixed effects.

Figure 5, Panel A summarizes the implied return to observed federal enrollment relative to the outside option. It plots the predicted potential outcome at each student’s observed degree of enrollment, $\hat{Y}_{ij}^{\text{obs}}$, and the predicted outside-option outcome, $\hat{Y}_{i0}$, against average ENEM score, separately by AA status. Three patterns stand out. First, non-targeted students have substantially better outside options at every score level. Second, at a given score, predicted graduation ranks are lower for targeted students, even though both groups enter programs of similar quality. This difference may reflect lower completion rates, graduation from lower-ranked programs, or both. Third, the return to federal enrollment is positive and large on average for both groups: enrollment value-added is 5.5 points for non-targeted students and 8.3 points for targeted students. This return narrows with score and falls to roughly zero at the top of the score distribution, where outcomes under the outside option nearly match those at a federal degree.

Parameter validation. We assess the in-sample fit of the preference estimates by comparing simulated and observed initial-offer cutoffs in Figure 5, Panel B. The model reproduces their ordering well, with rank correlations of 0.94 for open seats and 0.92 for reserved seats. For final cutoffs, which also reflect take-up decisions, the correlations are 0.77 for open seats and 0.80 for reserved seats.

7 The overall effects of affirmative action: Results

We use the model estimates to simulate student assignments and realized outcomes under different AA schedules. These simulations recover the components of the overall policy effect defined in Section 4.2. The analysis proceeds in three parts. First, we trace how each schedule changes admissions cutoffs, choice sets, and the assignments of targeted and non-targeted students. Second, we measure gains and losses among reallocated students, comparing their enrollment and graduation ranks under no AA and under the 50% schedule. Third, we aggregate over all students to measure how far AA closes the college-quality gap defined in Section 3.1 and how gains and losses are distributed between targeted and non-targeted students.

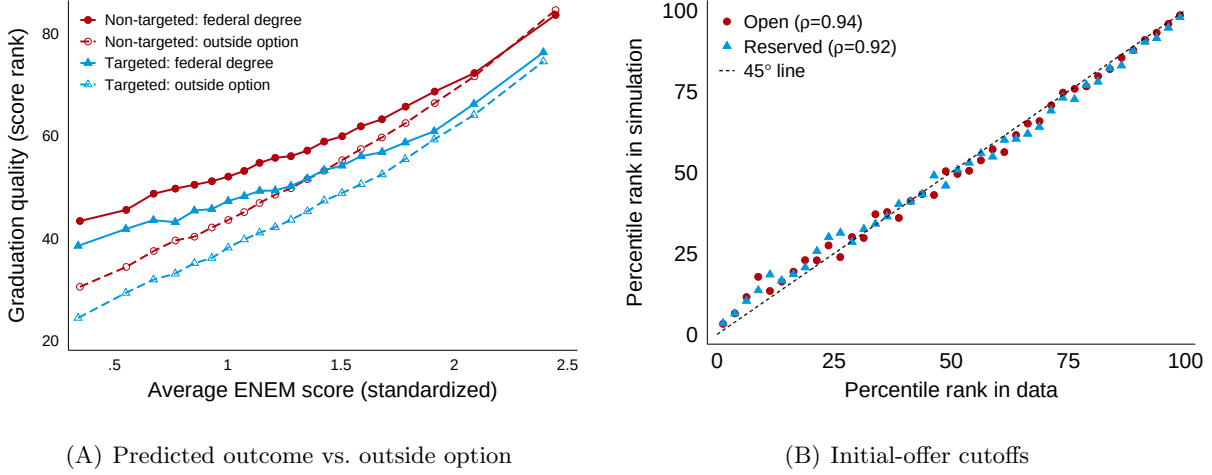


Figure 5: Model fit and predicted outcomes

Notes: Panel A plots the predicted potential outcome at the enrolled degree, $\hat{Y}_{ij}^{\text{obs}}$ (solid), and the predicted outside option, $\hat{Y}_{i0}$ (dashed), against standardized average ENEM score, for targeted (blue) and non-targeted (red) students. Predictions use the empirical-Bayes posterior means and set the selection terms to zero. Panel B plots the percentile rank of the simulated initial-offer cutoff against the percentile rank of the observed initial-offer cutoff across the 1,719 degree programs, for the open (red circles) and reserved (blue triangles) admissions tracks. Percentile ranks are computed within track, and each marker is the mean simulated rank within one of 40 equal-sized bins of the observed rank. The observed reserved cutoff is the seat-weighted average across the four reserved sub-tracks. Simulated cutoffs are means over 1,000 Monte Carlo replications, and the legend reports the rank correlations.

7.1 Estimating counterfactual assignments and outcomes

Because the rules of the mechanism φ are known, we can simulate the assignment using the matching function $\mu = \varphi(R, T, \mathbf{s}, \mathbf{t}; q, \omega)$. We are interested in allocations under different AA schedules ω that specify the number of seats reserved for targeted students in each degree. The remaining inputs are the reports R , take-up rules T , priority variables (scores and targeted status) $(\mathbf{s}, \mathbf{t})$, and degree capacities q . We hold capacities and the applicant pool $\mathcal{I}$ fixed at their observed values so that the simulations isolate the effect of changing the AA schedule.

From reports to preferences. Section 6 delivers estimates of preferences $\succ$ and take-up rules T . The mechanism takes reports R rather than preferences as inputs. Under student-proposing deferred acceptance with reservations, however, truthful reporting is a dominant strategy (Rios et al., 2021).³⁵ Thus, R_i represents the relevant restriction of $\succ_i$, and the initial offers constitute the student-optimal stable matching with respect to preferences (Gale and Shapley, 1962; Abdulkadiroğlu, 2005). The subsequent take-up and waitlist stages preserve this equivalence. Because initial-offer take-up is monotone in preferences, a student who would decline an offer would also decline any less-preferred offer; anticipating these decisions therefore creates no incentive to misreport the ranking of programs. We can consequently simulate the mechanism directly from preferences: $\varphi(R, T, \mathbf{s}, \mathbf{t}; q, \omega) = \varphi(\succ, T, \mathbf{s}, \mathbf{t}; q, \omega)$.

³⁵The counterfactuals use standard student-proposing deferred acceptance with unrestricted report length, under which truthful reporting is a dominant strategy. In the data, the initial-offer stage uses an iterative deferred acceptance mechanism with reports limited to two options, which we use for estimation in Section 6.

Counterfactual simulations. Each simulation uses the degree-preference, offer take-up, and potential-outcome models estimated in Section 6. We first draw preference shocks η_{ij} and take-up shocks ϵ_i^h . Given these draws and the estimated parameters, we recover each student’s preferences $\hat{\succ}_i$, take-up rule $\hat{T}_i$, and potential outcomes $\hat{Y}_{ij}$ under every assignment. We then simulate the allocation under each AA schedule ω and assign each student the potential outcome corresponding to that assignment. We hold the draws fixed across schedules so that differences in outcomes reflect the policy instead of simulation noise. We report point estimates averaged over 1,000 replications. Appendix J provides the equations and describes the procedure step by step.

7.2 How affirmative action reshapes student assignments

We first examine how AA changes admissions cutoffs, students’ feasible choice sets, and final assignments. We simulate the mechanism under the schedules $\omega' = \omega(x)$ for common reservation shares $x \in \{12.5, 25, 37.5, 50\}\%$, which mirror the policy’s phase-in, and compare each resulting allocation with the no-AA baseline ($\omega = \omega(0)$). Under no AA, all seats are open, so each degree has a single admissions cutoff. Introducing AA creates separate cutoffs for open and reserved seats. As x increases, seats shift from open to reserved capacity, raising the open-seat cutoff and lowering the reserved-seat cutoff. The resulting cutoff wedge widens with x . Appendix Figure A.9 compares the distribution of cutoffs under no AA and under 50% AA.

A degree is feasible for a student if their score clears the relevant admissions cutoff. The cutoff wedge therefore expands targeted students’ feasible choice sets and contracts those of non-targeted students. Figure 6, Panel A plots the change in the number of feasible degrees between no AA and the 50% reservation schedule against average ENEM score. On average, the choice set of targeted students expands from 175 to 315 degrees, while that of non-targeted students contracts from 463 to 320. These changes are concentrated among students with average ENEM scores between 600 and 750. Those below this range qualify for few degrees under either schedule, while those above it qualify for most.

To characterize the reallocations induced by AA, we compare each student’s assignment under a schedule ω' , $A_i(\mu')$, with their assignment under no AA, $A_i(\mu)$. As defined in Section 4.2, a student is a *switcher* if these assignments differ. Across the schedules we simulate, virtually every switcher belongs to one of two groups: *benefited* targeted students or *displaced* non-targeted students.³⁶ Benefited targeted students are defined as those who either enter the federal system from the outside option or move to a more-preferred degree within it. Displaced non-targeted students, in contrast, are those who either exit to the outside option or move to a less-preferred degree.

Panel B reports the number of benefited targeted students and displaced non-targeted students at each reservation share. These counts correspond to N_{AA}^S and N_{NA}^S , the numbers of switchers in each group defined in Equation (5). At $\omega' = \omega(0.5)$, approximately 34,000 targeted students benefit and 45,000 non-targeted students are displaced, equivalent to 12% of targeted applicants and 17% of non-

³⁶Knock-on effects can, in principle, displace targeted students or benefit non-targeted students. When moving from $\omega = \omega(0)$ to $\omega' = \omega(0.5)$, however, these cases affect only 0.05 students per simulation on average, compared with about 34,000 benefited targeted and 45,000 displaced non-targeted students. Because these cases are negligible, we exclude them from the analysis of benefited and displaced students, but the aggregate effects in Section 7.4 include them.

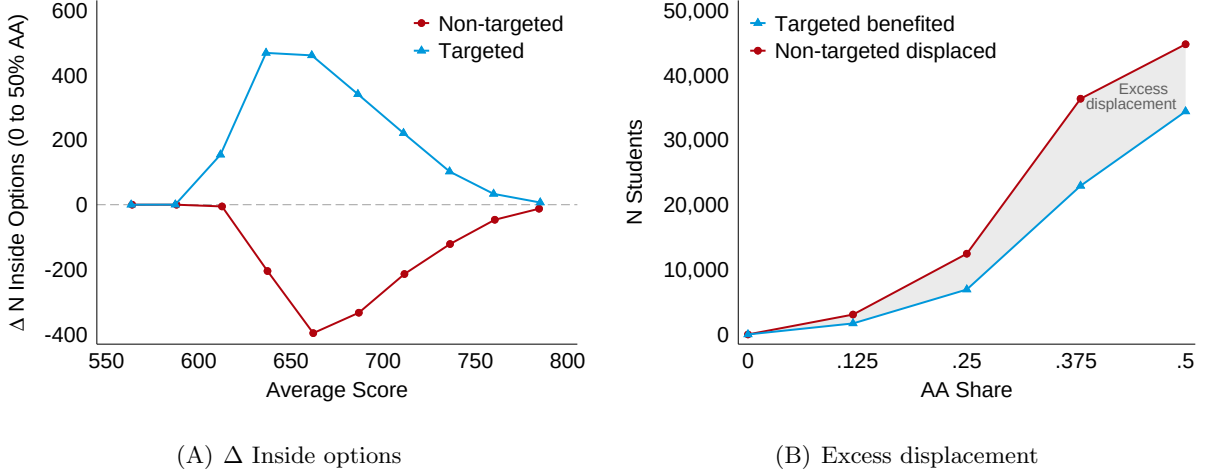


Figure 6: Assignment effects by AA share

Notes: Panel A plots the change in the number of feasible degrees against average ENEM score as the schedule rises from $\omega = \omega(0)$ to $\omega' = \omega(0.5)$. A degree is feasible if the student's score exceeds the relevant admissions cutoff. Panel B shows assignment changes relative to the $\omega(0)$ baseline at each reservation share. "Targeted benefited" counts targeted students who enter the federal system or switch to a more-preferred degree within it. "Non-targeted displaced" counts non-targeted students who exit the federal system or switch to a less-preferred degree. The gray area in Panel B shows the excess number of displaced students over benefited students. In both panels, targeted students are shown in blue and non-targeted students in red.

targeted applicants. Because federal seats are fixed, the number of benefited students who enter the system equals the number of displaced students who leave it, about 13,600. However, approximately 21,000 targeted students move to more-preferred federal degrees, compared with 31,000 non-targeted students moving to less-preferred ones. This imbalance reflects the knock-on effects described in Section 4.2. As a result, in our setting, about 30% more students are displaced than benefited.

7.3 Gains and losses for benefited and displaced students

We next quantify the gains and losses from these reallocations. For each switcher, we compare enrollment and graduation ranks under no AA ($\omega = \omega(0)$) and the 50% reservation schedule ($\omega' = \omega(0.5)$). Figure 7 plots average enrollment and graduation ranks under the two schedules against ENEM score on the left axis, separately for benefited targeted students in Panel A and displaced non-targeted students in Panel B. The gray histograms show the average number of switchers per simulation in each ENEM-score bin on the right axis. For each outcome, the gap between the two lines is the average, among switchers at that score, of the individual policy effect $\tau_i(\mu', \mu)$ defined in Equation (3).

The gray histograms show that reallocations are concentrated in the same score range in which AA most strongly changes students' feasible options (Figure 6, Panel A). Few students are affected at the bottom or top of the score distribution. Among switchers, lower-scoring benefited students see larger gains in access to higher-quality degrees, as measured by college enrollment rank (Figure 7, Panel A). Lower-scoring displaced students also face larger losses (Panel B). This score gradient is much weaker when we measure college quality at graduation.

Averaging across the score distribution gives the group-specific average effects among switchers,

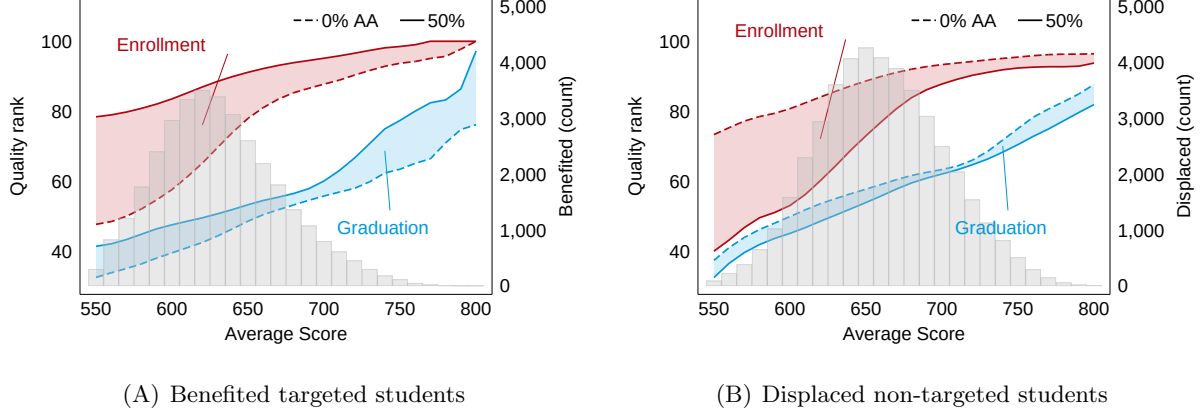


Figure 7: Access to and graduation from high-quality programs

Notes: Panel A shows benefited targeted students and Panel B displaced non-targeted students when moving from no AA to the 50% reservation schedule. Both panels plot outcomes against average ENEM score. Red lines show access to higher-quality programs, as measured by college enrollment rank. Blue lines show graduation from higher-quality programs, as measured by college graduation rank. Dashed lines show outcomes under no AA, and solid lines show outcomes under the 50% schedule. For each outcome, the shaded gap gives the average policy effect among switchers at that score. The gray histograms, measured on the right axis, show the average number of switchers per simulation in each score bin.

τ_t^S , defined in Equation (4). For enrollment rank, benefited students gain 18.8 points, while displaced students lose 14.4 points. This asymmetry reflects differences in fallback quality, with an average enrollment rank of 69.1 for benefited students without AA and 74.4 for displaced students under the 50% reservation schedule. Both gains and losses are smaller at graduation. Benefited students gain 6.5 graduation-rank points, while displaced students lose 3.3 points. This attenuation may partly reflect the higher dropout rates among targeted students and the greater propensity of non-targeted students to reapply documented in Section 3.4. Nevertheless, 34% of the enrollment-rank gain for benefited students persists seven years after application.

7.4 Aggregate effects and the college-quality gap

We next aggregate the gains and losses across all students. For each group, the total effect depends on how much the average switcher gains or loses, τ_t^S , and how many students switch, N_t^S . The product of these two terms gives the group-specific effect $\Delta_t(\mu', \mu)$ in Equation (6). Summing across groups gives the overall effect $\Delta(\mu', \mu)$ in Equation (7). We report the group-specific effects as changes in the average outcome within each group and the overall effect as the change in the average outcome for the full simulation sample.

Figure 8, Panel A reports changes in college quality at graduation, measured by average college graduation rank, as the reservation share rises from zero to 50%. At $\omega' = \omega(0.5)$, average graduation rank increases by 0.80 points among targeted students and decreases by 0.56 points among non-targeted students. Weighting these changes by the number of students in each group, total losses among non-targeted students offset 67% of total gains among targeted students. The resulting net improvement is modest: average graduation rank rises by 0.13 points on the 0–100 scale relative to no AA, averaging across the full simulation sample, including students whose assignments do not

change.³⁷

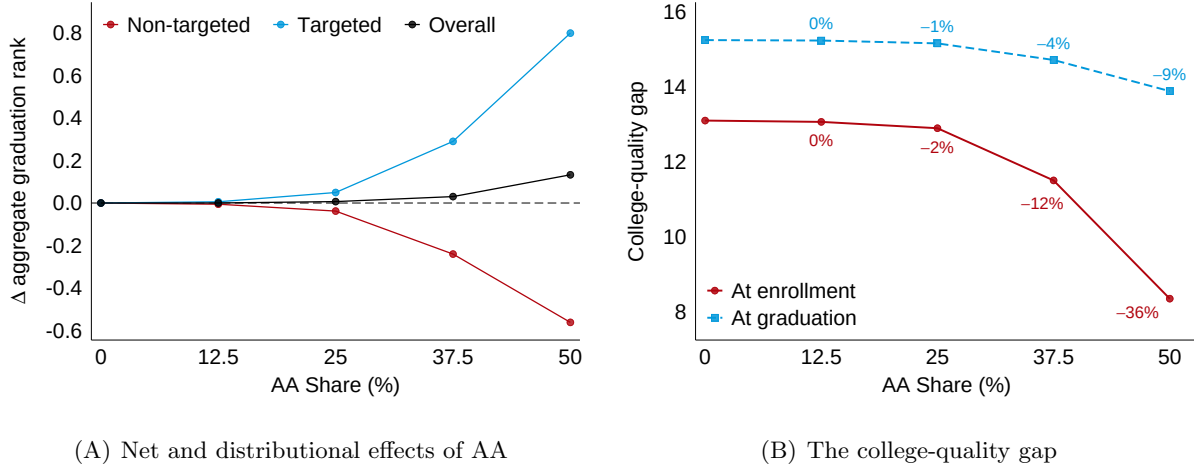


Figure 8: Aggregate and distributional effects of AA

Notes: Panel A plots the change in average graduation rank as the common reservation share x rises from zero to 50%, aggregating over all students in the simulation sample and separately over each group, including those whose assignments do not change, relative to the no-AA baseline. The “Overall” line averages over all students, giving each student the same weight. Panel B shows the implied college-quality gap between non-targeted and targeted students under each reallocation, in average enrollment rank (solid) and average graduation rank (dashed). Labels give the percent change in each gap relative to no AA.

Finally, we ask how much AA narrows the college-quality gaps documented in Section 3.1. Under no AA, the enrollment- and graduation-rank gaps are 13.1 and 15.2 points, respectively.³⁸ Figure 8, Panel B traces both gaps as the reservation share rises. Although the overall gain is small, AA narrows the college-quality gap by raising targeted students’ graduation ranks and lowering those of non-targeted students. At $\omega' = \omega(0.5)$, AA closes 36% of the enrollment-rank gap but only 9% of the graduation-rank gap. Much of the convergence achieved at enrollment therefore dissipates by graduation. This attenuation is consistent with the differences in dropout and reapplication discussed above. Overall, AA substantially narrows the college-quality gap at enrollment, produces a smaller reduction at graduation, and modestly raises average graduation rank across all students.

A one-for-one extrapolation of the discontinuity-based estimates overstates the net gains from AA. At the admissions margins, the 2SLS estimates τ_{AA}^M and τ_{NA}^M imply that benefited targeted students gain 9.8 graduation-rank points on average, while displaced non-targeted students lose 4.6 points. Across all switchers, the corresponding gain and loss, τ_{AA}^S and τ_{NA}^S , are 6.5 and 3.3 points. In both cases, the average benefited student gains about twice what the average displaced student loses. These individual effects, however, apply to unequal numbers of students. Knock-on effects cause the full policy to displace about 30% more students than it benefits. By assuming equal numbers of benefited and displaced students, a one-for-one extrapolation misses these additional

³⁷The simulation sample contains 277,449 targeted students and 266,365 non-targeted students. Using the unrounded estimates, these changes correspond to a total gain of 221,856 rank points among targeted students and a total loss of 149,641 rank points among non-targeted students. Thus, $149,641/221,856 \simeq 67\%$. The remaining net gain is 72,215 rank points, and $72,215/543,814 \simeq 0.13$ points per student.

³⁸Section 3.1 reports observed gaps among students finishing high school who took ENEM for the 2012 admissions cycle. Here, we report model-predicted gaps under no AA for the 2013 SISU estimation sample, restricted to federal applicants with average ENEM scores above 550.

losses and overstates the policy’s effect on average graduation rank.

7.5 Analysis limitations

These counterfactuals are informative, but their interpretation relies on three assumptions that are worth making explicit. First, we hold the applicant pool fixed across schedules, along with students’ preferences, entrance-exam scores, and AA status. Second, potential outcomes depend only on a student’s assignment and do not change directly with the policy. Third, the value of each student’s outside option remains fixed across counterfactuals. We discuss each in turn.

Behavioral responses. We rule out policy-induced changes in SISU participation, students’ degree rankings, entrance-exam preparation, and AA status. Holding degree rankings fixed as peer composition changes is standard in empirical school-choice models, which assume that utility depends on a student’s own assignment rather than on the full allocation, with rare exceptions (Agarwal and Somaini, 2020). AA may also change incentives to prepare for the entrance exam, although the direction of the response is theoretically ambiguous (Cotton et al., 2022; Akhtari et al., 2024; Tincani et al., 2026). Studies of two selective Brazilian public universities find little evidence that AA changed applicants’ exam preparation (Francis and Tannuri-Pianto, 2012; Estevan et al., 2019).³⁹ Finally, AA status is predetermined in our sample. Because eligibility required attending a public high school for all three years, the law was enacted too late for applicants in the 2013 cycle to strategically switch schools and qualify.⁴⁰

Stability of potential outcomes. We rule out policy-induced changes in graduation outcomes through peer composition, stigmatization, or grading standards. Because our data end in 2019, we cannot observe seven-year graduation outcomes for all cohorts in the policy’s rollout. In Appendix K, we instead study four-year dropout among the 2012–2016 entering cohorts. We compare dropout rates among students with similar entrance-exam scores in the same degree across cohorts facing different increases in reserved seats required by the law. Following Mello (2022, 2023), we instrument the reserved share with the required increase, controlling for degree and cohort effects. The law required larger increases in programs that initially reserved fewer seats and none in programs already reserving half. Even though our instrument predicts substantial increases in the share of reserved seats and the share of targeted entrants, we find no statistically significant change in dropout at a given score, in how dropout varies with scores, or in the dropout gap between targeted and non-targeted students with the same entrance-exam score. These findings are consistent with the assumption that a student’s potential outcome in a given degree does not change with the AA schedule, although the estimates are imprecise and concern dropout rather than graduation.

³⁹Estevan et al. (2019) interpret their null result as consistent with sufficiently convex effort costs, which imply a low elasticity of preparation effort to changes in admissions payoffs. They argue that this is plausible given the competitiveness of the entrance exam.

⁴⁰School switching can matter for later cohorts. Mello (2023) finds that the Law of Social Quotas increased moves from private to public high schools among ninth graders. These students moved to lower-SES and lower-performing schools in exchange for a higher probability of attending federal higher education.

Displacement beyond the federal system. Holding outside options fixed abstracts from additional displacement beyond the federal system. Displaced non-targeted students could enroll at private or other public institutions and displace students there, extending the policy’s effects beyond the market we simulate. Although we cannot observe these spillovers, capacity constraints appear limited in the private sector. Among private programs in the top quartile of quality, roughly 90% of seats are offered by programs operating below their reported capacity.⁴¹

8 Discussion

We study the distributional and aggregate effects of a large-scale AA reform in Brazil’s centralized college admissions system. The reform required federal institutions to reserve half the seats in every degree program for students from public high schools, with further reservations based on income and race. Our framework distinguishes the gains and losses for marginally benefited and displaced students from the overall effect of changing the reservation share.

Our findings are consistent with earlier evidence that students targeted by AA benefit from access to selective colleges. The average benefited student gains more than the average displaced student loses, but knock-on effects spread losses across more students and offset much of the total gain. Assessing other admissions policies that reallocate a fixed number of seats, such as geographic preferences, similarly requires accounting for gains and losses across all affected students.

Following students through graduation also matters for assessing AA’s distributional effects. The policy directly changes access to selective degrees, but its effect on the quality of degrees students complete depends on their subsequent educational trajectories. Evaluating AA at enrollment alone can therefore overstate how much it reduces inequality in college outcomes. Indeed, we find that much of the reduction in the college-quality gap at enrollment does not persist through graduation.

The relative gains and losses may differ in other settings because students face different outside options. Brazil’s large private sector provides alternatives to displaced non-targeted students and limits their losses. Displacement may be more costly where these alternatives are limited. At the same time, public scholarships through ProUni and subsidized student loans through FIES expand targeted students’ access to private institutions. Where financial aid is less generous, targeted students may have weaker outside options and gain more from federal enrollment. The distributional and aggregate effects of AA therefore depend on the broader higher-education system, including the availability of private alternatives and financial aid.

Our analysis follows students through graduation but does not measure how changes in educational trajectories affect earnings and employment. These effects may also depend on whether AA changes the signaling value of degrees to employers and whether graduates perceived to have benefited from reserved seats face stigma in the labor market. Understanding how these channels shape the longer-run distributional and aggregate effects of AA remains an important direction for future research.

⁴¹Institutions report both capacity and actual enrollment to the Higher Education Census, allowing us to identify programs operating at capacity. We use data from 2013.

References

- Abadie, A. (2002). Bootstrap tests for distributional treatment effects in instrumental variable models. *Journal of the American Statistical Association* 97(457), 284–292.
- Abdulkadiroğlu, A. (2005). College admissions with affirmative action. *International Journal of Game Theory* 33(4), 535–549.
- Abdulkadiroğlu, A., J. Angrist, and P. Pathak (2014). The elite illusion: Achievement effects at Boston and New York exam schools. *Econometrica* 82(1), 137–196.
- Abdulkadiroğlu, A., P. A. Pathak, J. Schellenberg, and C. R. Walters (2020). Do parents value school effectiveness? *American Economic Review* 110(5), 1502–1539.
- Agarwal, N., C. Hodgson, and P. Somaini (2025). Choices and outcomes in assignment mechanisms: The allocation of deceased donor kidneys. *Econometrica* 93(2), 395–438.
- Agarwal, N. and P. Somaini (2020). Revealed preference analysis of school choice models. *Annual Review of Economics* 12(1), 471–501.
- Akhtari, M., N. Bau, and J.-W. Laliberté (2024). Affirmative action and precollege human capital. *American Economic Journal: Applied Economics* 16(1), 1–32.
- Alon, S. (2015). *Race, Class, and Affirmative Action*. Russell Sage Foundation.
- Alon, S. and M. Tienda (2005). Assessing the “mismatch” hypothesis: Differences in graduation rates by institutional selectivity. *Sociology of Education* 78(4), 294–315.
- Arcidiacono, P. (2005). Affirmative action in higher education: How do admission and financial aid rules affect future earnings? *Econometrica* 73(5), 1477–1524.
- Arcidiacono, P., E. Aucejo, H. Fang, and K. Spenner (2011). Does affirmative action lead to mismatch? A new test and evidence. *Quantitative Economics* 2(3), 303–333.
- Arcidiacono, P., E. Aucejo, and J. Hotz (2016). University differences in the graduation of minorities in STEM fields: Evidence from California. *American Economic Review* 106(3), 525–562.
- Arcidiacono, P. and M. Lovenheim (2016). Affirmative action and the quality-fit trade-off. *Journal of Economic Literature* 54(1), 3–51.
- Arcidiacono, P., M. Lovenheim, and M. Zhu (2015). Affirmative action in undergraduate education. *Annual Review of Economics* 7(1), 487–518.
- Azevedo, E. and J. Leshno (2016). A supply and demand framework for two-sided matching markets. *Journal of Political Economy* 124(5), 1235–1268.
- Bagde, S., D. Epple, and L. Taylor (2016). Does affirmative action work? Caste, gender, college quality, and academic success in India. *American Economic Review* 106(6), 1495–1521.
- Barahona, N., C. Dobbin, and S. Otero (2025). The effects of widespread online education on market structure and enrollment. Working paper 34522, National Bureau of Economic Research.
- Barahona, N., C. Dobbin, and S. Otero (2026). Equilibrium price responses to targeted student financial aid. Working paper 33833, National Bureau of Economic Research.
- Bertrand, M., R. Hanna, and S. Mullainathan (2010). Affirmative action in education: Evidence from engineering college admissions in India. *Journal of Public Economics* 94, 16–29.
- Black, D. A. and J. A. Smith (2006). Estimating the returns to college quality with multiple proxies for quality. *Journal of Labor Economics* 24(3), 701–728.
- Black, S. E., J. T. Denning, and J. Rothstein (2023). Winners and losers? The effect of gaining and

- losing access to selective colleges on education and labor market outcomes. *American Economic Journal: Applied Economics* 15(1), 26–67.
- Bleemer, Z. (2022). Affirmative action, mismatch, and economic mobility after California’s Proposition 209. *The Quarterly Journal of Economics* 137(1), 115–160.
- Bleemer, Z. (2024). Top percent policies and the return to postsecondary selectivity. Working paper, UC Berkeley.
- Bó, I. and R. Hakimov (2020). Iterative versus standard deferred acceptance: Experimental evidence. *The Economic Journal* 130(626), 356–392.
- Bó, I. and R. Hakimov (2022). The iterative deferred acceptance mechanism. *Games and Economic Behavior* 135, 411–433.
- Brasil (2012). Lei nº 12.711, de 29 de agosto de 2012. Diário Oficial da União, 30 August 2012.
- Calonico, S., M. Cattaneo, M. Farrell, and R. Titiunik (2017). rdrobust: Software for regression-discontinuity designs. *The Stata Journal* 17(2), 372–404.
- Carlana, M., S. Chiuri, E. Miglino, and M. Tincani (2026). How far can inclusion go? The long-term impacts of preferential college admissions. Working paper 32525, National Bureau of Economic Research.
- Cattaneo, M. D., M. Jansson, and X. Ma (2020). Simple local polynomial density estimators. *Journal of the American Statistical Association* 115(531), 1449–1455.
- Chen, Y., C. Fu, H. Li, T. Li, and S. Tang (2026). College affirmative action and students’ outcomes: A matter of the dose. Working paper 35281, National Bureau of Economic Research.
- Chetty, R., D. J. Deming, and J. N. Friedman (2026). Diversifying society’s leaders? The determinants and causal effects of admission to highly selective private colleges. *The Quarterly Journal of Economics* 141(1), 51–145.
- Chetty, R., J. N. Friedman, E. Saez, N. Turner, and D. Yagan (2020a). Income segregation and intergenerational mobility across colleges in the United States. *The Quarterly Journal of Economics* 135(3), 1567–1633.
- Chetty, R., N. Hendren, M. R. Jones, and S. R. Porter (2020b). Race and economic opportunity in the United States: An intergenerational perspective. *The Quarterly Journal of Economics* 135(2), 711–783.
- Chetty, R., N. Hendren, P. Kline, and E. Saez (2014). Where is the land of opportunity? The geography of intergenerational mobility in the United States. *The Quarterly Journal of Economics* 129(4), 1553–1623.
- Concha-Arriagada, C. (2026). Strategic responses to college admission policies: Evidence from Chile. Working paper, Columbia University.
- Cotton, C., B. Hickman, and J. Price (2022). Affirmative action and human capital investment: Evidence from a randomized field experiment. *Journal of Labor Economics* 40(1), 157–185.
- Dahl, G. B. (2002). Mobility and the return to education: Testing a Roy model with multiple markets. *Econometrica* 70(6), 2367–2420.
- Dubin, J. and D. McFadden (1984). An econometric analysis of residential electric appliance holdings and consumption. *Econometrica* 52(2), 345–362.
- Dur, U., S. Kominers, P. Pathak, and T. Sönmez (2018). Reserve design: Unintended consequences and the demise of Boston’s walk zones. *Journal of Political Economy* 126(6), 2457–2479.

- Duryea, S., R. P. Ribas, B. Sampaio, G. R. Sampaio, and G. Trevisan (2023). Who benefits from tuition-free, top-quality universities? Evidence from Brazil. *Economics of Education Review* 95, 102423.
- Ellison, G. and P. Pathak (2021). The efficiency of race-neutral alternatives to race-based affirmative action: Evidence from Chicago’s exam schools. *American Economic Review* 111(3), 943–975.
- Estevan, F., T. Gall, and L.-P. Morin (2019). Redistribution without distortion: Evidence from an affirmative action programme at a large Brazilian university. *The Economic Journal* 129(619), 1182–1220.
- Fack, G., J. Grenet, and Y. He (2019). Beyond truth-telling: Preference estimation with centralized school choice and college admissions. *American Economic Review* 109(4), 1486–1529.
- Feinmann, J. and R. Hsu Rocha (2026). Social mobility and higher education: The role of elite public colleges. Working paper.
- Folha de S.Paulo (2022). Metade é a favor de cotas raciais em universidades. June 2022. <https://www1.folha.uol.com.br/educacao/2022/06/maioria-e-a-favor-de-cotas-raciais-em-universidades-34-sao-contra-diz-datafolha.shtml>.
- Francis, A. and M. Tannuri-Pianto (2012). Using Brazil’s racial continuum to examine the short-term effects of affirmative action in higher education. *The Journal of Human Resources* 47(3), 754–784.
- Francis-Tan, A. and M. Tannuri-Pianto (2018). Black movement: Using discontinuities in admissions to study the effects of college quality and affirmative action. *Journal of Development Economics* 135, 97–116.
- Francis-Tan, A. and M. Tannuri-Pianto (2024). Affirmative action in Brazil: Global lessons on racial justice and the fight to reduce social inequality. *Oxford Review of Economic Policy* 40(3), 642–655.
- Gale, D. and L. Shapley (1962). College admissions and the stability of marriage. *American Mathematical Monthly* 69(1), 9–15.
- Heckman, J. J., S. Urzúa, and E. Vytlacil (2008). Instrumental variables in models with multiple outcomes: The general unordered case. *Annales d’Économie et de Statistique* 91/92, 151–174.
- Hoekstra, M. (2009). The effect of attending the flagship state university on earnings: A discontinuity-based approach. *The Review of Economics and Statistics* 91(4), 717–724.
- Howell, J. (2010). Assessing the impact of eliminating affirmative action in higher education. *Journal of Labor Economics* 28(1), 113–166.
- IBGE (2011). *Censo Demográfico 2010: Características da População e dos Domicílios: Resultados do Universo*. Rio de Janeiro: Instituto Brasileiro de Geografia e Estatística.
- INEP (2011). Procedimento de cálculo das notas do Enem. Nota técnica, Instituto Nacional de Estudos e Pesquisas Educacionais Anísio Teixeira.
- INEP (2013). Censo Escolar da Educação Básica 2012: Resumo técnico. Technical report, Instituto Nacional de Estudos e Pesquisas Educacionais Anísio Teixeira, Brasília.
- Jacob, B. and J. Rothstein (2016). The measurement of student ability in modern assessment systems. *Journal of Economic Perspectives* 30(3), 85–108.
- Jenkins, L. and M. Moses (2017). Global: Affirmative action initiatives around the world. In *Understanding Global Higher Education. Global Perspectives on Higher Education*. SensePublishers, Rotterdam.

- Kapor, A. (2025). Transparency and percent plans. *Econometrica* 93(6), 2123–2157.
- Kapor, A., M. Karnani, and C. Neilson (2024). Aftermarket frictions and the cost of off-platform options in centralized assignment mechanisms. *Journal of Political Economy* 132(7), 2346–2395.
- Karnani, M. (2023). RCTs with interference in matching systems. Unpublished manuscript, MIT.
- Kirkeboen, L., E. Leuven, and M. Mogstad (2016). Field of study, earnings, and self-selection. *The Quarterly Journal of Economics* 131(3), 1057–1111.
- Klijn, F., J. Pais, and M. Vorsatz (2019). Static versus dynamic deferred acceptance in school choice: Theory and experiment. *Games and Economic Behavior* 113, 147–163.
- Larroucau, T. and I. Rios (2026). Dynamic college admissions. Working paper, ASU.
- Lee, L.-F. (1983). Generalized econometric models with selectivity. *Econometrica* 51(2), 507–512.
- Londoño-Vélez, J., C. Rodríguez, F. Sánchez, and L. E. Álvarez-Arango (2026). Financial aid and upward mobility: Evidence from Colombia’s Ser Pilo Paga. *Journal of Political Economy* 134(7), 2119–2165.
- Machado, C., G. Reyes, and E. Riehl (2025). The direct and spillover effects of large-scale affirmative action at an elite Brazilian university. *Journal of Labor Economics* 43(2), 391–431.
- MacLeod, W. B., E. Riehl, J. E. Saavedra, and M. Urquiola (2017). The big sort: College reputation and labor market outcomes. *American Economic Journal: Applied Economics* 9(3), 223–261.
- Mello, U. (2022). Centralized admissions, affirmative action, and access of low-income students to higher education. *American Economic Journal: Economic Policy* 14(3), 166–197.
- Mello, U. (2023). Affirmative action and the choice of schools. *Journal of Public Economics* 219, 104824.
- Melo, A. P. (2025). Affirmative action, college access and major choice: Redistribution with strategic behavior. *Economics of Education Review* 105, 102622.
- Mountjoy, J. (2022). Community colleges and upward mobility. *American Economic Review* 112(8), 2580–2630.
- Mountjoy, J. (2026). Marginal returns to public universities. *The Quarterly Journal of Economics* 141(1), 429–497.
- Munro, E. (2026). Causal inference under interference through designed markets. Working paper, Stanford University. arXiv:2504.07217.
- Rios, I., T. Larroucau, G. Parra, and R. Cominetti (2021). Improving the Chilean college admissions system. *Operations Research* 69(4), 1186–1205.
- Sander, R. (2004). A systemic analysis of affirmative action in American law schools. *Stanford Law Review* 57(2), 367–483.
- Singer, P. (2011). *Practical Ethics* (Third ed.). Cambridge University Press.
- Sönmez, T. and M. B. Yenmez (2022). Affirmative action in India via vertical, horizontal, and overlapping reservations. *Econometrica* 90(3), 1143–1176.
- Tincani, M. M., F. Kosse, and E. Miglino (2026). Beliefs and the incentive effects of preferential college admissions: Evidence from an experiment and a structural model. Discussion paper DP18564, Centre for Economic Policy Research.
- Tribunal de Contas da União (2022). Auditoria operacional na Lei de Cotas para ingresso no ensino superior nas instituições federais de ensino. Acórdão 2.376/2022-TCU-Plenário, Tribunal de Contas da União.

- Vieira, R. S. and M. Arends-Kuenning (2019). Affirmative action in Brazilian universities: Effects on the enrollment of targeted groups. *Economics of Education Review* 73, 101931.
- Walters, C. (2018). The demand for effective charter schools. *Journal of Political Economy* 126(6), 2179–2223.
- Zimmerman, S. (2014). The returns to college admission for academically marginal students. *Journal of Labor Economics* 32(4), 711–754.
- Zimmerman, S. (2019). Elite colleges and upward mobility to top jobs and top incomes. *American Economic Review* 109(1), 1–47.

Online Appendix for:

Affirmative Action in Centralized College Admissions Systems

Nano Barahona

Cauê Dobbin

Sebastián Otero

October 4, 2026

A Additional tables and figures

Table A.1: SISU and the affirmative-action phase-in, 2013–2016

	2013	2014	2015	2016
ENEM takers (millions)	3.6	4.3	5.1	4.8
SISU applicants (% of takers)	52.1	56.5	52.9	53.3
Targeted students (% of SISU applicants)	76.8	78.5	79.5	83.1
Federal SISU seats per entrant via selection (%)	54.8	73.4	80.6	83.4

Notes: The table reports, by year, the number of ENEM takers, the share of those takers who applied to SISU, the share of SISU applicants who are “targeted” (public high school graduates, the group eligible for AA), and the federal seats offered in SISU as a percentage of incoming in-person federal students admitted through a selection process (from the Higher Education Census).

Table A.2: Seats and admissions cutoffs by pool, 2013–2016

Admissions Pool	2013		2014		2015		2016	
	Spots	Gap	Spots	Gap	Spots	Gap	Spots	Gap
Open	75,726	–	90,915	–	103,170	–	101,701	–
Reserved, by law								
<i>Public HS</i>	6,479	13.0	11,516	17.5	14,940	19.1	18,380	21.5
<i>Public HS & low-income</i>	7,002	28.5	12,231	35.8	15,729	37.4	19,277	40.1
<i>Public HS & race</i>	10,558	32.0	18,680	35.3	23,990	38.5	27,727	42.3
<i>Public HS & race & low-income</i>	11,333	37.7	19,721	43.4	25,165	46.8	28,812	50.6
Other AA (institutional)	8,425	31.9	7,772	34.4	10,458	33.9	12,510	34.7
Total	119,523		160,835		193,452		208,407	

Notes: The sample is first-semester SISU applications to federal institutions, 2013–2016. For each year, the two columns report the number of seats (Spots) and the admissions-cutoff difference relative to open seats (Gap). The four reserved tracks are those mandated by the Law of Social Quotas, and “Other AA” collects institutional and state quotas offered outside the federal law. The cutoff difference is computed within each degree as the open cutoff minus the pool cutoff and averaged across degrees, weighting by the number of seats.

Table A.3: Covariate balance

	NA				AA			
	No-offer mean	Offer gap	s.e.	N	No-offer mean	Offer gap	s.e.	N
<i>Traits targeted by the policy</i>								
Public high school	0.000	—	—	150,950	0.984	-0.002	0.001	198,463
Black, Brown, Indig.	0.351	0.001	0.005	150,950	0.695	0.001	0.003	198,463
Household income	4,932.7	-44.95	50.003	150,950	1,411.7	-7.359	13.394	198,463
<i>SISU applications</i>								
# options ranked	1.980	0.001	0.002	150,667	1.976	0.001	0.001	197,972
<i>ENEM scores</i>								
Natural science	607.1	-0.332	0.435	150,950	525.8	-0.370	0.397	198,463
Social science	646.9	-0.617	0.431	150,950	585.4	-0.529	0.380	198,463
Language	594.2	-0.219	0.410	150,950	543.5	-0.663	0.358	198,463
Math	701.8	-0.541	0.661	150,950	587.5	-0.314	0.605	198,463
Essay	714.5	1.314	1.051	150,950	620.1	1.399	0.894	198,463
<i>Other student characteristics</i>								
Female	0.500	0.001	0.005	150,950	0.503	0.004	0.004	198,463
Age	20.20	-0.034	0.042	150,950	22.80	0.002	0.055	198,463
HS graduate in $t - 1$	0.393	0.009	0.006	150,667	0.281	-0.003	0.004	197,972
GED-equivalent	0.015	0.000	0.001	150,950	0.046	0.000	0.002	198,463
Married	0.033	0.001	0.002	150,950	0.119	0.002	0.003	198,463
Municipal income	2,326.1	3.806	6.538	150,950	1,918.9	0.059	5.239	198,463
<i>College enrollment in $t - 1$</i>								
Private	0.075	-0.006	0.003	150,667	0.120	-0.008	0.003	197,972
Federal	0.122	-0.007	0.004	150,950	0.087	0.000	0.003	198,463
Any college	0.210	-0.007	0.004	150,950	0.222	-0.008	0.004	198,463

Notes: This table reports covariate balance tests for SISU applications to federal universities in the first semester of 2013. We use the sample restrictions and graduation-rank bandwidths of Table 2, pooling the initial-offer and waitlist stages. Each covariate is regressed on crossing the admissions cutoff, admissions-list-stage fixed effects, and local-linear functions of the running variable that vary by degree, stage, and side of the cutoff, with triangular weights and no additional student covariates. Standard errors are clustered by student. The number of ranked options, HS graduate in $t - 1$, and prior private enrollment have slightly fewer observations because of missing values. Public-high-school attendance is constant at zero among non-targeted open-seat applicants, so its gap is not estimated. HS graduate in $t - 1$ indicates expected completion in 2012 reported at ENEM registration. Rows marked $t - 1$ refer to 2012.

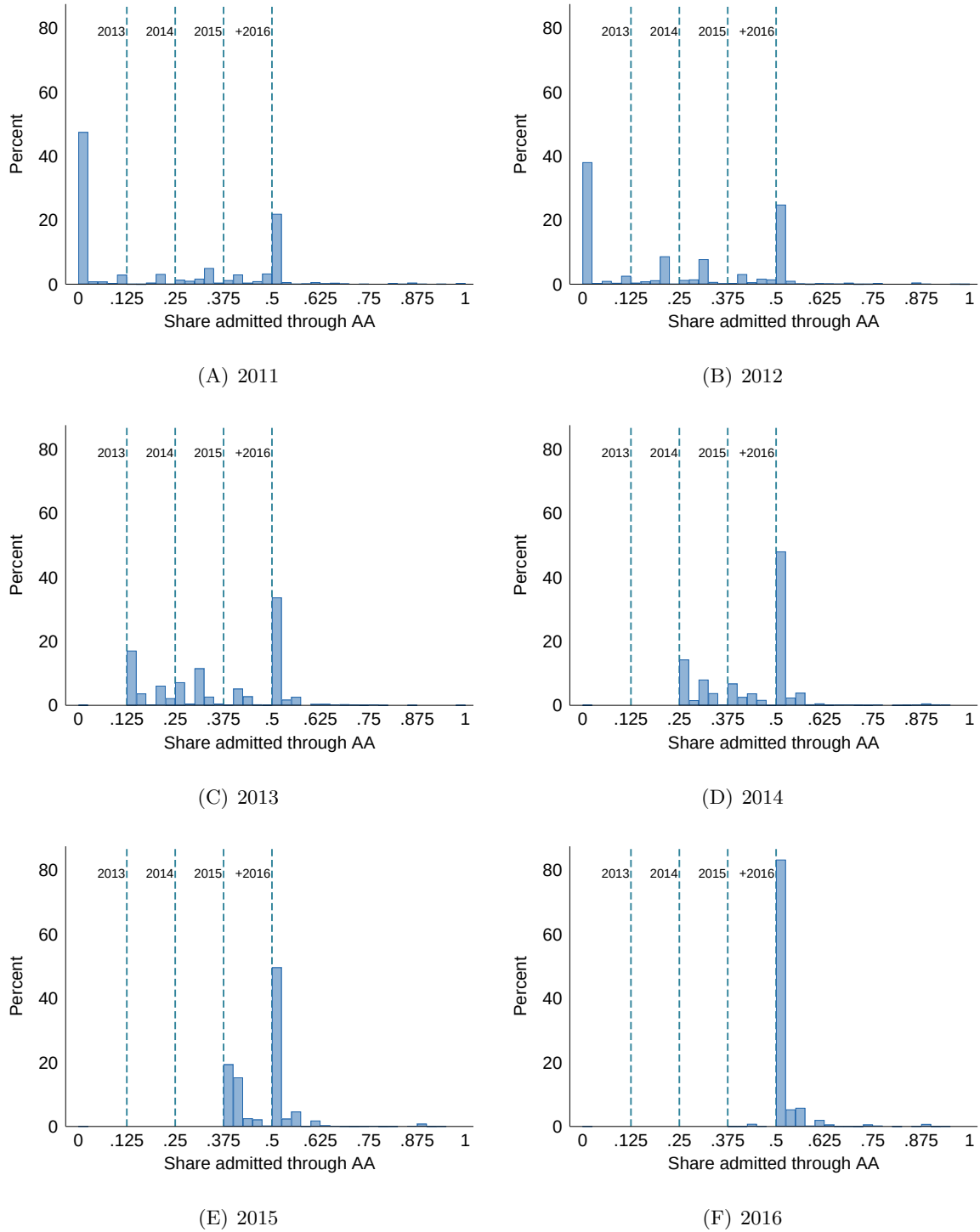


Figure A.1: Staggered implementation of the AA policy

Notes: This figure shows how the AA regulation was phased in across federal degree programs. Each panel corresponds to one year and plots the distribution of the share of seats a degree program reserved through an AA track, weighting programs by the number of seats they offered. The dashed lines mark the minimum reserved shares the law mandated as it phased in: 12.5% in 2013, 25% in 2014, 37.5% in 2015, and 50% from 2016 onward. The sample is federal institutions participating in the SISU admissions process. Before the law, many programs reserved no seats, while others already reserved half. From 2013 onward, reserved shares bunched at the minimum required each year, and by 2016 nearly all programs reserved about 50%.

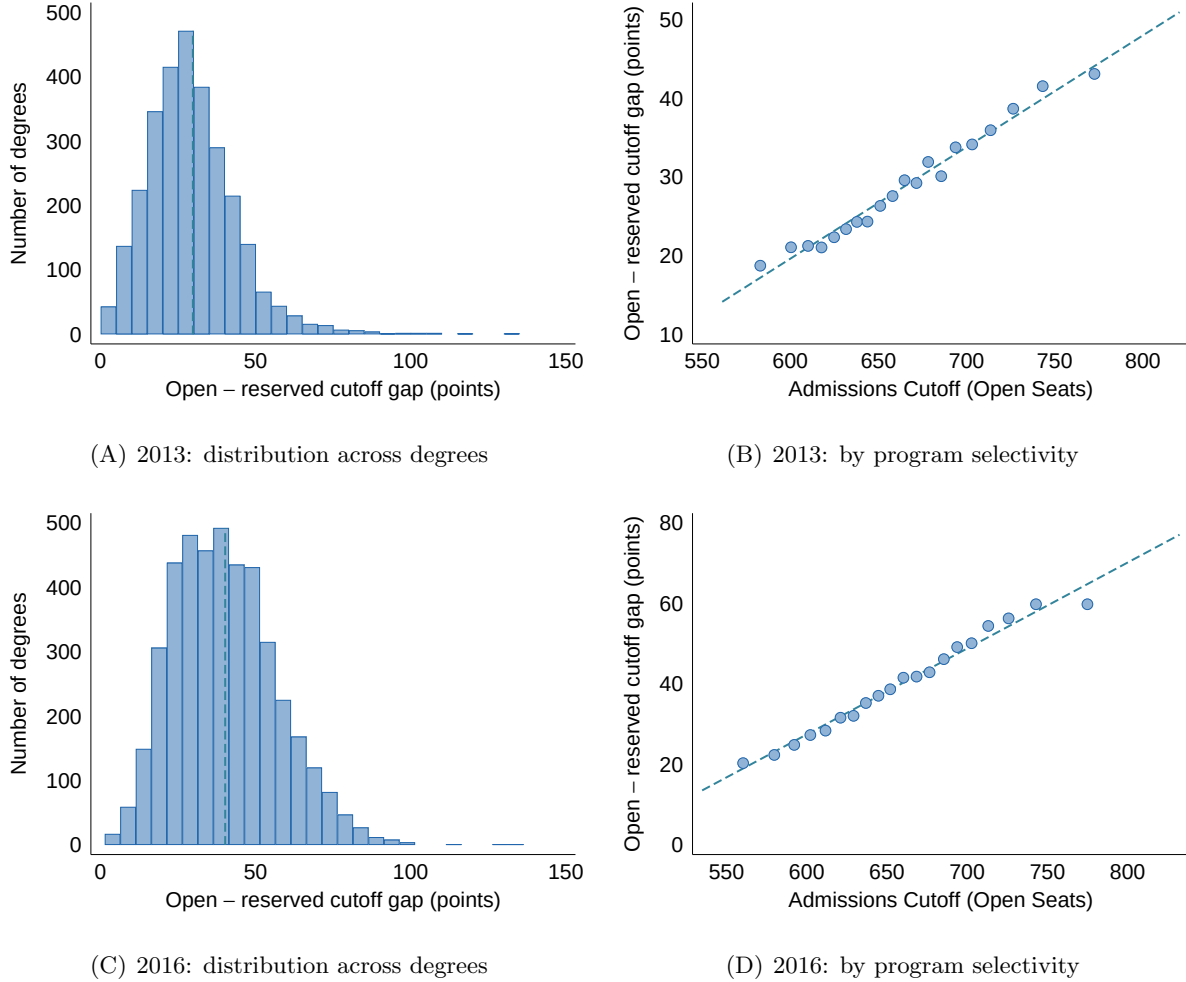


Figure A.2: Admissions-cutoff differences between open and reserved seats

Notes: Each row covers the federal degree programs of one SISU edition, 2013 (2,896 degrees) and 2016 (4,284 degrees). For each degree, the reserved-seat cutoff is the seat-weighted average cutoff across the four reserved tracks established by law, and the difference is the open-seat cutoff minus this reserved cutoff. Weighting degrees by their reserved seats, in 2013 the average open-seat cutoff is 663.6 points, the average reserved cutoff 633.9 points, and the average difference 29.8 points; in 2016 the three averages are 660.3, 620.1 and 40.2 points. Panels A and C show the distribution of the difference across degrees, with the average marked by the dashed line. Panels B and D are binned scatters (20 equal-sized bins) of the difference against the open-seat cutoff, our measure of program selectivity, with a fitted line; the difference rises with selectivity in both years (correlation 0.47 in 2013 and 0.71 in 2016).

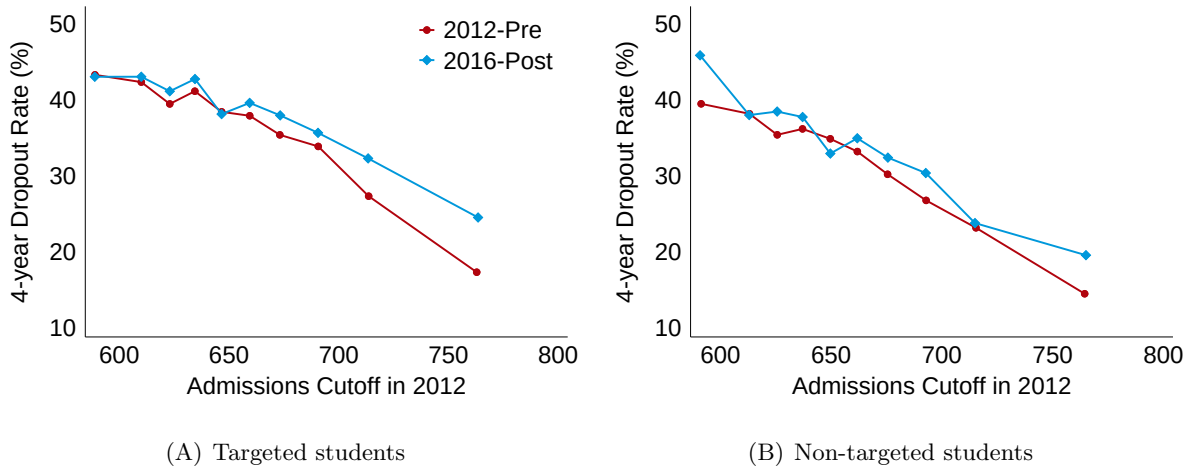


Figure A.3: Dropout and degree selectivity, by targeted status

Notes: Each panel plots the four-year dropout rate against the degree's admissions cutoff in the pre-policy 2012 SISU (an average of the cutoffs across all admissions tracks weighted by seats), binned into 10 equal-sized groups of degrees, for the 2012 (pre-policy, red) and 2016 (post-policy, blue) incoming cohorts. The sample is that of Figure 3, restricted to entrants whose targeted status is observed in the ENEM questionnaire. A student is counted as dropping out if, four years after admission, they have left college or are enrolled in a degree ranked lower on the quality scale of Section 3.1, tracked across the whole higher-education system. Targeted students are those who attended only public high school.

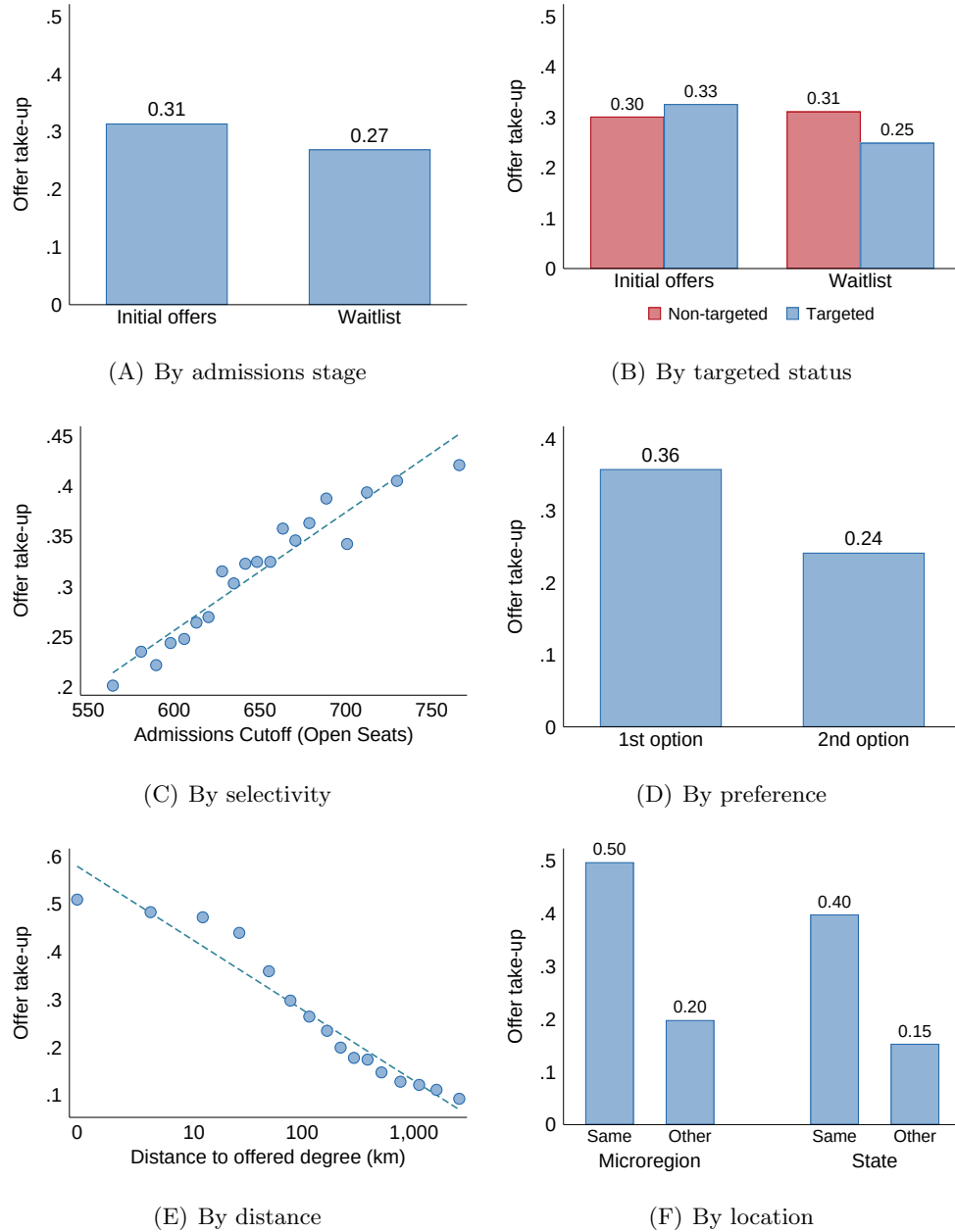


Figure A.4: Offer take-up by admissions stage, student status, and program characteristics

Notes: Take-up is the share of offered students who enroll in the offered degree. The sample covers federal degrees in the first semester of 2013, including open seats and the four reserved-seat pools. Targeted students are defined as in Table 1. In every panel, the outcome of interest is the take-up of offers. Panel A compares stages, and Panel B compares targeted and non-targeted students. Panels C–F use initial offers: Panel C measures selectivity by the open-seat cutoff in the initial-offer stage, Panel D compares first and second choices, Panel E uses the distance between municipality centroids (in kilometers, log scale), and Panel F compares offers within and outside the student’s microregion and state. In Panels C and E, dots are means within 20 equal-sized bins, and dashed lines are linear fits.

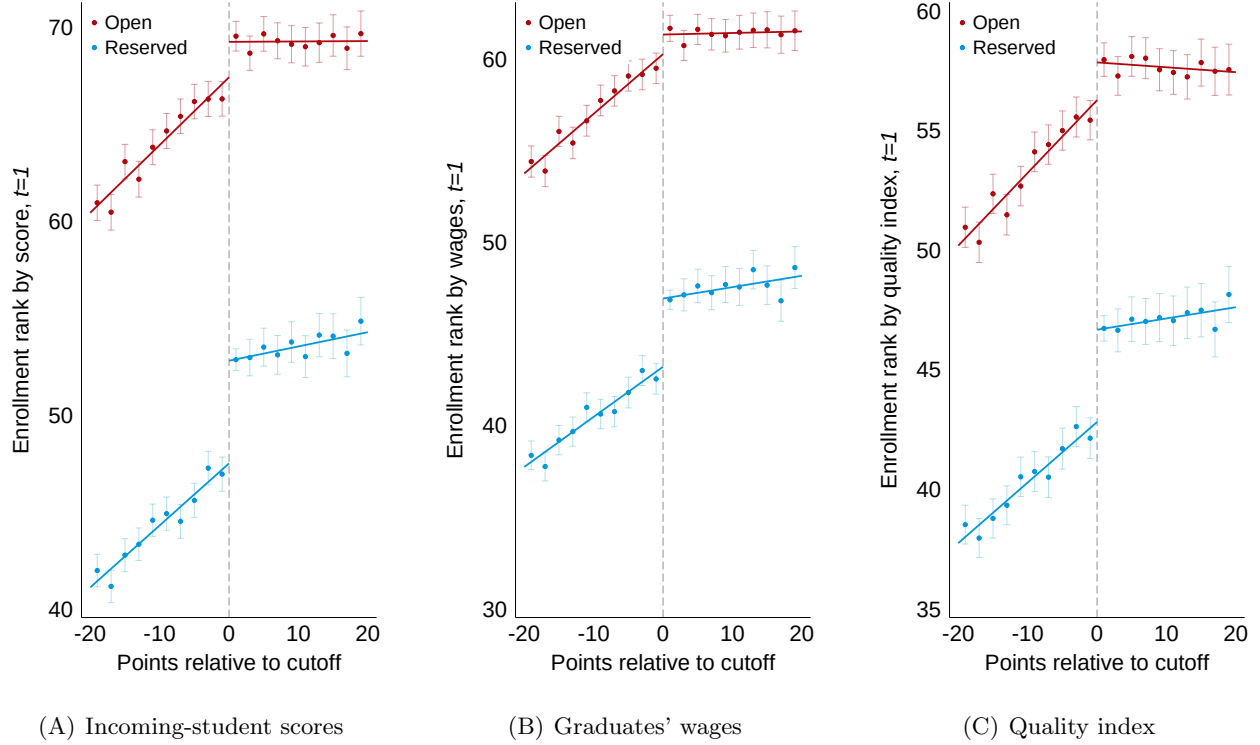


Figure A.5: College enrollment rank under alternative quality measures

Notes: We pool applications across admissions lists and the initial-offer and waitlist stages, combining the four reserved-seat categories. Application scores are centered at the relevant cutoff. Red and blue show non-targeted applicants to open seats and targeted applicants to reserved seats, respectively. Panels A, B, and C rank programs by incoming-student scores, graduates' wages, and the Ministry of Education's quality index, respectively. Enrollment rank is the attended program's national percentile (1–100), with zero for students who do not enroll. Dots show means in two-point bins after removing admissions-list-stage fixed effects. We use the same sample as Figure 4, including applications within 20 points of the cutoff. Lines are linear fits on each side, weighted by the number of applications per bin. Error bars show approximate 95% confidence intervals for the bin means, using unclustered standard errors. The sample covers federal-university applications in the first semester of 2013. Appendix C defines the quality measures.

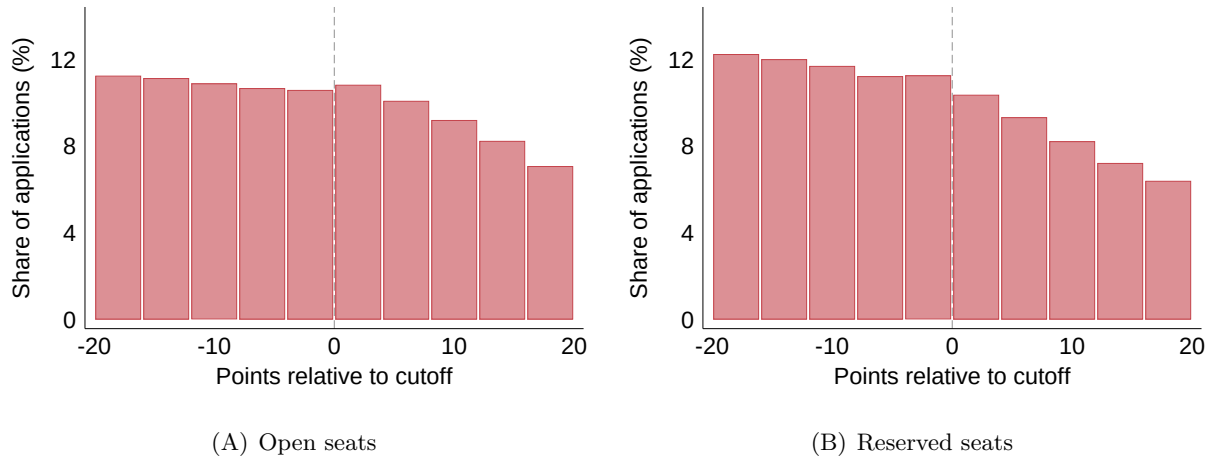


Figure A.6: Application score density

Notes: This figure shows the distribution of application scores around the admissions cutoff for non-targeted students applying through open seats (Panel A) and targeted students applying through reserved seats (Panel B). We pool admissions lists across the initial-offer and waitlist stages and include observations within 20 points on either side of the cutoff, in four-point bins. We exclude observations exactly at the cutoff, since the last admitted student's score defines it, and keep lists with at least five offers in the initial-offer stage. The robust p -values from the local-polynomial manipulation test (Cattaneo et al., 2020) are 0.20 and 0.68 for open and reserved seats, respectively.

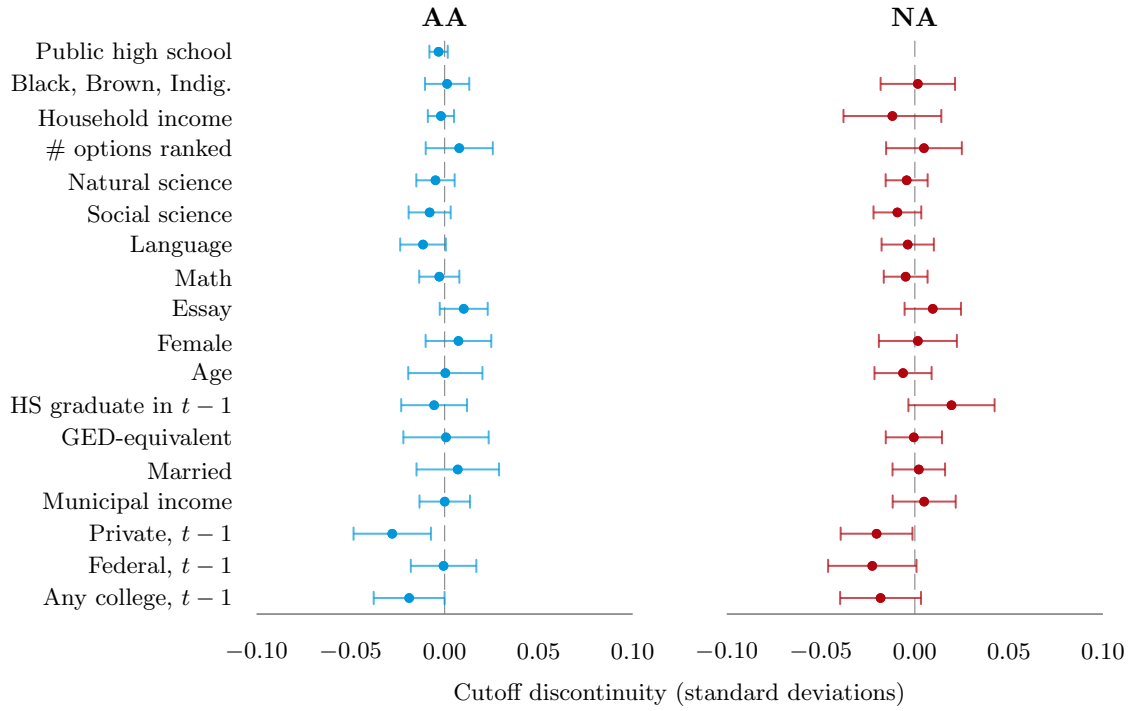


Figure A.7: Regression discontinuity estimates, predetermined characteristics

Notes: This figure reports the discontinuities in predetermined characteristics for AA applicants (left, blue) and NA applicants (right, red). We pool the initial-offer and waitlist stages and use the specification and graduation-rank bandwidths in Table 2, without additional student covariates. Dots show estimates divided by each characteristic's standard deviation in the pooled AA and NA student sample within 30 points of the cutoffs. Bars show 95% confidence intervals, with standard errors clustered by student. The characteristics and labels follow Table 1, excluding application score. Public-high-school attendance is constant among NA applicants, so no coefficient is reported for that group. Period $t - 1$ refers to 2012.

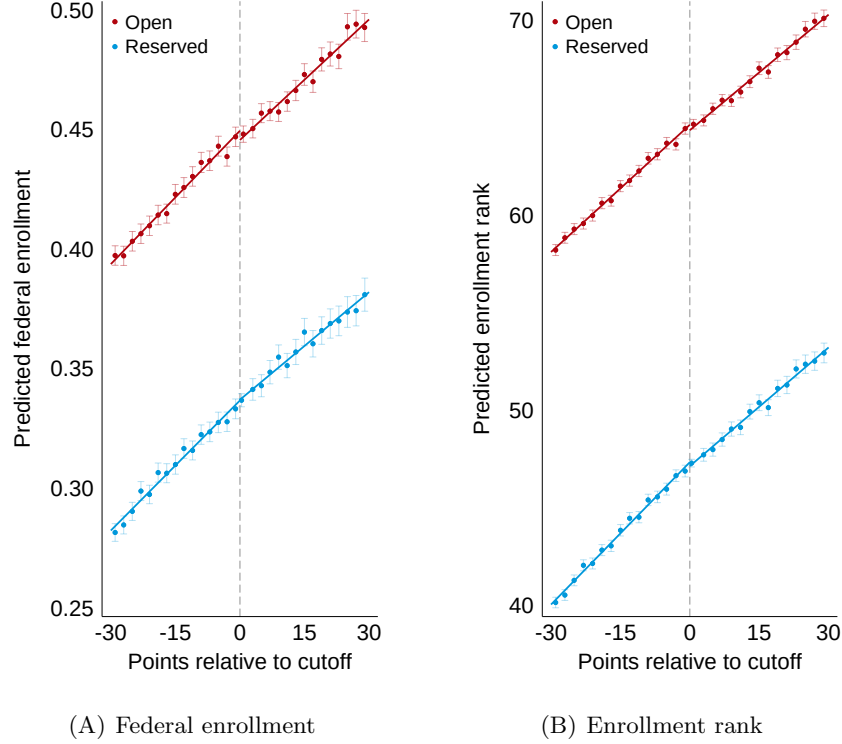


Figure A.8: Regression discontinuity plots, predicted outcomes

Notes: This figure plots predicted federal enrollment (Panel A) and predicted college enrollment rank (Panel B) around the admissions cutoffs. We obtain these predictions from a logit regression for federal enrollment and a linear regression for enrollment rank, using the student characteristics reported in Table 1, except application score, and targeted status. Both models are fitted in sample, and enrollment rank is zero for students who do not enroll. We pool applications across admissions lists and the initial-offer and waitlist stages, combining the four reserved-seat categories. Red and blue show NA applicants to open seats and AA applicants to reserved seats, respectively. Scores are centered at the relevant cutoff. Each panel shows binned means adjusted for admissions-list-stage fixed effects, using applications within 30 points of the cutoff, with linear fits on each side. Error bars show approximate 95% confidence intervals for the bin means, using unclustered standard errors. The sample covers federal-university applications in the first semester of 2013.

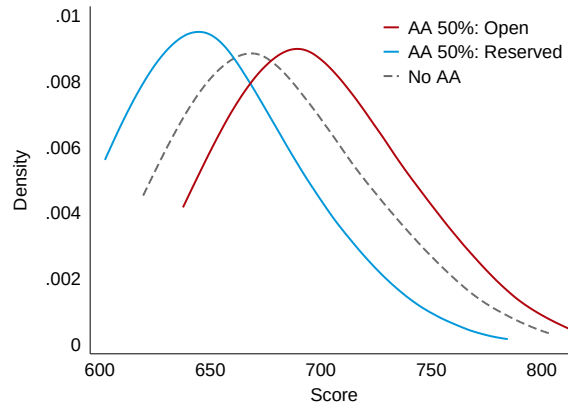


Figure A.9: Distribution of simulated cutoffs under no AA and 50% AA

Notes: This figure shows kernel densities of degree-level admissions cutoffs after take-up and the waitlist, averaged across simulations. The dashed gray line shows the distribution under no AA ($\omega = \omega(0)$), where all students face a single cutoff. Under 50% AA, the cutoff splits into an open-seat cutoff (red) and a reserved-seat cutoff (blue).

B Dropout and test scores

This appendix defines our measure of dropout among students entering federal universities. Section B.1 documents dropout rates and how they evolve after admission. Section B.2 then examines how ENEM scores predict dropout.

B.1 Dropout rates

We define dropout as leaving college altogether or moving to a lower-ranked degree on the quality scale of Section 3.1. Because students often switch degrees and retake the ENEM, we follow them across the entire higher-education system and do not count moves to degrees of equal or higher rank as dropout.¹ Among students who entered a federal university in 2012, before the AA policy was implemented, 43.8% left their initial degree within four years, but about a quarter of those students moved to a degree ranked as high or higher (Table B.1). In total, 32.4% of entrants dropped out, with 19.8% leaving college altogether and 12.6% moving to a lower-ranked degree.

Table B.1: Academic trajectories four years after admission

	2012 cohort	2016 cohort
Enrolled in or graduated from the initial degree	56.2	56.0
Left the initial degree	43.8	44.0
Moved to a degree ranked as high or higher	11.4	9.3
Moved to a lower-ranked degree	12.6	10.9
Left college	19.8	23.8
Dropout (lower-ranked degree or left college)	32.4	34.7
Entrants	108,538	113,124

Notes: This table reports where students who entered a federal university in 2012 and 2016 are four years after admission, as a percentage of entrants. A student is counted in the first row if they are enrolled in or have graduated from the degree they were admitted to. The remaining students are classified by the rank of the best degree they are enrolled in or have graduated from on the quality scale of Section 3.1, relative to the degree they entered, tracked across the whole higher-education system. Dropout, the measure used in Section 3.4, is the sum of the two rows above it. The sample is that of Figure 3.

Figure B.1 plots dropout by years since admission. It rises steeply in the first three years and slowly thereafter, reaching 42% after eight years. Between the 2012 and 2016 cohorts, before and after the policy, dropout at four years rose slightly, from 32% to 35%.

B.2 The role of test scores

Next, we examine how well ENEM scores predict dropout. We use the same measure of dropout as in Section B.1, four years after admission. For the 2012 cohort, we estimate

$$\text{Dropout}_{ij} = \alpha_j + \beta \text{Score}_i + \varepsilon_{ij}, \quad (\text{B.1})$$

¹Larroucau and Rios (2026) also document high dropout and reapplication rates in Chile’s centralized admissions system.

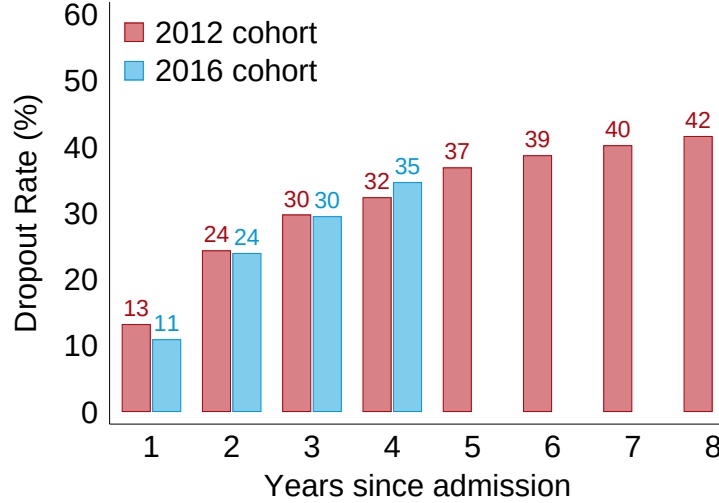


Figure B.1: Dropout by years since admission

Notes: This figure plots dropout rates for students who entered federal universities in 2012 (red) and 2016 (blue). We track enrollment and graduation across the entire higher-education system, using the degree ranks defined in Section 3.1. At each year, students count as dropouts if they are not enrolled in a degree of equal or higher rank than the degree they entered and have not completed one. The admission year counts as year 1. Both cohorts are observed through 2019, giving eight years of observations for the 2012 cohort and four for the 2016 cohort.

where $\text{Dropout}_{i,j}$ indicates that student i , admitted to degree j , dropped out within four years, Score_i is the student's average ENEM score in tens of points, and α_j is a degree fixed effect. The fixed effects hold the degree constant, so β compares students admitted to the same degree who entered with different scores. The sample is the 2012 entrants of Section B.1 matched to their ENEM scores, in degrees with at least 30 such students, 105,256 students in 1,826 degrees. A second specification adds a targeted indicator and its interaction with the score, for the 97,476 students whose targeted status is observed in the ENEM questionnaire. Standard errors are clustered by degree.

Table B.2 reports the estimates. Within a degree, a 10-point higher ENEM score lowers the probability of dropping out by 0.36 percentage points (column 1), against a mean dropout rate of 32%. The open and reserved cutoffs differ by 30 to 40 points within a degree (Section 3.3), so a student admitted at the reserved cutoff is predicted to drop out roughly 1.1 to 1.4 percentage points more often than one admitted at the open cutoff. Column 2 adds a targeted indicator and its interaction with the score. The gradient is the same for targeted and non-targeted students, 0.32 and 0.32 percentage points per 10 points, and targeted students drop out 2.9 percentage points more often than non-targeted students with the same score in the same degree. Combining both, a targeted student admitted at the reserved cutoff is predicted to drop out roughly 4 percentage points more often than a non-targeted student admitted at the open cutoff.

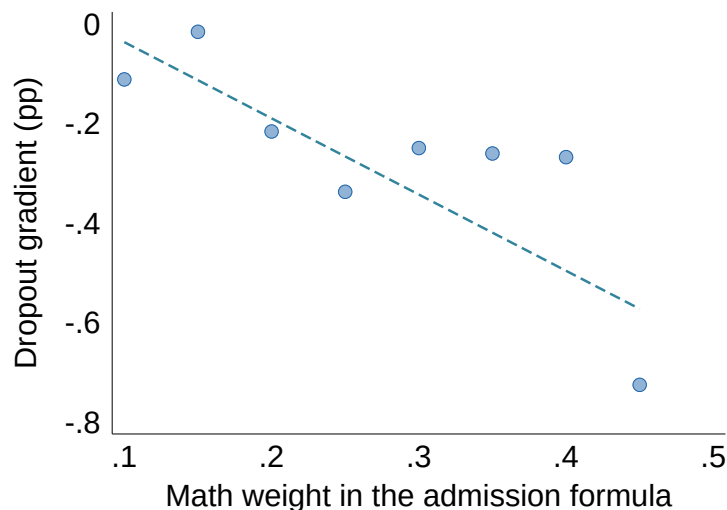
The relationship between scores and dropout may reflect academic preparation, but also financial constraints or other student characteristics. We examine the role of preparation by comparing degrees that place different weights on mathematics in their SISU admissions formula. Degrees that give mathematics more weight are likely to require more of it in their coursework, so scores should predict

Table B.2: Test scores and dropout, within degree

	(1)	(2)
ENEM score	-0.0036*** (0.0004)	-0.0032*** (0.0005)
ENEM score $\times$ Targeted		-0.0000 (0.0005)
Targeted		0.0285*** (0.0036)
Observations	105,256	97,476
Within- R^2	0.001	0.002
Mean dropout rate	0.316	0.310

Notes: This table reports within-degree regressions of dropout on the student's average ENEM score. The score enters in units of 10 points, so each coefficient is the change in dropout per 10 points of the score, and it is centered at the sample mean of 615 points. The dependent variable is an indicator for dropping out within four years of admission, defined as leaving college or moving to a lower-ranked degree (Section B.1). Column 2 adds a targeted indicator and its interaction with the score. Both specifications include degree fixed effects. The sample is the 2012 entrants of Figure B.1 matched to their ENEM scores, in degrees with at least 30 such students. Standard errors clustered by degree are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

dropout more strongly in these programs. We estimate Equation (B.1) separately for 1,803 degrees and shrink the slopes using empirical Bayes. Figure B.2 shows that a 10-point higher average ENEM score is associated with 0.05 percentage points lower dropout in the lowest quintile of math weight, compared with 0.34 in the highest.

**Figure B.2:** Per-degree score gradient of dropout, by math weight

Notes: This figure plots the empirical-Bayes-shrunk per-degree slope of dropout on the student's average ENEM score, in percentage points per 10 points, for the 2012 cohort of Table B.2 in degrees with at least 30 students and a within-degree standard deviation of the score of at least 15 points (1,803 degrees). The slopes are plotted against the share of the total weight in the degree's SISU admission formula assigned to mathematics, rounded to 0.05, with one point per value. The weights are the declared SISU weights averaged over the 2013, 2014, and 2016 editions. Points are precision-weighted means and lines are precision-weighted linear fits.

C Degree program quality measures

We rank degree programs by quality as measured by the average test scores of their incoming students. We validate this measure by comparing it with graduates’ wages, the Ministry of Education’s quality index, and tuition fees for private programs.

C.1 Construction

Scores. We rank each program by the average entrance-exam score of its incoming students. We average the four multiple-choice ENEM areas (natural science, social science, language, and math) over the 2012–2015 entering cohorts, and exclude the essay because of the prevalence of zero scores. We keep each student’s first attempt. We then assign each program its national percentile, from 1 to 100, across all 52,869 degree programs in the country, public and private, in person and online. We compute the average directly for 62% of programs, which account for 84% of enrolled students nationally and 94% of enrolled students at in-person federal programs.

Wages. Our second measure is the average wage of a program’s graduates. We use data from [Feinmann and Hsu Rocha \(2026\)](#), who link graduates in the Higher Education Census to the matched employer–employee records (RAIS). For each program we take the 2010 and 2011 graduating cohorts and compute the mean December wage in their main job in 2020 and 2021, weighting by the number of graduates. We rank programs by the log of this mean. Programs with five or fewer graduates are treated as missing. The wage is observed for 44% of programs, which account for 54% of enrolled students nationally and 55% of enrolled students at in-person federal programs.²

Quality index. Our third measure is the CPC (*Conceito Preliminar de Curso*), the index the Ministry of Education uses to assess program quality. It combines students’ value-added on the national college exit exam with faculty credentials and infrastructure. We use each program’s most recent evaluation between 2012 and 2019, which falls in 2017–2019 for 94% of evaluated programs, and rank programs by the continuous score, which ranges from 0 to 5. The index is available for 49% of programs, which account for 77% of enrolled students nationally and 73% of enrolled students at in-person federal programs.

Tuition fees. For private programs we also measure annual tuition. We combine four sources of monthly list prices at the program level for 2010–2019: the tuition that programs report to the ProUni scholarship program, the tuition in FIES loan contracts, the Hoper survey of private institutions, and Quero Bolsa listings, converted to list prices. We harmonize prices across sources as in [Barahona et al. \(2026\)](#) and calculate each program’s average annual tuition in 2016 reais. Public programs charge no tuition. Tuition is observed for 87% of private programs, which account for 96% of students enrolled in private programs.

Imputation. For programs missing a measure, we predict it from a regression on the university, the detailed field of study, the program’s start year (with second- and third-order polynomials),

²A program needs graduates in 2010 or 2011, so it must have existed since about 2007. Among programs created after 2007, 91% have an imputed wage, and among online programs 89%. Among in-person programs that existed by 2007, 14% are imputed, and 7% when weighted by enrollment.

enrollment (with polynomials), the number of hours required to complete the program, shift dummies (morning, afternoon, night, and full-time), graduating-class size, and an indicator for online delivery. We then rank programs using both observed and imputed values.

C.2 Validation

Figure C.1 plots each of the other measures against the score ranking, with programs weighted by enrollment. The score ranking uses entrance-exam scores, wages come from employer records, the index uses exit exams and program inputs, and tuition comes from price listings. The score ranking has a rank correlation of 0.61 with graduates' wages, 0.31 with the quality index, and, among private programs, 0.46 with tuition. The correlation with the quality index is lower in part by construction, since its value-added component nets out students' entering scores.

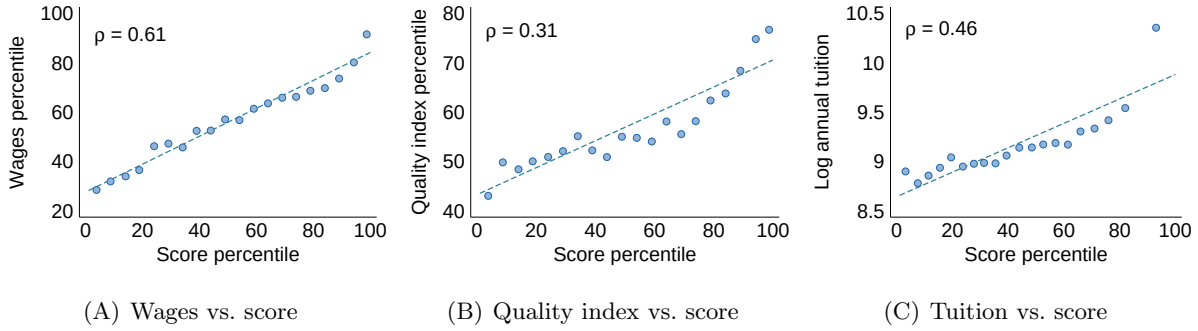


Figure C.1: Validation of the score-based quality ranking

Notes: Each panel plots one measure against the score percentile. Wages and the quality index are national percentiles, and tuition is log annual tuition in 2016 reais. The tuition panel uses private degree programs; the others use all programs. Programs are grouped into 20 bins of the score percentile. Each marker is the enrollment-weighted mean within a bin, and the dashed line is the enrollment-weighted linear fit. The number in each panel is the enrollment-weighted rank correlation.

D Two-stage assignment mechanism

This appendix describes the two-stage assignment mechanism φ used to simulate the counterfactual allocations in Section 7. We use the notation from Section 4. In the first stage, a student-proposing deferred acceptance algorithm generates initial offers, which students accept or decline. In the second stage, a college-proposing waitlist reallocates unfilled seats.

D.1 Setup and notation

Each degree $j \in \mathcal{J}$ has $q_j^R = \omega_j$ reserved seats and $q_j^O = q_j - q_j^R$ open seats. Non-targeted students are eligible only for open seats, while targeted students are eligible for both. Applicants are ranked by their application scores s_{ij} . The *precedence rule* $p \in \{0, 1\}$ determines whether open or reserved seats are filled first (Sönmez and Yenmez, 2022). Under $p = 0$, open seats go first to the highest-scoring applicants, and reserved seats then go to the highest-scoring targeted applicants not already admitted. Under $p = 1$, reserved seats go first to the highest-scoring targeted applicants, and open seats then go to the highest-scoring remaining applicants. Our empirical implementation uses $p = 0$.

D.2 Initial offers: Student-proposing deferred acceptance

Offers. The initial-offer stage implements the deferred acceptance algorithm with reservations of Rios et al. (2021). Each degree j is split into two virtual programs with capacities q_j^O and q_j^R . Targeted students' reports are expanded so that each degree appears as the ordered pair (j, O) , (j, R) under precedence $p = 0$, or in the reverse order under $p = 1$, while non-targeted students see only the open programs. A single student-proposing deferred acceptance algorithm then runs on these expanded reports and virtual programs, producing an initial offer $D_i^1 \in \mathcal{J} \cup \{0\}$ for each student.

Take-up. Each student with $D_i^1 = j > 0$ then evaluates the offer with the initial-offer take-up rule T_{ij}^1 : if $T_{ij}^1 = 1$, the seat is accepted and held, and if $T_{ij}^1 = 0$, the offer is rejected and j is recorded as declined.

D.3 Aftermarket: College-proposing waitlist

Waitlists. A student i is on the waitlist of pool $P \in \{O, R\}$ at degree j if i is eligible for that pool, j is ranked strictly above i 's initial offer (if any) and strictly above any seat i currently holds, and i has not previously declined j . Students who received an initial offer can be waitlisted only for degrees they strictly prefer to the one offered, even if they declined that offer. Those who received no initial offer and hold no seat are eligible for any degree they have not declined.

Offers. The aftermarket reallocates unfilled seats through offers from degree programs. Open pools make offers first, followed by reserved pools. Each pool (j, P) offers seats to students on its waitlist from highest to lowest application score s_{ij} . Offers continue until the pool is full or its list

is exhausted. When a student switches degrees, their previous seat returns to its original pool. This may free a seat in a pool that has already finished making offers. After all pools have made their offers, we repeat the process if any student has freed a seat by switching degrees. Each pool considers students it has not previously considered, in the same score order. We repeat the process until no further seats are freed.

Take-up. The student evaluates the offer with the waitlist take-up rule T_{ij}^2 : if $T_{ij}^2 = 1$, the student moves to j and any previously held seat is returned to its pool; if $T_{ij}^2 = 0$, j is recorded as declined and the student's current holding is unchanged.

Termination. The aftermarket ends after finitely many offers because each pool has a finite list and considers each student only once.

D.4 Output

Let A_i denote the degree student i holds at the end of the mechanism, with $A_i = 0$ if the student is unassigned; this is the realized enrollment $A_i(\mu)$ of Section 4. We also record the initial offer D_i^1 and the last aftermarket offer D_i^2 , with $D_i^h = 0$ if the student receives no offer in stage h , as in Section 4. For each stage in which the student receives an offer, we record the take-up decision $T_{iD_i^h}^h$ for their last offer. The mechanism also produces initial-offer and final cutoffs for each pool $P \in \{O, R\}$ at degree j . The initial-offer cutoff $c_j^{P,(1)}$ is the lowest application score among students offered a seat in that pool, regardless of take-up. The final cutoff $c_j^P(\mu)$ is the lowest application score among students holding a seat in that pool at the end of the mechanism. The initial-offer cutoffs by AA status are therefore $c_j^{1,NA} = c_j^{O,(1)}$ for non-targeted students and $c_j^{1,AA} = \min\{c_j^{O,(1)}, c_j^{R,(1)}\}$ for targeted students.

Algorithm 1 collects these steps.

Algorithm 1 Two-stage assignment mechanism $\mu = \varphi(R, T, \mathbf{s}, \mathbf{t}; q, \omega)$

Require: student types Θ (each θ_i with preferences $\succ_i$, take-up rules T_{ij}^1, T_{ij}^2 , scores $s_i = (s_{i1}, \dots, s_{iJ})$, and AA status t_i), utilities u_{ij} representing $\succ_i$, capacities q , reservation schedule ω , precedence $p \in \{0, 1\}$ (fixed at $p = 0$ in our implementation).

- 1: Split capacities: $q_j^R \leftarrow \omega_j$, $q_j^O \leftarrow q_j - q_j^R$ for all j (for the common-share schedules $\omega(x)$ with $0 < x < 1$, any pool that rounds to zero seats is given one seat, so that both pools exist).
- 2: Initialize $\tilde{q}_j^O \leftarrow q_j^O$, $\tilde{q}_j^R \leftarrow q_j^R$, $A_i \leftarrow 0$, $P_i \leftarrow 0$, $\underline{u}_i \leftarrow -\infty$, $\mathcal{R}_i \leftarrow \emptyset$; $D_i^2 \leftarrow 0$.

Initial offers: student-proposing deferred acceptance

- 3: Run student-proposing deferred acceptance on $(\succ_i, s_i, t_i)$ with pool capacities (q_j^O, q_j^R) and precedence p . Let D_i^1 be the resulting offer and $P_i^{(1)} \in \{O, R\}$ the pool through which it was issued.
- 4: **for** each student i with $D_i^1 = j > 0$ **do**
- 5: $\underline{u}_i \leftarrow u_{ij}$ ▷ raise utility floor regardless of take-up
- 6: **if** $T_{ij}^1 = 1$ **then**
- 7: $\tilde{q}_j^{P_i^{(1)}} \leftarrow \tilde{q}_j^{P_i^{(1)}} - 1$; $A_i \leftarrow j$; $P_i \leftarrow P_i^{(1)}$ ▷ accept: lock seat
- 8: **else**
- 9: $\mathcal{R}_i \leftarrow \mathcal{R}_i \cup \{j\}$ ▷ decline: record refusal
- 10: **end if**
- 11: **end for**

Aftermarket: college-proposing waitlist

- 12: For each pool (j, P) let L_{jP} be its priority list, the students with t_i permitting P ordered by s_{ij} , and set its pointer $k_{jP} \leftarrow 1$.
- 13: **repeat**
- 14: $freed \leftarrow \text{false}$
- 15: **for** each pool (j, P) , open pools first and then reserved pools **do**
- 16: **while** $\tilde{q}_j^P > 0$ and $k_{jP} \leq |L_{jP}|$ **do**
- 17: $i \leftarrow L_{jP}[k_{jP}]$; $k_{jP} \leftarrow k_{jP} + 1$
- 18: **if** $u_{ij} \leq \underline{u}_i$ or $j \in \mathcal{R}_i$ **then** ▷ i is not on the waitlist of (j, P)
- 19: **continue**
- 20: **end if**
- 21: $D_i^2 \leftarrow j$ ▷ record the offer
- 22: **if** $T_{ij}^2 = 1$ **then** ▷ accept: move up to j
- 23: $\tilde{q}_j^P \leftarrow \tilde{q}_j^P - 1$
- 24: **if** $A_i \neq 0$ **then**
- 25: $\tilde{q}_{A_i}^{P_i} \leftarrow \tilde{q}_{A_i}^{P_i} + 1$; $freed \leftarrow \text{true}$ ▷ return previously held seat
- 26: **end if**
- 27: $A_i \leftarrow j$; $P_i \leftarrow P$; $\underline{u}_i \leftarrow u_{ij}$
- 28: **else** ▷ decline: record refusal, keep $\underline{u}_i$
- 29: $\mathcal{R}_i \leftarrow \mathcal{R}_i \cup \{j\}$
- 30: **end if**
- 31: **end while**
- 32: **end for**
- 33: **until** $freed = \text{false}$

- 34: Let $c_j^{P, (1)}$ be the minimum score across students offered a seat in pool P of degree j in the initial-offer stage, and $c_j^P(\mu)$ the minimum score across students with $A_i = j$ and $P_i = P$ at termination.
 - 35: **return** enrollments A_i , offers D_i^1, D_i^2 and the take-up decisions $T_{iD_i^1}^1, T_{iD_i^2}^2$, initial-offer cutoffs $c_j^{P, (1)}$, and final cutoffs $c_j^P(\mu)$.
-

E Average and marginal effects in a single-degree example

In this appendix we use a simple example to show, within the framework of Section 4, what shapes the gains and losses from AA and how the average effect among reallocated students differs from the effect at the margin.

For expositional clarity, assume a single selective degree ($j = 1$) and an outside option ($j = 0$), a single admissions stage, and full take-up, so that $A_i(\mu) = D_i(\mu)$. The AA policy ω' produces admissions cutoffs c^{NA} for non-targeted and c^{AA} for targeted students. With a single degree, a displaced student can only exit to the outside option, so the knock-on effects of Section 4.2 are absent and each benefited student displaces exactly one other. The example therefore isolates the first of the two differences in that section and holds the second fixed.

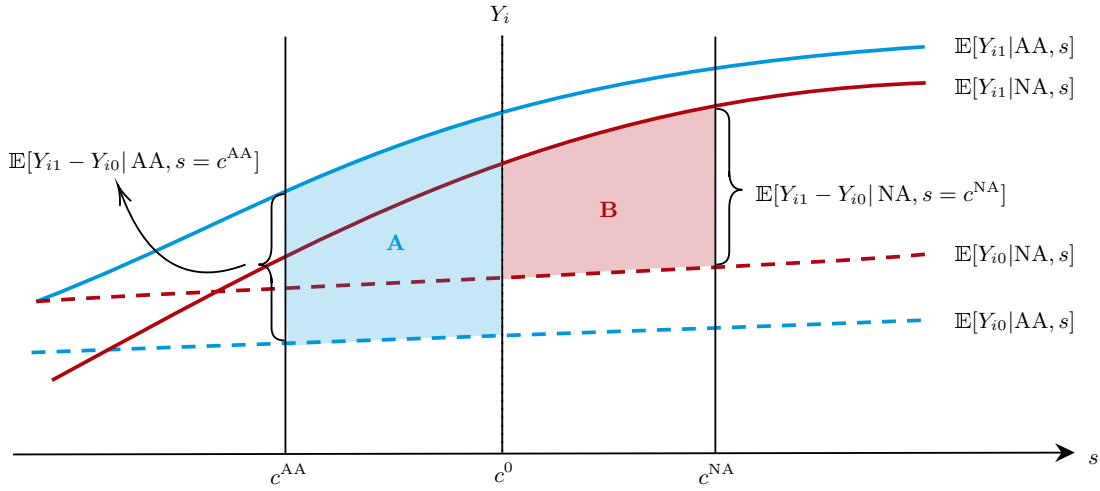


Figure E.1: Marginal and average gains from AA

Notes: Each line denotes the mean potential outcome for targeted (blue) and non-targeted (red) individuals. The solid line presents the expected outcome of attending the selective degree, and the dashed one the expected outcome under the outside option.

Figure E.1 plots outcomes by application score and AA status. The solid lines show the outcome of enrolling in the selective degree, $\mathbb{E}[Y_{i1}|t, s]$, and the dashed lines the outcome of the outside option, $\mathbb{E}[Y_{i0}|t, s]$. The figure shows why the average effect among reallocated students need not equal the effect at the margin. The effect on the marginal student compares the gain to the marginally benefited targeted student, $\mathbb{E}[Y_{i1} - Y_{i0}|AA, s = c^{\text{AA}}]$, with the loss to the marginally displaced non-targeted student, $\mathbb{E}[Y_{i1} - Y_{i0}|NA, s = c^{\text{NA}}]$. The average effect, in contrast, covers every student the schedule reallocates. Without AA, a single cutoff c^0 would apply to all students, so AA admits targeted students with $s_i \in [c^{\text{AA}}, c^0]$ and displaces non-targeted students with $s_i \in [c^0, c^{\text{NA}}]$; Areas A and B are their aggregate gains and losses.

Three forces determine the size of these areas, and so the average gains and losses from the policy. First, the solid lines slope upward. The steeper the relationship between application scores and outcomes, the larger the loss from moving a seat from a higher- to a lower-scoring student. Second, this relationship can differ across groups. If, for instance, application scores are biased against the targeted group, a targeted student has a higher outcome than a non-targeted student

with the same score, reducing the loss. Third, the dashed lines show that outside options may differ. The larger the gap in favor of non-targeted students, the less a displaced non-targeted student loses.

Because the return to enrolling—the distance between a group’s solid and dashed lines—widens with the application score, the average gain among reallocated targeted students exceeds the gain to the student at the cutoff.

F Cutoff dynamics in the iterative mechanism

In SISU’s iterative mechanism, students can revise their rank-order lists throughout the application period. At the end of each day, the system publishes a provisional cutoff for each degree and admissions track, equal to the lowest score that would receive an initial offer under the current lists. The administrative data retain only the final initial-offer cutoffs. To examine how the cutoffs converge and what information students have before finalizing their lists, we scraped the daily cutoffs for the Federal University of Minas Gerais (UFMG) during the 2021 admissions period. We observe 450 degree-track cutoffs (90 degrees with five admissions tracks each) on each of the nine application days.

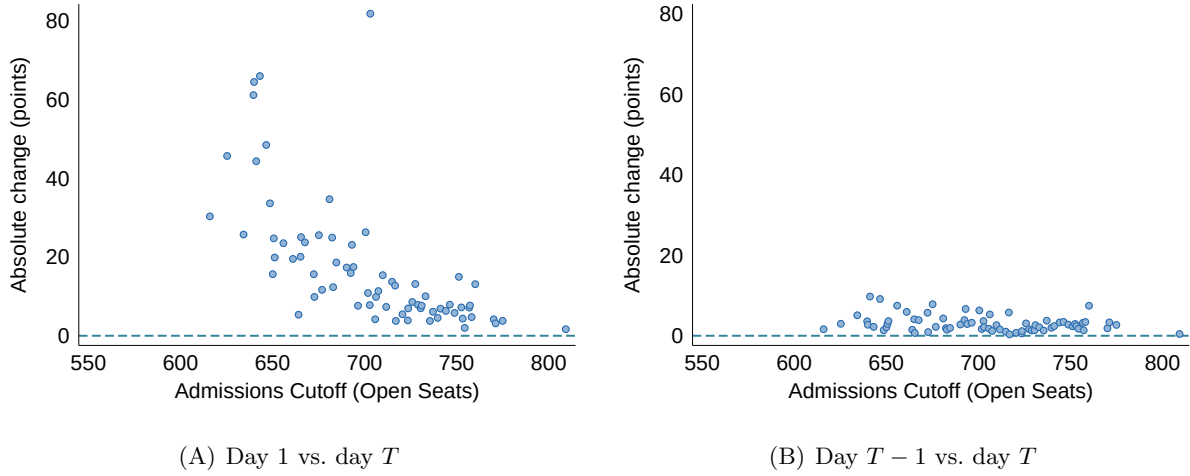


Figure F.1: Absolute change in admissions cutoffs

Notes: This figure plots the absolute difference between provisional and final initial-offer cutoffs at UFMG in 2021 against each degree’s final open-seat cutoff. Panel A uses the provisional cutoffs on the first day and Panel B those on the penultimate day, $T - 1$, where T is the final application day. Each point is a seat-weighted mean within one of 100 equal-sized bins of degree-track pairs. The dashed line marks zero.

Figure F.1 shows that cutoffs converge over the application period. On the first day, they can differ substantially from their final values, especially for less selective degrees (Panel A). By the penultimate day, however, most cutoffs are within a few score points of their final values (Panel B). This convergence suggests that students face little uncertainty about which degrees are feasible when they finalize their lists, supporting the stability assumption in Section 6.1. The small final-day changes also provide little indication of the manipulation described by Bó and Hakimov (2022), in which high-scoring students temporarily inflate cutoffs by listing degrees and then withdraw their applications at the last moment.

G Derivation of conditional expectations

This appendix derives the control-function terms that enter the outcomes estimating Equations (23)–(24),

$$\lambda_{iak} \equiv \mathbb{E}[\tilde{\eta}_{ik} \mid \Omega_i, R_i, \mathcal{H}_i, A_i = a], \quad m_{ia}^h \equiv \mathbb{E}[\epsilon_i^h \mid \Omega_i, R_i, \mathcal{H}_i, A_i = a],$$

where Ω_i is the student's initial-offer feasible set, $R_i = (R_{i1}, R_{i2})$ is the submitted rank-order list, and A_i is the final assignment. The offer and take-up history is $\mathcal{H}_i = \{(D_i^h, T_{iD_i^h}^h) : h = 1, 2, D_i^h > 0\}$. The variable D_i^h denotes the degree offered in stage h , with zero indicating no offer, and $T_{iD_i^h}^h$ indicates whether the offer was taken up. The demeaned preference shock is $\tilde{\eta}_{ik} \equiv \eta_{ik} - \bar{\eta}$, where $\bar{\eta}$ is the mean of a standard Type I Extreme Value (EV1) random variable. The take-up shock in stage h is ϵ_i^h . These objects follow the definitions in Sections 4 and 6.

The offer and take-up history records which offers the student received and accepted, and therefore determines the final assignment A_i . We omit A_i from the conditioning set below because it adds no information beyond $\mathcal{H}_i$. We keep the subscript a to match Equations (23)–(24). In the data, the history contains at most one offer per stage, since students can join the waitlist of their first choice only (see Section 2). The take-up decision on an offer from degree j in stage h depends on the preference shock $\tilde{\eta}_{ij}$ and on the take-up shock ϵ_i^h . Given (Ω_i, R_i) , the offers are determined by the mechanism and by other students' decisions, so the history is informative about the shocks only through the take-up decisions. As an approximation, we treat each take-up decision as informative only about the shock of the offered degree and the take-up shock of that stage.³ The λ terms correct for selection on degree preferences, and the m terms for selection on take-up shocks. We derive each term in turn, after introducing notation.

Choice probabilities. The revealed-preference information comes from the rank-order list R_i given the feasible set Ω_i . Let $C_i \equiv \Omega_i \cup \{R_{i1}, R_{i2}\}$ be the comparison set of Section 6.1. The report probability p_i^R is the product of the logit probabilities of ranking R_{i1} first from C_i and R_{i2} first from $C_i \setminus \{R_{i1}\}$,

$$\hat{P}_{iR_{i1}}^{(1)} = \frac{\exp(\hat{V}_{iR_{i1}})}{\sum_{m \in C_i} \exp(\hat{V}_{im})}, \quad \hat{P}_{iR_{i2}}^{(2)} = \frac{\exp(\hat{V}_{iR_{i2}})}{\sum_{m \in C_i \setminus \{R_{i1}\}} \exp(\hat{V}_{im})},$$

so that $p_i^R = \hat{P}_{iR_{i1}}^{(1)} \hat{P}_{iR_{i2}}^{(2)}$, where $\hat{V}_{ij}$ is V_{ij} evaluated at the preference estimates $\hat{\kappa}^t$.

For a set S of degrees, let $\mathcal{P}_i(k \mid S)$ be the logit probability that degree k is ranked first from S , and let $\mathcal{I}_i(S)$ be the inclusive value of S ,

$$\mathcal{P}_i(k \mid S) = \frac{\exp(\hat{V}_{ik})}{\sum_{m \in S} \exp(\hat{V}_{im})}, \quad \mathcal{I}_i(S) = \bar{\eta} + \ln \sum_{m \in S} \exp(\hat{V}_{im}).$$

³A decision on a listed degree is also informative about the shocks of the other degrees, because the list orders R_{i1} above R_{i2} and R_{i2} above every other feasible degree. This information is indirect, and it matters mainly for students with offers in both stages. Using it would make every term depend on the full history, with double integrals for the feasible unlisted degrees and for students with two offers. We ignore it, so that each term depends on the history only through the offer of the degree in question.

For a subset $S' \subseteq S$, $\mathcal{P}_i(S' | S) = \sum_{k \in S'} \mathcal{P}_i(k | S)$ is the probability that the first-ranked degree from S lies in S' . The report probabilities above are $\hat{P}_{iR_{i1}}^{(1)} = \mathcal{P}_i(R_{i1} | C_i)$ and $\hat{P}_{iR_{i2}}^{(2)} = \mathcal{P}_i(R_{i2} | C_i \setminus \{R_{i1}\})$.

Preference shocks given the list. We need the density of $\tilde{\eta}_{ij}$ given (Ω_i, R_i) for each listed degree j . Offers are only for listed degrees, so these are the only densities the history requires. Write $f_{\text{EV1}}(\tilde{\eta}; \chi)$ for the EV1 density with location χ and unit scale. For the first-ranked degree, the list reveals that R_{i1} beats every other degree in C_i , and the shock has density $f_{\text{EV1}}(\tilde{\eta}; \chi_i^{(1)})$ with $\chi_i^{(1)} = -\ln \hat{P}_{iR_{i1}}^{(1)} - \bar{\eta}$ (Dubin and McFadden, 1984).

For the second-ranked degree, the list reveals that R_{i2} loses to R_{i1} and beats the rest of C_i . Consider two events. The first is that R_{i2} beats every other degree in $C_i \setminus \{R_{i1}\}$. It has probability $\hat{P}_{iR_{i2}}^{(2)}$, and on this event the shock has density $f_{\text{EV1}}(\tilde{\eta}; \chi_i^{(2)})$ with $\chi_i^{(2)} = -\ln \hat{P}_{iR_{i2}}^{(2)} - \bar{\eta}$. The second is that R_{i2} beats every other degree in C_i , including R_{i1} . It has probability $\mathcal{P}_i(R_{i2} | C_i) = (1 - \hat{P}_{iR_{i1}}^{(1)}) \hat{P}_{iR_{i2}}^{(2)}$, and on this event the shock has density $f_{\text{EV1}}(\tilde{\eta}; \chi_i^C)$ with $\chi_i^C = -\ln \mathcal{P}_i(R_{i2} | C_i) - \bar{\eta}$. The second event is contained in the first, and the list is the first event without the second. The joint density of the list and the shock is therefore $\hat{P}_{iR_{i2}}^{(2)} f_{\text{EV1}}(\tilde{\eta}; \chi_i^{(2)}) - \mathcal{P}_i(R_{i2} | C_i) f_{\text{EV1}}(\tilde{\eta}; \chi_i^C)$. Dividing by the probability of the list, $\hat{P}_{iR_{i1}}^{(1)} \hat{P}_{iR_{i2}}^{(2)}$, gives

$$f(\tilde{\eta} | \Omega_i, R_i) = \frac{f_{\text{EV1}}(\tilde{\eta}; \chi_i^{(2)}) - (1 - \hat{P}_{iR_{i1}}^{(1)}) f_{\text{EV1}}(\tilde{\eta}; \chi_i^C)}{\hat{P}_{iR_{i1}}^{(1)} \hat{P}_{iR_{i2}}^{(2)}}. \quad (\text{G.1})$$

These densities are the $f_{\kappa^t}(\tilde{\eta} | \Omega_i, R_i)$ of Equation (17).

Offer take-up. The take-up decision introduces a second source of selection, through the take-up shock ϵ_i^h . Equation (14) can be written as $T_{ij}^h = \mathbb{1}\{T_{ij}^{h*} > 0\}$ with

$$T_{ij}^{h*} = V_{ij} + \tilde{\eta}_{ij} - \nu_i^h - \sigma_e^t \epsilon_i^h,$$

where ν_i^h is the deterministic component of the outside option defined in Equation (15).

Let $\zeta(\tilde{\eta}) \equiv (V_{ij} + \tilde{\eta} - \nu_i^h) / \sigma_e^t$, so that take-up occurs when $\epsilon_i^h < \zeta(\tilde{\eta}_{ij})$, as in Equation (16). Conditioning the λ and m terms on take-up requires the following integrals:

$$\begin{aligned} \mathcal{Q}_1(V_{ij}, \nu_i^h, \sigma_e^t) &= \int \Phi(\zeta(\tilde{\eta})) f(\tilde{\eta} | \Omega_i, R_i) d\tilde{\eta}, \\ \mathcal{Q}_2(V_{ij}, \nu_i^h, \sigma_e^t) &= \int \phi(\zeta(\tilde{\eta})) f(\tilde{\eta} | \Omega_i, R_i) d\tilde{\eta}, \\ \mathcal{Q}_3(V_{ij}, \nu_i^h, \sigma_e^t) &= \int \tilde{\eta} \Phi(\zeta(\tilde{\eta})) f(\tilde{\eta} | \Omega_i, R_i) d\tilde{\eta}, \end{aligned} \quad (\text{G.2})$$

where Φ and ϕ are the standard normal CDF and PDF. The density $f(\tilde{\eta} | \Omega_i, R_i)$ is the one derived in the previous paragraph, the EV1 density with location $\chi_i^{(1)}$ if $j = R_{i1}$ and Equation (G.1) if $j = R_{i2}$. Let $\mathcal{Q}_0 \equiv \int \tilde{\eta} f(\tilde{\eta} | \Omega_i, R_i) d\tilde{\eta} = \mathbb{E}[\tilde{\eta}_{ij} | \Omega_i, R_i]$ be the mean of $\tilde{\eta}_{ij}$ given the list, before conditioning on take-up. It equals $-\ln \hat{P}_{iR_{i1}}^{(1)}$ if $j = R_{i1}$ and the mean of the density in Equation (G.1) if $j = R_{i2}$. $\mathcal{Q}_1$ is the take-up probability p_{ih}^T of Equation (17). $\mathcal{Q}_2$ is the normal density of ϵ_i^h at the take-up threshold, averaged over $\tilde{\eta}_{ij}$. $\mathcal{Q}_3$ is the mean of $\tilde{\eta}_{ij}$ weighted by the take-up probability.

1. Preference selection at the offered degree (λ_{iaj}). Consider a degree j offered in stage h , so that (j, T_{ij}^h) is in the history. Under the approximation above, a take-up decision on another offer is uninformative about $\tilde{\eta}_{ij}$. The history is then informative about $\tilde{\eta}_{ij}$ only through this offer and its take-up, and

$$\lambda_{iaj} = \mathbb{E}[\tilde{\eta}_{ij} \mid \Omega_i, R_i, D_i^h = j, T_{ij}^h].$$

Without the take-up decision, $\mathbb{E}[\tilde{\eta}_{ij} \mid \Omega_i, R_i, D_i^h = j] = \mathcal{Q}_0$. Since take-up depends on $\tilde{\eta}_{ij}$ through the take-up index, observing whether student i takes up the offer is informative about $\tilde{\eta}_{ij}$. By Bayes' rule, the conditional density of $\tilde{\eta}_{ij}$ given the offer and the take-up decision is proportional to $\Phi(\zeta(\tilde{\eta})) f(\tilde{\eta} \mid \Omega_i, R_i)$ for takers and $[1 - \Phi(\zeta(\tilde{\eta}))] f(\tilde{\eta} \mid \Omega_i, R_i)$ for non-takers. The resulting conditional expectations are

$$\lambda_{iaj} = \begin{cases} \frac{\mathcal{Q}_3}{\mathcal{Q}_1} & \text{if } T_{ij}^h = 1, \\ \frac{\mathcal{Q}_0 - \mathcal{Q}_3}{1 - \mathcal{Q}_1} & \text{if } T_{ij}^h = 0. \end{cases}$$

2. Preference selection at other degrees (λ_{iak}). Consider a degree k not offered in either stage. Under the approximation above, the take-up decisions are uninformative about $\tilde{\eta}_{ik}$. Since the offers themselves reveal nothing about the shocks, λ_{iak} does not depend on the history, and

$$\lambda_{iak} = \mathbb{E}[\tilde{\eta}_{ik} \mid \Omega_i, R_i] \equiv \lambda_{ik}.$$

This includes students with no offer, whose history is empty.

We now derive λ_{ik} . Because C_i includes the two listed degrees, the list is a two-option rank-ordered logit over C_i whether or not R_{i1} and R_{i2} lie in Ω_i . The four cases below are exhaustive and mutually exclusive. Table G.1 summarizes them.

Table G.1: Summary of λ_{ik} formulas by case, for degrees not offered in either stage

Case	Conditions	Equation
1	$k = R_{i1}$	(G.3)
2	$k = R_{i2}$	(G.4)
3	$k \in \Omega_i \setminus \{R_{i1}, R_{i2}\}$	(G.5)
4	$k \notin \Omega_i \cup \{R_{i1}, R_{i2}\}$	(G.6)

Case 1: $k = R_{i1}$. The first ranking reveals that R_{i1} is preferred to every alternative in C_i . Conditional on R_{i1} being ranked first, the ranking of the remaining degrees is independent of $u_{iR_{i1}}$, a property of independent EV1 utilities, so the second ranking is uninformative about $\tilde{\eta}_{ik}$. By the [Dubin and McFadden \(1984\)](#) formula for the chosen alternative,

$$\lambda_{ik} = -\ln \mathcal{P}_i(k \mid C_i). \quad (\text{G.3})$$

Case 2: $k = R_{i2}$. Degree k is unchosen from C_i (since R_{i1} was ranked first) and chosen from $C_i \setminus \{R_{i1}\}$. This is the second-ranked alternative in a standard rank-ordered logit. By [Abdulkadiroğlu et al. \(2020, Appendix B.1.1\)](#):

$$\lambda_{ik} = -(\hat{V}_{ik} + \bar{\eta}) + \frac{\mathcal{I}_i(C_i \setminus \{R_{i1}\}) - \mathcal{P}_i(C_i \setminus \{R_{i1}\} \mid C_i) \cdot \mathcal{I}_i(C_i)}{\mathcal{P}_i(R_{i1} \mid C_i)}. \quad (\text{G.4})$$

This expression is also the mean of the density in Equation (G.1).

Case 3: $k \in \Omega_i \setminus \{R_{i1}, R_{i2}\}$. Degree k appears in both C_i and $C_i \setminus \{R_{i1}\}$ and is unchosen from both. By Abdulkadiroğlu et al. (2020, Appendix B.1.2, list length $l(i) = 2$):

$$\lambda_{ik} = -(\hat{V}_{ik} + \bar{\eta}) + \frac{\mathcal{N}_{ik}}{\mathcal{D}_{ik}}, \quad (\text{G.5})$$

where

$$\begin{aligned} \mathcal{N}_{ik} &= [1 - \mathcal{P}_i(C_i \setminus \{R_{i1}, k\} \mid C_i \setminus \{k\})] (\bar{\eta} + \hat{V}_{ik}) - \mathcal{P}_i(k \mid C_i \setminus \{R_{i1}\}) \mathcal{I}_i(C_i \setminus \{R_{i1}\}) \\ &\quad + \mathcal{P}_i(C_i \setminus \{R_{i1}, k\} \mid C_i \setminus \{k\}) \mathcal{P}_i(k \mid C_i) \mathcal{I}_i(C_i), \\ \mathcal{D}_{ik} &= \frac{\mathcal{P}_i(R_{i1} \mid C_i) \cdot \mathcal{P}_i(R_{i2} \mid C_i \setminus \{R_{i1}\})}{\mathcal{P}_i(R_{i2} \mid C_i \setminus \{R_{i1}, k\})}. \end{aligned}$$

Case 4: $k \notin \Omega_i \cup \{R_{i1}, R_{i2}\}$. Degree k does not appear in C_i . By independence of the EV1 shocks, both rankings are uninformative about $\tilde{\eta}_{ik}$:

$$\lambda_{ik} = 0. \quad (\text{G.6})$$

3. Take-up selection (m_{ia}^h). The shock ϵ_i^h is independent of the preference shocks and of the take-up shock of the other stage. Under the approximation above, the history is informative about ϵ_i^h only through the stage- h offer and its take-up. For the degree j offered in stage h , $m_{ia}^h = \mathbb{E}[\epsilon_i^h \mid \Omega_i, R_i, D_i^h = j, T_{ij}^h]$, and $m_{ia}^h = 0$ if the student received no offer in stage h . Since $\epsilon_i^h \sim N(0, 1)$ and take-up occurs when $\epsilon_i^h < \zeta(\tilde{\eta}_{ij})$, given $\tilde{\eta}_{ij}$ the conditional expectation of ϵ_i^h is the inverse Mills ratio $-\phi(\zeta)/\Phi(\zeta)$ for takers and $\phi(\zeta)/[1 - \Phi(\zeta)]$ for non-takers, with $\zeta = \zeta(\tilde{\eta}_{ij})$. Averaging over $\tilde{\eta}_{ij}$ with the taker and non-taker densities of part 1 gives

$$m_{ia}^h = \begin{cases} -\frac{\mathcal{Q}_2}{\mathcal{Q}_1} & \text{if } T_{ij}^h = 1, \\ \frac{\mathcal{Q}_2}{1 - \mathcal{Q}_1} & \text{if } T_{ij}^h = 0. \end{cases}$$

In Equations (23)–(24), λ_{iak} for a degree k offered in the history is λ_{iaj} of part 1 with $j = k$, evaluated at its take-up decision. For every other degree, it is λ_{ik} of part 2. The terms m_{ia}^h come from part 3 in each stage with an offer and are zero in a stage without one.

H Full parameter estimates

Tables H.1–H.3 report the full parameter estimates of the offer choice model, the take-up model, and the outcomes model, one table per model. Table 4 in the main text reports a subset of these parameter estimates. Degree-level parameters are summarized by their mean, standard deviation, and 10th and 90th percentiles across degrees. For other parameters, we report the estimate with its standard error in parentheses. The sample and the definition of targeted students are those of Section 6.4.

Table H.1: Offer choice model parameters

	Targeted				Non-targeted			
	Mean	SD/(SE)	P10	P90	Mean	SD/(SE)	P10	P90
Degree FE (δ_j^t)	1.172	2.522	-2.085	4.267	1.633	3.377	-2.660	5.540
Log distance (κ_d^t)	-1.219	(0.005)			-1.015	(0.005)		
Log distance $\times$ same CZ	0.276	(0.007)			0.117	(0.008)		
Log distance $\times$ same state	0.224	(0.002)			0.182	(0.002)		
Log distance $\times$ income	0.054	(0.002)			0.055	(0.002)		
Log distance $\times$ score	-0.088	(0.005)			-0.106	(0.004)		
Same municipality (κ_{muni}^t)	-2.363	(0.032)			-2.066	(0.031)		
STEM $\times$ language (κ_ℓ^t)	-0.237	(0.012)			-0.382	(0.012)		
Medicine $\times$ language ($\kappa_{\text{Med},\ell}^t$)	1.706	(0.047)			1.389	(0.039)		
STEM $\times$ math (κ_m^t)	0.574	(0.013)			0.530	(0.014)		
Medicine $\times$ math ($\kappa_{\text{Med},m}^t$)	1.982	(0.044)			2.090	(0.037)		
Application score (κ_s^t)	-3.326	(0.062)			-4.221	(0.060)		
Quality $\times$ score (κ_{qs}^t)	29.187	(0.250)			42.742	(0.291)		
Share of students	51.0%				49.0%			
Number of students	543,814							
Number of degrees	1,719							

Notes: Coefficients of the offer choice model, Equation (12). δ_j^t are degree fixed effects. The distance block κ_d^t contains the coefficients on log distance, on log distance interacted with indicators for a campus in the student’s commuting zone and state and with centered log household income and average ENEM score, and on an indicator for a campus in the student’s municipality. The score block κ_s^t contains the coefficients on the application score, on average ENEM score interacted with the degree’s quality rank, and on language and math scores interacted with STEM and medicine indicators. All other fields form the omitted base. Distance is in log kilometers, scores in s.d., and the quality rank in $[0, 1]$.

Table H.2: Enrollment take-up model parameters

	Targeted				Non-targeted			
	Mean	SD/(SE)	P10	P90	Mean	SD/(SE)	P10	P90
Intercept (π_0^t)	11.483	(3.792)			36.700	(5.478)		
Female ($\pi_{w,1}^t$)	0.996	(0.203)			2.328	(0.241)		
Log age ($\pi_{w,2}^t$)	-5.739	(0.548)			-15.621	(1.008)		
Log HH income (survey) ($\pi_{w,3}^t$)	0.100	(0.105)			2.015	(0.145)		
Log HH income (municipal) ($\pi_{w,4}^t$)	1.088	(0.431)			0.968	(0.561)		
PPI ($\pi_{w,5}^t$)	1.364	(0.220)			-0.114	(0.252)		
Prior federal enrollment ($\pi_{w,6}^t$)	1.158	(0.309)			1.324	(0.361)		
Prior private enrollment ($\pi_{w,7}^t$)	4.258	(0.368)			1.130	(0.446)		
HS graduate in $t - 1$ ($\pi_{w,8}^t$)	1.588	(0.262)			2.858	(0.315)		
Average score ($\pi_{w,9}^t$)	25.136	(0.616)			39.424	(0.596)		
Local supply & competition								
Log federal degrees	-3.045	(0.274)			-2.662	(0.434)		
Log non-federal degrees in CZ	3.494	(0.290)			5.364	(0.493)		
Log high-scoring applicants	-0.243	(0.259)			-1.014	(0.345)		
Score bin 2	-4.080	(0.407)			-8.071	(0.645)		
Score bin 3	-5.282	(0.617)			-11.894	(0.792)		
Score bin 4	-4.633	(0.840)			-12.446	(0.974)		
Federal $\times$ bin 2	0.028	(0.228)			0.201	(0.418)		
Federal $\times$ bin 3	0.725	(0.256)			0.457	(0.420)		
Federal $\times$ bin 4	0.780	(0.266)			0.316	(0.407)		
Non-federal $\times$ bin 2	-0.561	(0.208)			-1.310	(0.391)		
Non-federal $\times$ bin 3	-1.857	(0.246)			-2.669	(0.408)		
Non-federal $\times$ bin 4	-1.375	(0.253)			-2.480	(0.396)		
Waitlist shift (π_h^t)	3.957	(0.248)			0.270	(0.263)		
Utility scale (σ_e^t)	19.898	(0.869)			23.432	(1.084)		
Students offered		75,125				83,926		
Initial offer		38,179				44,605		
Waitlist		36,946				39,321		
Enrollment rate		34.6%				35.7%		
Waitlist offers		29.7%				35.2%		

Notes: Coefficients of the take-up model, Equations (14)–(15). π_0^t is the intercept of the outside-option value ν_i^h , π_w^t are the coefficients on student characteristics, and π_z^t the coefficients on local college supply and competition in the student's commuting zone (CZ). PPI indicates a Black, Brown, or Indigenous student. HS graduate in $t - 1$ indicates high-school graduation in the year before application. Score bins 2 to 4 indicate an average ENEM score in the 75th–90th percentile, the 90th–95th percentile, or the top 5% of the applicant distribution, with lower scores as the base. π_h^t is the waitlist shift and σ_e^t the scale of the take-up shock ϵ_i^h . A larger π raises the outside-option value and lowers take-up. The bottom rows report the number of students offered a seat, by stage, and the enrollment rate, overall and among waitlist offers.

Table H.3: Outcomes model parameters

	Targeted				Non-targeted			
	Mean	SD/(SE)	P10	P90	Mean	SD/(SE)	P10	P90
<i>Panel A. OLS estimates</i>								
Degree FE (α_j^t)	719.3	49.9	690.0	754.4	728.6	100.7	703.2	760.7
STEM $\times$ language (β_ℓ^t)	-0.820	(0.898)			-1.435	(0.988)		
Medicine $\times$ language ($\beta_{\text{Med},\ell}^t$)	3.045	(4.547)			-4.616	(3.349)		
STEM $\times$ math (β_m^t)	9.364	(1.020)			4.581	(1.152)		
Medicine $\times$ math ($\beta_{\text{Med},m}^t$)	1.840	(5.087)			1.830	(4.489)		
Application score (β_s^t)	-14.748	(4.307)			-12.097	(4.735)		
Quality $\times$ score (β_{qs}^t)	4.816	(17.766)			-6.262	(25.185)		
Selection on gains (ρ^t)	0.527	(0.267)			-0.124	(0.314)		
Selection through take-up (ξ^t)	0.883	(1.421)			3.900	(7.600)		
Outside option (ξ_0^t)	1.296	(0.380)			0.475	(0.391)		
Selection on levels (ψ_j^t)	-0.403	13.936	-8.306	7.450	0.081	11.834	-6.281	5.505
Outside option (ψ_{0j}^t)	-0.460	2.234	-3.052	2.261	-0.589	2.298	-3.413	2.078
<i>Panel B. Empirical-Bayes posterior means</i>								
Degree FE (α_j^t)	721.7	9.8	711.2	732.1	733.3	8.8	724.4	741.8
STEM $\times$ language (β_ℓ^t)	-0.820	(0.898)			-1.435	(0.988)		
Medicine $\times$ language ($\beta_{\text{Med},\ell}^t$)	3.045	(4.547)			-4.616	(3.349)		
STEM $\times$ math (β_m^t)	9.364	(1.020)			4.581	(1.152)		
Medicine $\times$ math ($\beta_{\text{Med},m}^t$)	1.840	(5.087)			1.830	(4.489)		
Application score (β_s^t)	-14.748	(4.307)			-12.097	(4.735)		
Quality $\times$ score (β_{qs}^t)	4.816	(17.766)			-6.262	(25.185)		
Selection on gains (ρ^t)	0.527	(0.267)			-0.124	(0.314)		
Selection through take-up (ξ^t)	0.883	(1.421)			3.900	(7.600)		
Outside option (ξ_0^t)	1.296	(0.380)			0.475	(0.391)		
Selection on levels (ψ_j^t)	-0.548	0.651	-1.315	0.268	-0.230	0.300	-0.389	-0.043
Outside option (ψ_{0j}^t)	-0.485	0.501	-1.105	0.154	-0.612	0.657	-1.484	0.187

Notes: Coefficients of the outcomes model, Equations (19)–(20), estimated from Equations (23)–(24). The outcome is the college graduation rank defined in Section 3.1. α_j^t are degree fixed effects, and the score block β_s^t is defined as the score block of Table H.1. ρ^t is selection on gains, ξ^t and ξ_0^t selection through take-up for the inside and outside options, and ψ_j^t and ψ_{0j}^t selection on levels. Coefficients on the student covariates $\mathbf{x}_i$ are not reported. Panel A reports OLS estimates. Panel B reports the empirical-Bayes posterior means of the degree-level parameters α_j^t , ψ_j^t , and ψ_{0j}^t (Appendix I); the other coefficients are not shrunk and repeat Panel A.

I Empirical-Bayes shrinkage

This appendix describes the empirical-Bayes procedure behind the posterior means in Tables 4 and H.3. For each degree j and AA status t , the outcomes model has three degree-level parameters, the degree fixed effect α_j^t and the selection-on-levels coefficients ψ_j^t and ψ_{0j}^t . Collect them in $\varphi_j^t = (\alpha_j^t, \psi_j^t, \psi_{0j}^t)'$ and let $\hat{\varphi}_j^t$ be the OLS estimate from Equations (23)–(24), with sampling covariance matrix Ψ_j^t . These estimates are noisy for degrees with few students. Following Abdulkadiroğlu et al. (2020), we shrink them toward a prior mean that depends on degree characteristics.

The hierarchical model is

$$\begin{aligned}\hat{\varphi}_j^t \mid \varphi_j^t &\sim N(\varphi_j^t, \Psi_j^t), \\ \varphi_j^t &\sim N(\Delta^t \mathbf{z}_j, \Sigma^t),\end{aligned}$$

where $\mathbf{z}_j = (1, r_j, \mathbf{f}_j)'$ collects a constant, the degree's quality rank r_j , and the STEM and medicine indicators $\mathbf{f}_j$ of Section 6.1. The matrix Δ^t maps degree characteristics into the prior mean, and Σ^t is the prior covariance. Both are common across degrees for a given AA status t . We treat Ψ_j^t as known and replace it by its OLS estimate $\hat{\Psi}_j^t$. Integrating out φ_j^t , the estimate $\hat{\varphi}_j^t$ is normal with mean $\Delta^t \mathbf{z}_j$ and covariance $\hat{\Psi}_j^t + \Sigma^t$. Treating the estimates as independent across degrees, the log-likelihood of the hyperparameters is, up to a constant,

$$\ell(\Delta^t, \Sigma^t) = \sum_j \left[-\frac{1}{2} \ln |\hat{\Psi}_j^t + \Sigma^t| - \frac{1}{2} (\hat{\varphi}_j^t - \Delta^t \mathbf{z}_j)' (\hat{\Psi}_j^t + \Sigma^t)^{-1} (\hat{\varphi}_j^t - \Delta^t \mathbf{z}_j) \right],$$

and $(\hat{\Delta}^t, \hat{\Sigma}^t)$ maximize it. The posterior mean of φ_j^t given $\hat{\varphi}_j^t$ is

$$\hat{\varphi}_j^{t, \text{EB}} = \left((\hat{\Psi}_j^t)^{-1} + (\hat{\Sigma}^t)^{-1} \right)^{-1} \left((\hat{\Psi}_j^t)^{-1} \hat{\varphi}_j^t + (\hat{\Sigma}^t)^{-1} \hat{\Delta}^t \mathbf{z}_j \right),$$

a precision-weighted average of the estimate and the prior mean. Imprecise estimates are pulled toward the prior mean, and precise estimates stay close to their OLS values. We restrict Σ^t to be diagonal and use only the diagonal of $\hat{\Psi}_j^t$, so each of the three parameters is shrunk separately. We apply the procedure separately by AA status. The other coefficients of the outcomes model are not shrunk.

J Simulation procedure

We describe the procedure used to simulate assignments and outcomes under an AA schedule ω . Each simulation draws from the three parts of the model. The preferences of Section 6.1 depend on the preference shocks η_{ij} , the take-up rules of Section 6.2 on the take-up shocks ϵ_i^h , and the potential outcomes of Section 6.3 on the assignment and on both sets of shocks. All parameters are evaluated at the estimates of Section 6.4, and we omit the hats.

Let B denote the number of replications. Replication b under schedule ω proceeds as follows.

1. **Draw shocks.** Draw $\eta_{ij}^b \sim \text{EV1}$ for all (i, j) and $\epsilon_i^{h,b} \sim N(0, 1)$ for all i and $h \in \{1, 2\}$, independently. Let $\tilde{\eta}_{ij}^b = \eta_{ij}^b - \bar{\eta}$.
2. **Preferences.** Compute $u_{ij}^b = V_{ij} + \eta_{ij}^b$ for all (i, j) and the induced preference order $\succ_i^b$ over $\mathcal{J}$.
3. **Take-up rules.** For each stage h and each (i, j) , set

$$T_{ij}^{h,b} = \mathbb{1}\{V_{ij} + \tilde{\eta}_{ij}^b > \nu_i^h + \sigma_e^t \epsilon_i^{h,b}\},$$

as in Equation (14). Let $T_i^b = (T_i^{1,b}, T_i^{2,b})$ collect the two stage-specific rules.

4. **Types.** Form $\theta_i^b = (\succ_i^b, T_i^b, s_i, t_i)$ from the preferences, the take-up rules, the application scores, and the AA status, as in Section 4.
5. **Mechanism.** Run the mechanism of Appendix D, $\varphi(\succ^b, T^b, \mathbf{s}, \mathbf{t}; q, \omega)$, with students reporting $\succ_i^b$ truthfully (Section 7.1). It returns the assignment A_i^b , the offers $D_i^{1,b}$ and $D_i^{2,b}$, the initial-offer cutoffs, and the final cutoffs.
6. **Potential outcomes.** For every degree j and for the outside option, compute

$$Y_{ij}^b = \alpha_j^t + \beta_s^t s_{ij} + \beta_x^t \mathbf{x}_i + \sum_{k=1}^J \psi_k^t \tilde{\eta}_{ik}^b + \rho^t \tilde{\eta}_{ij}^b + \xi^t (\epsilon_i^{1,b} + \epsilon_i^{2,b}),$$

$$Y_{i0}^b = \alpha_0^t + \beta_{0x}^t \mathbf{x}_i + \sum_{k=1}^J \psi_{0k}^t \tilde{\eta}_{ik}^b + \xi_0^t (\epsilon_i^{1,b} + \epsilon_i^{2,b}),$$

which are Equations (19)–(20) with the residuals replaced by their conditional means in Equations (21)–(22).

7. **Realized outcome.** The outcome under the simulated assignment is $Y_i^b = Y_{iA_i^b}^b$.

We use the same draws for every schedule ω . Student i 's expected outcome under ω is the average over the $B = 1,000$ replications,

$$\bar{Y}_i(\omega) = \frac{1}{B} \sum_{b=1}^B Y_i^b.$$

K Stability of potential outcomes under the policy

The counterfactuals in Section 7 assume that potential outcomes do not change with the reservation schedule ω . This appendix examines that assumption. It would fail if, for example, universities relaxed grading standards in response to the policy, or if a student's outcomes depended on the composition of their peers. Ideally, we would compare the relationship between dropout and scores for comparable students in two degrees, one exposed to more AA than the other. We leverage variation in the policy's rollout to approximate this comparison.

We proceed in two steps. First, we estimate the relationship between dropout and scores in every degree and cohort. Second, we regress these estimates on the share of reserved seats in that degree. Because universities chose how many seats to reserve, and those that reserved more may also have treated targeted students differently, we instrument the reserved share with the increase required by the law for each degree and cohort. The law required every degree to reserve half of its seats by 2016, phased in from 2013, so it forced a large increase in degrees that reserved few seats in 2012 and no increase in degrees that already reserved half.

For this exercise, we use data on students who entered federal degree programs between 2012 and 2016 and whom we can match to their ENEM scores. We keep only programs that participated in SISU in both 2012 and 2016. We keep programs with at least 30 such students in the 2012 cohort, so that inclusion does not depend on changes in enrollment after the policy. These restrictions leave us with 8,051 degree-cohort cells. The two-step estimates use the 7,903 cells, across 1,632 degree programs and 69 institutions, in which Equation (K.1) can be estimated, which requires at least ten students.

Relationship between dropout, scores, and targeted status. For each degree j and entering cohort t , we estimate

$$\text{Dropout}_{ijt} = \alpha_{jt} + \beta_{jt} \text{Score}_i + \gamma_{jt} \text{Targeted}_i + \varepsilon_{ijt}, \quad (\text{K.1})$$

where Dropout_{ijt} indicates whether student i , who entered degree j in cohort t , dropped out within four years. Targeted_i equals one for targeted students, and ε_{ijt} is an error term. We subtract the 2012 mean from each student's average ENEM score and divide by 10 to define Score_i . The intercept α_{jt} gives the predicted dropout rate for non-targeted students at the 2012 mean score. We use this mean for every cohort so that we compare dropout levels at the same score over time. The coefficient β_{jt} measures the change in dropout, in percentage points, per 10 ENEM points, while γ_{jt} measures the dropout gap between targeted and non-targeted students with the same score. We then test whether these parameters change with the share of reserved seats. Because estimates are noisy in degree-cohort cells with few students, we report the second step using both the raw estimates, weighted by the inverse of their sampling variances, and empirical-Bayes posterior means.

Reserved shares and the dropout parameters. Let x_{jt} be the share of degree j 's seats reserved in the admission year of cohort t , in units of 10 percentage points. For each of the three parameters

of Equation (K.1), $\phi \in \{\alpha, \beta, \gamma\}$, we estimate

$$\phi_{jt} = \delta_j^\phi + \delta_t^\phi + \kappa_\phi x_{jt} + v_{jt}^\phi, \quad (\text{K.2})$$

where δ_j^ϕ and δ_t^ϕ are degree and cohort fixed effects and v_{jt}^ϕ is an error term, with one observation per degree and cohort. The coefficient κ_α is the change in dropout at the 2012 mean score per 10 percentage points of reserved share, κ_β is the change in the score gradient, and κ_γ is the change in the targeted gap. Multiplying by five gives the change for a degree going from no reserved seats to 50%.

If universities relaxed grading standards in response to the policy, we would expect lower dropout at a given test score ($\kappa_\alpha < 0$). The score gradient would remain unchanged if dropout fell equally at every score, but would flatten if lower-scoring students benefited more ($\kappa_\beta > 0$). Standards could also change differently for the two groups. If they were relaxed only for targeted students, their dropout would fall relative to non-targeted students with the same score ($\kappa_\gamma < 0$). AA also changes peer composition within degrees. If classmates affect learning, dropout could change even for students who stay in the same degree. Peer effects could raise or lower dropout, so they imply no particular sign for κ_α . Hence, we test whether the dropout level, score gradient, and gap between targeted and non-targeted students change with the reserved-seat share.

Universities chose x_{jt} , and those that reserved more seats may also have changed how they teach, support, or grade their students, so x_{jt} may be correlated with v_{jt}^ϕ . Following Mello (2022, 2023), we instrument x_{jt} with $G_{jt} = \max(0, \text{Policy}_t - Q_{j,2012})$, the increase the law required of degree j for cohort t , where Policy_t is the minimum reserved share the law set for the cohort's admission year, 12.5, 25, 37.5, and 50% for the 2013 to 2016 cohorts and zero before, and $Q_{j,2012}$ is the share of the degree's SISU seats reserved in 2012, in percentage points. Appendix Figure A.1 shows that many degrees already reserved seats for targeted students before the law, with some reserving half their seats. The identifying assumption is that, absent the law, dropout at a given score would have followed the same cohort trend in degrees that reserved different shares in 2012.⁴

First stage. Table K.1 reports the first-stage regression of the reserved share x_{jt} on the instrument G_{jt} . In a degree required to go from no reserved seats to 50%, the reserved share rose by 49.4 percentage points relative to degrees the law did not force, close to the required increase. The share of targeted entrants rose by 15.0 percentage points and the share of Black, Brown, and Indigenous (PPI) entrants by 7.4 percentage points.

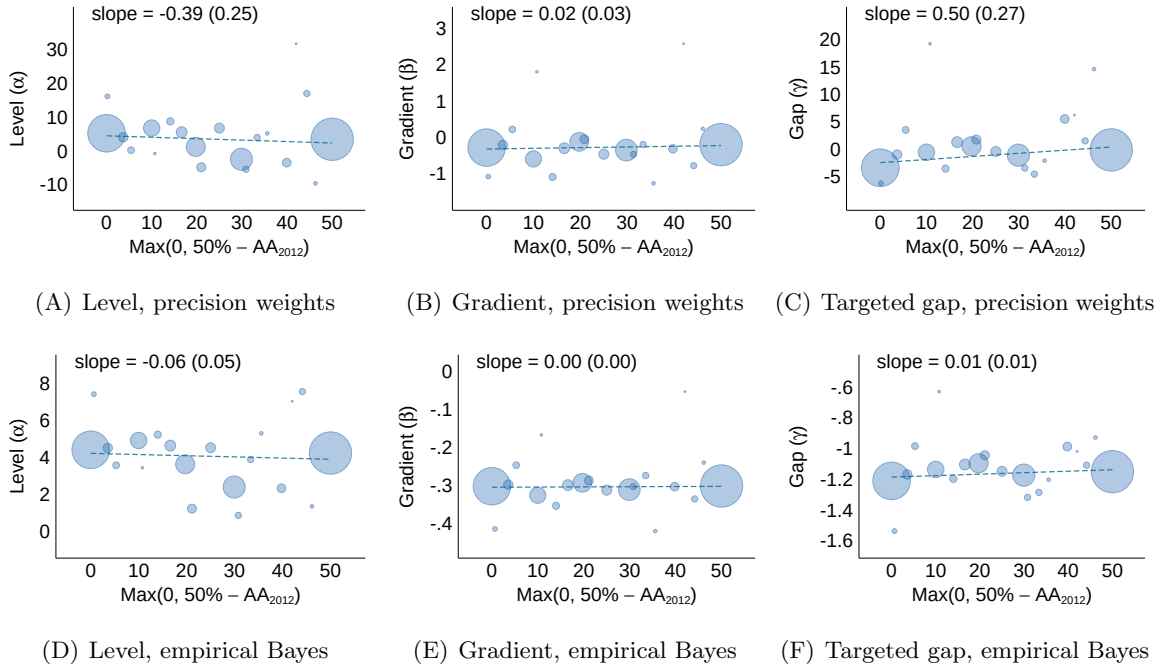
⁴One caveat is that, as the reserved share rises, the students who enroll at a given score change for both groups, so the estimates combine changes in dropout with changes in unobserved selection at a given score.

Table K.1: Reserved seats and composition of entrants, first stage

	Reserved share	Targeted share	PPI share
	(1)	(2)	(3)
Required increase (G_{jt})	9.87*** (0.32)	3.00*** (0.62)	1.49*** (0.24)
Mean, 2012 cohort	25.9	57.4	47.0
Observations	8,051	8,051	8,051

Notes: We regress degree-cohort means on the instrument G_{jt} , measured in units of 10 percentage points. Column 1 uses the share of seats reserved in the admission year, and columns 2 and 3 the shares of targeted and Black, Brown, and Indigenous (PPI) entrants. All outcomes are in percentage points. Multiplying coefficients by five gives the predicted changes for a degree required to move from no reserved seats to 50%. Regressions include degree and cohort fixed effects and weight cells by their number of students. Standard errors are clustered by institution. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reduced form. Figure K.1 presents the reduced form in long differences, plotting the change in each parameter between the 2012 and 2016 cohorts against the increase the law required. Overall, we find no strong correlation between the changes in the three parameters and the required increase in the reserved share.

**Figure K.1:** Changes in the dropout parameters and AA shares

Notes: Each panel plots the change in one parameter of Equation (K.1) between the 2012 and 2016 entering cohorts against the increase in the reserved share the law required of the degree by 2016, $\text{max}(0, 50 - Q_{j,2012})$, in percentage points. Degrees are grouped in bins of 2.5 percentage points, with the degrees at 0 and at 50 forming one bin each, and the size of each marker is proportional to the number of degrees in the bin. The first row uses the raw estimates, weighting each degree by the inverse of the sampling variance of its change; the second row uses empirical-Bayes posterior means with equal weights. The dashed line is the fitted regression across degrees, and the slope, per 10 percentage points of required increase, is reported with its standard error clustered by degree.

Results. Table K.2 reports the IV estimates. None of the coefficients is significant, whether the cells are weighted by precision or shrunk by empirical Bayes, and we cannot reject that κ_α , κ_β , and κ_γ are all zero. The level estimate κ_α is negative but small. The point estimate would imply that, for a degree that went from no reserved seats to 50%, dropout at the 2012 mean score changes by -1.4 percentage points. Consistent with no meaningful change in average dropout rates, the score gradient of dropout does not flatten (κ_β) and the targeted gap does not narrow (κ_γ). Columns 3 and 4 repeat the exercise for non-targeted students, whose eligibility the policy does not change, so a change in their dropout at a given score would reflect a degree-wide change in standards or peers rather than the changing composition of targeted admits. The estimates are similar. Overall, we find no evidence that dropout at a given score and degree changed where the reserved share rose, whether through grading standards or peer composition.

Table K.2: Reserved share and the score-dropout relationship

	All students		Non-targeted students	
	Precision weights (1)	Empirical Bayes (2)	Precision weights (3)	Empirical Bayes (4)
Level (κ_α)	-0.276 (0.518)	-0.113 (0.107)	-0.189 (0.500)	-0.159 (0.116)
Score gradient (κ_β)	0.0026 (0.0311)	-0.0001 (0.0024)	-0.0012 (0.0481)	0.0000 (0.0000)
Targeted gap (κ_γ)	0.310 (0.240)	0.008 (0.006)		
Joint test (p -value)	0.60	0.58	0.93	0.18
First-stage F	671	1,008	450	800
Observations	7,903	7,903	5,923	5,923
Number of degrees	1,632	1,632	1,299	1,299
Number of institutions	69	69	61	61

Notes: This table reports the coefficients κ_ϕ of Equation (K.2), the change in each parameter of Equation (K.1) per 10 percentage points of the degree's reserved share, instrumented by the increase the law required, G_{jt} , with degree and cohort fixed effects. Columns 1 and 3 weight cells by the inverse of the sampling variance of the first-step estimate. Columns 2 and 4 use empirical-Bayes posterior means, shrunk toward a prior mean given by degree and cohort effects. Columns 3 and 4 use non-targeted students only. The joint test is of $\kappa_\alpha = \kappa_\beta = \kappa_\gamma = 0$ in columns 1 and 2 and of $\kappa_\alpha = \kappa_\beta = 0$ in columns 3 and 4, and the first-stage F is the smallest across the equations. Standard errors clustered by institution are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.